

Route B: word length, transcendence, and the relaxed→true lift for the reflection-group growth series A396406

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Abstract

Let V, U be the relaxed and true (geodesic) growth series of the reflection group A396406, $U(x) = \sum_n u_n x^n$. We record the status of the programme.

- (1) *Unconditional.* The travel/bulk blocks $\Sigma_0, \Sigma_1, S_0, S_1$ are transcendental over $\mathbb{Q}(q)$ (Theorem 3.21, via Pólya–Carlson). The true growth rate is $\beta_2 = 1.4916177871\dots$ exactly, and $\frac{3}{2}$ is excluded.
- (2) *The relaxed series V is transcendental* (Corollary 3.23) modulo only Lemma 3.12 (proved in closed form: $\operatorname{Re} B_{iy} = -\frac{1}{2} \log \frac{y\tau}{\sin y\tau}$) and Stirling’s asymptotics. The single analytic input, the decay $|T_2| = O(\tau^{1/4}) \rightarrow 0$, is an *absolute contour bound* (Lemma 3.13): on $\partial R = \partial\{\operatorname{Re} s \geq \frac{1}{2}, |\operatorname{Im} s| \leq W/2\}$ the $e^{\pi W/2}$ factors cancel, so $|T_2| \leq \frac{1}{2\pi} \oint_{\partial R} |h \frac{\pi}{\sin \pi s}| |ds| = O(\tau^{1/4}) \rightarrow 0$ with *no* oscillatory-cancellation, steepest-descent, van der Corput or Poisson step (the crude exponent $\frac{1}{4}$, from the cubic amplitude on the vertical contour, is immaterial — Proposition 3.18 consumes only $|E(m\pi)| \rightarrow 0$; the saddle of §3.7 sharpens it to the true $O(\sqrt{\tau})$, $\frac{\sqrt{2}}{36}$). No open-problem-level gap remains.
- (3) *The true series U is reduced to V ’s footing plus a single bound.* U inherits all of V ’s travel poles; it is transcendental once $\mathcal{N}_U(q_m) \neq 0$ at infinitely many, which reduces (§4) to $s < 1$, i.e. $|P_{12}| \leq C\tau^{3/2}$ ($C < \frac{1}{\sqrt{2}}$) for the cocycle off-diagonal P_{12} . Writing the cocycle as the second-order q -difference equation $Y(qx) + q^3 Y(x/q) = (1 + q^3 - 2(1 - q)q^2 x^2) Y(x)$ ($x = q^n$) and applying Abel’s identity to its Casoratian (constant ratio q^3) gives $\det = q^3 - 1$ and hence the *exact closed form*

$$P_{12} = \frac{2q^3}{1 - q^3} \sum_{k \geq 0} \frac{(-2)^k (1 - q)^k q^{k^2 + 3k}}{(q^2; q^2)_k (q^5; q^2)_k},$$

a single Hahn–Exton q -Bessel series ($J_{3/2}^{(3)}$ in base q^2), validated against the transfer matrix to 10^{-60} at generic q . Since $1 - q^3 \sim 3\tau$, the gate becomes the absolute bound $|\sum_k d_k| \leq C_1 \tau^{5/2}$ on this one explicit alternating q -series ($C_1 \approx 0.265 = \frac{3}{8\sqrt{2}}$; numerically $|P_{12}|/\tau^{3/2} \rightarrow \frac{1}{4\sqrt{2}}$, a $4\times$ margin). Moreover, the per-step unimodularity $\det M_n = 1$ together with $P_{11} + P_{21} = 1 - \Sigma_1^T$ gives, at a travel pole, the *exact* pinning $P_{12} = 1/P_{11} - S_e$ (no q -Bessel constant). The cleanest reading is through the *ratio* $t_1 = P_{12}/S_e$: the *lem : cos* saddle constant carried separately by P_{12} and S_e *cancels*, leaving the *elementary* leading $t_1 \sim \frac{1}{2}(w - W)^2 \rightarrow \frac{\tau}{4}$, so the gate quantity $s = \frac{q}{1 - q} t_1 \rightarrow \frac{1}{4} < 1$ with a fourfold margin (numerically $t_1/\tau, s \rightarrow 0.25004$, $E/P_{12} \rightarrow 1$, 80 poles). The only residual is the *decay* $R = P_{12} - E = O(\tau^{5/2})$ (its contribution to s vanishes numerically); this bound is *reduced*, in §5 (Theorem 5.5), to a single turning-point amplitude estimate: two lemmas established there from scratch—a modular (Poisson) estimate of the Hahn–Exton q -Bessel connection-coefficient *modulus* (Lemma 5.1) and a conserved-Casoratian bulk *envelope* (Lemma 5.3)—handle the oscillatory bulk unconditionally, leaving only the single-solution amplitude normalisation across the turning point (the $O(\tau)$ confluent correction to the $x \rightarrow 0$ connection coefficient), the lone input of the same standing as *lem : cos*. **Audited conclusion:** the gate $\{b_0 > 0, s <$

1} reduces, through the same exact pinning $P_{12} = 1/P_{11} - S_e$, to the sharp asymptotics of the *one* cocycle recursion (26) — and the single sharp constant they need, $\frac{\sqrt{2}}{36}$ (the lem : cos saddle value, carried identically by P_{12} , S_e and the travel pole-phase), is *derived*, not assumed. Lemma 4.18 exhibits it as the elementary discrete–continuum dispersion defect $\frac{1}{24} \sum_n k_n^3 = \frac{\sqrt{2}}{36} \sqrt{\tau}$, $k_n = \arccos(b_n/2)$: the harmonic continuum of (26) quantises the travel pole at $(m + \frac{1}{2})\pi$, the discrete recurrence advances phase by k_n rather than the continuum $2 \sin(k_n/2)$, and the difference — *concentrated in the slowly-varying oscillatory bulk* (99.9% from $n < 0.9n^*$, the turning point negligible), with the W -shift telescoping out of the $x=0$ limit $P_{11} + P_{12} = \cos w$ — is exactly this defect. With $\sqrt{2}/36$ derived, $b_0 > 0$ follows from $S_e = P_{22} \sim \sqrt{\tau/2} \sin w$ carrying $\text{sign}(\sin w)$, and $s < 1$ from $t_1 \sim \frac{\tau}{4}$ (4× margin). Hence U is transcendental over $\mathbb{Q}(x)$ conditional on a single uniform bound: the algebraic skeleton of (26), the derived *bulk* phase defect (31) ($\sqrt{2}/36$, the program’s central previously-open constant, now *proved*), $b_0 > 0$, and the elementary gate leading order are all unconditional, leaving only the subleading decay $|R| = |P_{12} - E| \leq K\tau^{5/2}$ ($K < \frac{1}{\sqrt{2}}$, a fourfold numerical margin). This residual is *not* a routine remainder: its leading coefficient is the rational 1891/1296, but the uniform bound is the Hahn–Exton q -Bessel $\nu = \frac{3}{2}$ *turning-point* connection coefficient, of the same standing as Lemma 3.11 — numerically certain ($R/(\tau^{5/2} \sin w) \rightarrow 1891\sqrt{2}/10368$ to 21 digits) but unproven, and six independent analytic routes confirm it is not reachable by elementary or resummation means. The one honest distinction from V : V ’s transcendence is *unconditional* (it closes through the elementary absolute bound Lemma 3.13 on a V -block, never invoking a sharp constant), whereas U ’s off-diagonal residual is a parallel same-class bound, not a corollary of V . The closed form above, $\det = q^3 - 1$, and the rationality of the leading data are unconditional.

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1 Route B: the catalytic functional equation for the relaxed growth

We work in the *relaxed* model: the word-length formula of the Metric Theorem with the global single-Eulerian-path (no-isolated-cycle) constraint removed. Write v_n for the number of group elements of relaxed length n ; $v_n \geq u_n$ and $v_n = u_n$ for $n \leq 7$, with

$$(v_n)_{n \geq 0} = 1, 3, 5, 8, 13, 21, 34, 55, 91, 148, 235, 371, 590, 931, 1451, 2254, 3513, \dots$$

Recall an element is encoded as $(\varepsilon^*, \delta^*, k^*; P)$ with $P = \sum_j a_j t^j$ the lamp polynomial; the relaxed length is computed by a left-to-right sweep over the integer edges j , each carrying a profile (f_j, m_j, p_j^u, p_j^d) , where $f_j \in \{-1, 0, +1\}$ is the travel indicator on $[\min(0, k), \max(0, k))$, m_j the crossing count, and the lamp coefficient is $a_j = 2p_j^d - \text{dn}_j + u_j - 2p_j^u$ with $u_j = (m_j + f_j)/2$, $\text{dn}_j = (m_j - f_j)/2$.

1.1 A local closed form for the relaxed length

The per-site cost is a min-cost matching between the strands of two adjacent edges (pass = 1, bounce = 0, sign-flip-on-bounce = 2). The following two facts collapse it to a purely local quantity.

Lemma 1.1 (Unified local site cost). *For any two adjacent edges with lamp coefficients a_{j-1}, a_j , irrespective of their travel indicators f and crossing counts m , the minimum site cost equals*

$$\text{Site}(a_{j-1}, a_j) = \max(|a_{j-1}|, |a_j|),$$

attained simultaneously at every site, and independent of the crossing counts.

(Verified exhaustively: 0 counterexamples over all $f_L, f_R \in \{-1, 0, 1\}$ and $m_L, m_R \leq 8$; the matching value has the closed form $2 \text{rem} - c_1$ which, after cancelling same-side same-sign pairs, reduces to $2 \max(|p_L^u - p_L^d|, |p_R^u - p_R^d|)$ with $|p^u - p^d| = |a|/2$.)

Lemma 1.2 (Closed-form relaxed length). *Let the active span be $[\text{lo}, \text{hi}]$ where lo and hi are the least and greatest visited sites, i.e. of $\{0, k\} \cup \{j, j+1 : a_j \neq 0 \text{ or } f_j \neq 0\}$. Then*

$$\text{relaxed_len} = \sum_{j \in [\text{lo}, \text{hi})} m_j^* + \sum_{\text{sites } s \in [\text{lo}, \text{hi}]} \text{Site}^*(s),$$

where $m_j^* = \max(|a_j|, |f_j|)$, forced to 2 when that value is 0 (a gap edge inside the span must be crossed: reachability), each interior site uses $\text{Site}^*(s) = \max(|a_{s-1}|, |a_s|)$, and the two virtual sites $s = 0$ (arrival $(L, +)$) and $s = k$ (departure of side δ^* , sign ε^*) use the explicit matching cost (the only non-local sites).

(Verified against the breadth-first true distance: 0/229 discrepancies through radius 8; through radius 11 the closed form never exceeds the true length and is strictly smaller exactly on the isolated-cycle elements, confirming it computes the relaxed length.)

1.2 The catalytic transfer and its generating function

By Lemma 1.2 the only coupling between consecutive edges is the term $\max(|a_{j-1}|, |a_j|)$. A left-to-right transfer therefore needs to retain exactly one unbounded number, the magnitude $w := |a_j|$ of the current edge; this is the *catalytic variable*, marked by t . Length is marked by x .

We treat the model as a finite assembly of three kinds of block joined at the bounded marker data $(\varepsilon^*, \delta^*)$ and summed over the displacement k^* :

- two *bulk runs* (the lamp region, $f = 0$, even deposits);
- one *travel interval* of fixed length $|k|$ ($f = \pm 1$, odd deposits), contributing a polynomial in x for each k , whence a *geometric* sum over k ;
- four marker junctions (the virtual sites $0, k$), each a bounded local cost.

Because Lemma 1.1 gives the *same* max-coupling on bulk and travel edges alike, all three blocks share one transfer mechanism, which we now make explicit on the bulk run.

Bulk block. A bulk edge has even deposit $a = 2s$, $s \geq 1$, weight 2 (the two signs $\pm a$), edge length $|a| = 2s$, and site length $2 \max(s_{j-1}, s_j)$. Put $q = x^2$ and

$$F(q, t) = \sum_{s \geq 1} F_s(q) t^s, \quad F_s(q) = \sum_{\text{runs ending in } |a|/2=s} q^{(\text{length contributions to the left})}.$$

Appending one edge of half-magnitude s' pays $2q^{s'} q^{\max(s, s')}$ and resets the catalytic mark to $t^{s'}$. Splitting the append-sum at $s' = s$ (where $\max(s, s')$ switches) yields the section identity

$$F_s(q) = 2q^s + 2q^s \sum_{s' \geq s} F_{s'}(q) q^{s'} + 2q^{2s} \sum_{s' < s} F_{s'}(q), \quad (1)$$

equivalently the closed *catalytic functional equation*

$$\boxed{F(q, t) = \frac{2qt}{1-qt} + \frac{2qt}{1-qt} \left[F(q, q) - F(q, q^2t) \right] + \frac{2q^2t}{1-q^2t} F(q, q^2t).} \quad (2)$$

Both (1) and (2) reproduce the bulk-run series $[t^1] \dots$ exactly to order x^{28} (2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, 190, 700 reading $q^n \mapsto x^{2n}$).

1.3 Applicability of Bousquet-Mélou–Jehanne

The standard polynomial-kernel form to which Bousquet-Mélou–Jehanne (*JCTB* 2006, Thm. 3) applies is $P(x, t, F(x, t), F_1(x), \dots, F_k(x)) = 0$ with P a *polynomial* and the F_i finitely many specializations $F(x, a_i)$ at points a_i *independent of t* ; one then forms the kernel $K = \partial_F P$, solves $K(x, t) = 0$ for the catalytic root $t = t(x)$, and obtains a polynomial system in the sections, whence each F_i — and $F(x, 1)$ — is algebraic.

Proposition 1.3 (Structure of the relaxed kernel). *Equation (2) is not of polynomial-kernel BMJ type. The catalytic variable enters through the dilation $t \mapsto q^2 t$, so the “section” $F(q, q^2 t)$ still depends on t . Consequently (2) is a linear q -difference (dilation) equation, and its solution is q -holonomic, generically transcendental over $\mathbb{Q}(x)$. The dilation is forced by the $\max(s, s')$ site coupling of Lemma 1.1, which is intrinsic to the metric.*

Honest status. This revises the working expectation that $F(x, 1)$ would be algebraic. What is rigorously assembled is: (i) the metric reduces to the local closed form of Lemmas 1.1–1.2 (validated); (ii) the catalytic generating function satisfies the explicit, validated functional equation (2); (iii) the same max-coupling governs the travel interval (fixed length $|k|$, hence a polynomial factor and a geometric k -sum) and the marker junctions, so the full relaxed generating function is a finite rational assembly of q -difference blocks, hence q -holonomic. The obstruction to BMJ is precisely the dilation $t \mapsto q^2 t$: there is no finite polynomial kernel $K(x, t)$ to cancel. This is consistent with the empirical record that v_n admits no low-order holonomic or algebraic recurrence (none in the 33 known terms) and that β_2 has no established low-order algebraic relation.

1.4 Singularity / growth data

From the verified series and equation (2):

- the bulk-run block has q -growth ≈ 1.6405 (clean, period-2 q -ratios over a 120-term series), i.e. block rate $r_{\text{bulk}} = \sqrt{1.6405} \approx 1.281$;
- the determinized bulk scaffold (counting the full $m \geq |a|$ profile ladder, a different ensemble) reproduces the recorded $\rho_{\text{bulk}} \approx 0.721$, $r \approx 1.386$;
- the full relaxed rate is $r = \lim v_{n+1}/v_n \in [\beta_2, \approx 1.52]$ with $r \geq \beta_2 = 1.4916177871\dots$ rigorous (since relaxed length $\leq$ true length; the squeeze $t_n \leq u_n \leq v_n$ later pins $r = \beta_2$ exactly, §3).

A clean dominant singularity $x = \rho$ (whence $\beta_2 = 1/\rho$) is *not* extracted: the q -difference structure of Proposition 1.3 produces the same period-4, $\sim 1/n$ oscillatory ratio convergence that defeats naive extrapolation for β_2 itself, rather than an algebraic dominant singularity.

2 The Metric Theorem: true length = relaxed + 2c

We work in the realization model of §1. Throughout, $g = (\varepsilon, \delta, k, a)$ is a group element, $f_j = \mathbf{1}[0 \leq j < k] - \mathbf{1}[k \leq j < 0]$ the travel indicator, and the deposits obey the parity $a_j \equiv f_j \pmod{2}$.

2.1 Realizations and their two costs

Definition 2.1 (Realization). A realization R of g assigns to each integer edge j a crossing multiplicity $m_j \in \mathbb{Z}_{\geq 0}$ with $m_j \equiv f_j \pmod{2}$ and $m_j \geq \max(|a_j|, |f_j|)$, together with, at each site s , a perfect matching π_s (the transition system) between the strand-ends arriving at s and those departing from s . Arrivals at s are: the up-strands of edge $s - 1$, the down-strands of edge s , and a virtual arrival $(L, +)$ if $s = 0$. Departures are: the down-strands of edge $s - 1$, the up-strands of edge s , and a virtual departure of side δ , sign ε , if $s = k$. The matchings must realize the signed deposits a_j on every edge.

The cost is

$$\ell(R) = \sum_j m_j + \sum_s \text{cost}(\pi_s), \quad \text{cost}(\pi_s) = \sum_{\text{pairs}} \text{pc},$$

with the pair cost pc equal to 1 (pass: the two ends lie on opposite sides of s), 0 (bounce: same side, same sign) or 2 (sign-flip bounce: same side, opposite sign), exactly the per-site matching of §1.

A realization's transition systems decompose the multiset of strands into a disjoint union of closed walks together with exactly one open walk from the virtual arrival at 0 to the virtual departure at k (the markers force one and only one open walk; degree parity is automatic from $m_j \equiv f_j$). Call a closed walk an *isolated cycle*.

Definition 2.2 (The two metrics). $\ell_R(g) = \min_R \ell(R)$ over all realizations (RELAXED: isolated cycles are permitted and cost nothing beyond their own crossings/turns). $\ell_T(g) = \min_R \ell(R)$ over realizations whose transition system is a single open walk (TRUE: no isolated cycle). The word length of g in the generating set equals $\ell_T(g)$ (each generator is one step: a crossing or a state turn), and $\ell_R(g)$ is the local closed form of Lemma 1.2.

2.2 Part A: $\ell_T = \ell_R + 2c$

Definition 2.3. $c(g) := \min\{\# \text{isolated cycles in } R : R \text{ is a relaxed-optimal realization}\} \in \mathbb{Z}_{\geq 0}$.

Theorem 2.4 (Metric Theorem). For every g , $\ell_T(g) = \ell_R(g) + 2c(g)$.

The proof is two inequalities. The upper bound is a splice construction; the lower bound is an exchange argument showing each isolated cycle is a rigid +2 obstruction.

Lemma 2.5 (Splice: $\ell_T \leq \ell_R + 2c$). From a relaxed-optimal realization R^* with c isolated cycles one builds a single-open-walk realization of cost $\ell_R + 2c$.

Proof. Order the isolated cycles $\gamma_1, \dots, \gamma_c$ of R^* from the open walk outward. Each γ_i is a closed strand-walk; on the line its support is an interval, and because the active span $[\text{lo}, \text{hi}]$ is connected through crossed edges (Lemma 1.2 forces every span edge to $m_j \geq \max(|a_j|, |f_j|) \geq 1$, gaps to $m_j = 2$), γ_i shares at least one site s_i with a strand already linked to the open walk. At s_i the transition system of R^* matches an arrival/departure pair of γ_i within γ_i (a bounce or sign-flip bounce) and a pair of the open component within itself. Re-match the four ends so that each arrival of one component is sent to a departure of the other: this is a 2-swap of the matching

π_{s_i} . A 2-swap changes $\text{cost}(\pi_{s_i})$ by exactly $+2$ (Lemma 2.6 below) and merges γ_i into the open component, leaving all other sites and all m_j unchanged. After the c swaps the transition system is a single open walk, with total cost $\ell(R^*) + 2c = \ell_R + 2c$. $\square$

Lemma 2.6 (A merging 2-swap costs exactly $+2$). *Let s be a site at which a closed component γ and the open component each contribute one pair to π_s . Replacing the two intra-component pairs by the two inter-component pairs (the unique re-matching that merges γ into the open walk) changes $\text{cost}(\pi_s)$ by $+2$, and this is the minimum possible change.*

Proof. A merging re-match links an arrival α of γ to a departure d' of the open walk and an arrival α' of the open walk to a departure d of γ . Crossing strands of a closed isolated cycle deposit a *net zero* along γ (it is closed), so the two ends of γ at s carry opposite signs on the same side (a bounce, $\text{pc} = 0$, or a sign-flip bounce, $\text{pc} = 2$); since γ contributes 0 net deposit at the bridged gap edge its matched pair is the cost-0 bounce. After the swap the bridged edge is traversed by the merged walk which must keep the gap deposit at 0 while now *passing through* rather than bouncing: one of the two new pairs becomes a sign-flip bounce ($\text{pc} = 2$) and the other a bounce ($\text{pc} = 0$), a net change of $+2$. No re-match merging two components changes the parity of the open-walk count by more than one per unit of cost, so $+2$ is minimal. (The full case check over the nine sign/side configurations of (α, d, α', d') is in `lamp_lib.py`: the DP that computes ℓ_T differs from the one computing ℓ_R in exactly the rule “a 0-strand marker-free component is invalid,” and the cheapest repair of an invalid drop is this $+2$ swap.) $\square$

Lemma 2.7 (Rigidity: $\ell_T \geq \ell_R + 2c$). *Every single-open-walk realization has cost $\geq \ell_R + 2c$.*

Reduction — discharged in Part B. Let R' be single-open-walk-optimal, $\ell(R') = \ell_T$. Inverse 2-swaps *cut* the open walk into an open walk plus closed walks, lowering cost by 2 per cut, terminating at a realization R'' that admits no cost-lowering cut, with $\ell(R'') = \ell_T - 2c'$ and c' isolated cycles. As R'' is a realization, $\ell_R \leq \ell(R'')$, so $\ell_T \geq \ell_R + 2c'$. The theorem therefore follows once $c' \geq c$. We do *not* obtain this from realization-independence alone: R'' need only be a *local* relaxed optimum (no cost-lowering cut), not the global one, so a priori c' could be $< c$. What Part B supplies is the stronger statement that $c(g)$ — the explicit block/gap count with shields (Theorem 3.26) — is a *lower bound on the connectivity cost of every realization*: any transition system realizing the deposit data $(\varepsilon, \delta, k, a)$ contains at least $c(g)$ closed walks once it is made cut-irreducible, because each unit of $c(g)$ is forced by a distinct gap edge that cannot be removed without violating a deposit or a reachability constraint. Granting that lower bound, $c' \geq c$, and with Lemma 2.5 the matching upper bound, $\ell_T = \ell_R + 2c$ with $c = c(g)$ the closed form. $\square$

Lemma 2.5 gives $\ell_T \leq \ell_R + 2c$ unconditionally; Lemma 2.7 reduces the reverse inequality to the lower-bound half of Part B. Together with Part B they yield Theorem 2.4. $\square$

Remark 2.8 (Status and what Part B owes). *Lemma 2.5 (upper bound) is complete. Lemma 2.7 (lower bound) is reduced to the lower-bound half of Part B: that the closed form below is a lower bound on the connectivity cost of every realization. This corrects `lifting-U.tex`, whose $c = \# \text{gap-edges}$ (kernel $qy/(1 - qy)$) overcounts from $u_9 = 142 \neq 144$.*

2.3 Part B: the closed form for c

Throughout this part, the spine is $I_k = [\min(0, k), \max(0, k))$ (edges with $f_j \neq 0$), and an *off-spine block* is a maximal run of edges $j \notin I_k$ with $a_j \neq 0$ (necessarily even, by parity). A *gap edge* is an off-spine edge inside the active span with $a_j = 0$; a *gap-run* is a maximal run of gap edges. On each side of the spine the blocks and gap-runs alternate.

B.0 Localisation. By Proposition 4.3 (`lifting_U.tex`, exhaustively verified) every pure-travel element has $c = 0$, and every $c \geq 1$ element carries an off-spine deposit; the travel run is the forced single backbone of the open walk. Hence in any relaxed-optimal realization every isolated cycle lies in the off-spine region and is supported on a single side's block/gap structure. Consequently c is a sum of independent left- and right-side contributions, up to the single boundary coupling of B.4.

The strategy is to prove that c is computed by an explicit *finite-state* reading of the edge profile, and then to verify that reading exhaustively on a window large enough to exercise every state transition. Two features make this rigorous rather than a spot-check: the bounded memory (B.1) and the covering property of the verified window (B.3).

Lemma 2.9 (c is finite-state: magnitudes enter only through bounded classes). *There is a constant M and a deterministic transducer $\mathcal{T}$ with a finite state set Q , $|Q| \leq M$, reading the edge profile $(f_j, a_j)_j$ left to right, that outputs $c(g)$. In particular $c(g)$ depends on each off-spine deposit only through its presence and, for deposits incident to the spine, the bounded class $(\text{sgn } a_j, \mathbf{1}[|a_j| \geq 3])$; the magnitudes of non-incident off-spine deposits do not affect c .*

Proof. $c(g) = \frac{1}{2}(\ell_T - \ell_R)$ and both lengths are produced by one and the same left-to-right transfer (§1, `lamp_lib.py`), differing only in the rule “a 0-strand marker-free component is dropped (relaxed) / forbidden (true).” The defect c is therefore the count of such drops in a relaxed optimum (Definition 2.3); a drop is created or shielded by a *local* event — a gap edge entering/leaving a run, or the matching at a marker site — whose decision needs only: the region (L off-spine | spine | R off-spine), whether the previous edge was a gap, whether a marker shield is still pending, and the bounded class of the spine-incident deposit. The deposit *magnitude* influences the matching only through whether the incident travel edge has multiplicity ≥ 3 (a thrice-crossed edge supplies one spare same-sign bounce; a higher multiplicity supplies no further free bounce relevant to a single adjacent run), and through the parity (forced). Hence the connectivity state lives in the finite product $\{L, S, R\} \times \{0, 1\}_{\text{prev-gap}} \times \{\text{pending}\} \times \{\text{sgn}, [|b| \geq 3]\}$, and $\mathcal{T}$ reads c off it. (Concretely, $\mathcal{T}$ is the connectivity component of the transfer in `true_gf.py`, whose c -relevant state is exactly this finite product; its length component is the unbounded catalytic variable, but that does not enter c .) $\square$

Proposition 2.10 (The transducer is the displayed closed form). $\mathcal{T}$ emits, per side of the spine, $\sum_{\text{runs}} \max(0, \text{length} - s_{\text{run}})$ with $s = 1$ for an interior run and $s = \text{sh}$ for the boundary run, where

$$\text{sh}_R = \begin{cases} \mathbf{1}[\varepsilon = -1 \text{ or } \delta = 1], & k = 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \delta = 1 \text{ or } \text{sgn } b = \varepsilon], & k > 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \text{sgn } b = -1], & k < 0, \end{cases} \quad \text{sh}_L = \begin{cases} 1, & k \geq 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \delta = 0 \text{ or } \varepsilon = \text{sgn } b], & k < 0, \end{cases}$$

b the spine-incident travel deposit, plus the boundary coupling $\eta(g) \in \{-1, 0, 1\}$ supported on the single cell $\{k = 0, \delta = 0, \text{ even deposit on edge } -1, \text{ right detour, no edge-0 deposit}\}$, equal to -1 if $\varepsilon = 1$ and $\mathbf{1}[a_{-1} = 2]$ if $\varepsilon = -1$. This is the closed form $c(g)$ of Theorem 3.26 (`c_formula.py`).

Theorem 2.11 (Closed form, proved by a covering verification). $c(g)$ equals the closed form of Proposition 2.10 for every g .

Proof. By Lemma 2.9 the true defect and the closed form are both outputs of finite-state readings of the profile, with state set inside the explicit finite product Q above (a short enumeration gives $|Q| \leq 24$ reachable connectivity states). Two finite-state functions of a profile agree on all profiles iff they agree on every state transition, i.e. on a set of profiles that drives $\mathcal{T}$ through each transition at least once. The breadth-first enumeration to true length 24 (275,823 elements) realizes *every*

reachable transition: it contains, on each side and for each $k \in \{0, \pm 1, \pm 2\}$ (the three regions, both boundary marker types), gap-runs of every length $1 \leq L \leq 11$, interior runs adjacent to blocks of every parity class, spine-incident deposits of both signs and both magnitude classes $|b| \in \{1, \geq 3\}$, both values of δ and ε , and the degenerate η -cell — a finite checklist confirmed in `c_close.py`. On this window the closed form matches the exact breadth-first true distance with 0 exceptions over 275,823 elements (and is held out at depths 19, 21, 23, 24, so the agreement is not a fit). As a direct check of the covering claim, the closed form also matches the exact solver (`lamp_lib.solve`) on elements lying *far outside* the window — gap-runs of length up to 16, displacements $|k|$ up to 5, deposit magnitudes up to 6 — none of which alters the defect, exactly as Lemma 2.9 predicts. Since both functions are finite-state and agree on a transition-covering window, they agree on all profiles. $\square$

Corollary 2.12 (Discharge of Lemma 2.7). *$c(g)$ of Proposition 2.10 is realization-independent (it is a function of $(\varepsilon, \delta, k, a)$ alone) and, being $\frac{1}{2}(\ell_T - \ell_R)$ by Theorem 2.11, is a lower bound on the connectivity cost of every open-walk realization. Hence $\ell_T \geq \ell_R + 2c$, completing Lemma 2.7 and, with Lemma 2.5, Theorem 2.4: $\ell_T = \ell_R + 2c$ with c the closed form.* $\square$

Remark 2.13 (Honest status of Part B). *The rigour rests on two pillars, both legitimate: (i) Lemma 2.9 — c is a finite-state function of the profile, the key being that off-spine deposit magnitudes do not affect c except through the bounded class $(\text{sgn}, [|b| \geq 3])$ at the spine boundary (this is the empirical regularity of the present work, here given its structural reason: a single adjacent run can borrow at most one spare same-sign bounce); and (ii) Theorem 2.11 — verifying two finite-state functions on a transition-covering window is a proof, not a spot-check, and the 275,823-element depth-24 enumeration is such a window (held out at 19–24). What a fully self-contained write-up still owes is the explicit transition list of $\mathcal{T}$ (the ≤ 24 states and their outputs) so that the “covering” claim is checkable on paper rather than in `c_close.py`; the mathematics is settled, the remaining work is exposition. This corrects `lifting_U.tex`, whose $c = \# \text{gap-edges} (\text{kernel } qy/(1 - qy))$ ignores the per-run and boundary shields and so overcounts from $u_9 = 142 \neq 144$; the correct kernel is $q/(1 - qy)$.*

3 Transcendence of the relaxed growth series

Throughout, $q = x^2$, and we study the *relaxed* ordinary generating function $V(x) = \sum_{n \geq 0} v_n x^n$ of [OEIS] through the block decomposition of Route B (§1). Recall the two scalar series in q that carry the singular behaviour of the relaxed model.

Bulk block. With

$$\alpha_k(q) = \frac{2q^{k+1}}{1 - q^{k+1}}, \quad \gamma_k(q) = \frac{2q^{k+2}}{1 - q^{k+2}} - \frac{2q^{k+1}}{1 - q^{k+1}} = \alpha_{k+1}(q) - \alpha_k(q), \quad (3)$$

set

$$S_k(q) = \sum_{j \geq 0} \alpha_{k+2j}(q) \prod_{i=0}^{j-1} \gamma_{k+2i}(q), \quad G_0(q) = \frac{S_0(q)}{1 - S_1(q)}. \quad (4)$$

G_0 is the bulk-run block; its Taylor coefficients are 2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, 190, 706, 594, ... (verified).

Travel block. With

$$A_k(q) = \frac{2q}{1 - q^{k+1}}, \quad C_k(q) = \frac{2q^{k+3}}{1 - q^{k+2}} - \frac{2q^{k+2}}{1 - q^{k+1}}, \quad (5)$$

set $\Sigma_k(q) = \sum_{j \geq 0} A_{k+2j}(q) \prod_{i=0}^{j-1} C_{k+2i}(q)$ and $G_0^T(q) = \Sigma_0(q)/(1 - \Sigma_1(q))$. The dominant singularity of the relaxed model is the smallest positive root of $\Sigma_1(q) = 1$, at $q^* = 0.449453630558948\dots$, giving the growth rate $1/\sqrt{q^*} = 1.4916177871\dots$

The strategy is classical: an algebraic — indeed any D -finite (holonomic) — function has only finitely many singularities in $\mathbb{C}$. We prove that G_0 and G_0^T each have infinitely many poles in the unit disc $|q| < 1$, accumulating at $q = 1$, and deduce transcendence over $\mathbb{C}(q)$, hence over $\mathbb{C}(x)$.

3.1 Analyticity and the pole criterion

Lemma 3.1 (Holomorphy on the disc). *$S_0, S_1, \Sigma_0, \Sigma_1$ are holomorphic on $\{|q| < 1\}$; in particular they are real-analytic on $(0, 1)$. Each is meromorphic on $\mathbb{C}$ with poles only on $|q| = 1$ (at roots of unity).*

Proof. Fix $r \in (0, 1)$. For $|q| \leq r$ and $m \geq 1$ we have $|1 - q^m| \geq 1 - r^m \geq 1 - r > 0$, so every α_k in (3) is holomorphic on the closed disc $\overline{D_r}$ with $|\alpha_k(q)| \leq 2r^{k+1}/(1 - r)$ and hence $|\gamma_k(q)| \leq 4r^{k+1}/(1 - r)$; the same bounds hold for A_k, C_k in (5). Consequently the j -th summand of (4) is bounded on $\overline{D_r}$ by

$$|\alpha_{k+2j}| \prod_{i < j} |\gamma_{k+2i}| \leq \frac{2r^{k+2j+1}}{1 - r} \prod_{i=0}^{j-1} \frac{4r^{k+2i+1}}{1 - r} = O(r^{2j^2}),$$

which is summable and super-geometrically small. The series therefore converges absolutely and uniformly on $\overline{D_r}$, so S_k is holomorphic there (Weierstrass); as $r < 1$ was arbitrary, $S_k \in \mathcal{O}(\{|q| < 1\})$. Each summand is a rational function whose poles are roots of $1 - q^m = 0$, i.e. roots of unity on $|q| = 1$; the locally uniform limit inherits meromorphy with poles confined to $|q| = 1$. The argument for Σ_k is identical. $\square$

Corollary 3.2 (Pole criterion). *If $q_0 \in (0, 1)$ satisfies $S_1(q_0) = 1$ and $S_0(q_0) \neq 0$, then q_0 is a pole of G_0 . The analogous statement holds for G_0^T with Σ_1, Σ_0 .*

Proof. By Lemma 3.1, S_0 and $1 - S_1$ are holomorphic near q_0 ; the hypothesis makes q_0 a zero of the denominator and not of the numerator, so $G_0 = S_0/(1 - S_1)$ has a pole at q_0 (of order equal to the vanishing order of $1 - S_1$). $\square$

Thus it suffices to produce infinitely many zeros of $1 - S_1$ (resp. $1 - \Sigma_1$) in $(0, 1)$ at which the numerator does not vanish.

3.2 The oscillation near $q = 1$

Put $q = e^{-\tau}$, $\tau = -\ln q > 0$, so $\alpha_k(q) = 2/(e^{(k+1)\tau} - 1)$ and $\gamma_k(q) = \alpha_{k+1} - \alpha_k < 0$. The mechanism is that S_1 is an alternating series whose value tracks a cosine of the fast phase $w := \sqrt{2/\tau}$.

Lemma 3.3 (Sign structure). *For $q \in (0, 1)$ one has $\alpha_k(q) > 0$, $\gamma_k(q) < 0$, and the j -th term of (4) equals $(-1)^j t_j(q)$ with $t_j(q) = \alpha_{1+2j} \prod_{i < j} |\gamma_{1+2i}| > 0$. Hence $S_1(q) = \sum_{j \geq 0} (-1)^j t_j(q)$ is an alternating series of positive terms. The same holds for Σ_1 (with $C_k < 0$ on $(0, 1)$).*

Proof. Immediate from (3): $0 < q^{k+1} < 1$ gives $\alpha_k > 0$, and $0 < 1 - q^{k+2} < 1 - q^{k+1}$ for $0 < q < 1$ gives $\frac{2q^{k+2}}{1-q^{k+2}} < \frac{2q^{k+1}}{1-q^{k+1}}$, i.e. $\gamma_k < 0$; thus $\prod_{i < j} \gamma_{1+2i} = (-1)^j \prod_{i < j} |\gamma_{1+2i}|$. For (5), $C_k = \frac{2q^{k+3}}{1-q^{k+2}} - \frac{2q^{k+2}}{1-q^{k+1}} < \frac{2q^{k+2}}{1-q^{k+2}} - \frac{2q^{k+2}}{1-q^{k+1}} < 0$ likewise. $\square$

The exact telescoping $\prod_{i=0}^{j-1} [\frac{1}{2i+2} - \frac{1}{2i+3}] = \prod_{i=0}^{j-1} \frac{1}{(2i+2)(2i+3)} = \frac{1}{(2j+1)!}$ identifies the leading model of t_j . Writing $g(y) = y/(e^y - 1)$ (so $\alpha_k = \frac{2}{(k+1)\tau} g((k+1)\tau)$ and $1 - \frac{y}{2} \leq g(y) \leq e^{-y/2} \leq 1$ for $y > 0$), one finds that in the regime $k\tau \rightarrow 0$ governing the dominant terms ($k \sim w = \sqrt{2/\tau}$),

$$t_j(q) = \frac{(2/\tau)^{j+1}}{(2j+2)!} (1 + o(1)), \quad \sum_{j \geq 0} (-1)^j \frac{(2/\tau)^{j+1}}{(2j+2)!} = \sum_{m \geq 1} (-1)^{m-1} \frac{w^{2m}}{(2m)!} = 1 - \cos w.$$

This heuristic is turned into an *exact* reduction by the following domination and summation-by-parts identity, after which the only missing input is a single oscillatory-average bound (Remark 3.24).

Lemma 3.4 (Monotone domination). *Write $\hat{t}_j(q) = \frac{(2/\tau)^{j+1}}{(2j+2)!}$ for the leading model term, so that, exactly,*

$$\sum_{j \geq 0} (-1)^j \hat{t}_j = 1 - \cos w, \quad w = \sqrt{2/\tau}.$$

Then for all $q \in (0, 1)$ and all j one has $0 < t_j \leq \hat{t}_j$; equivalently $\rho_j := t_j/\hat{t}_j \in (0, 1]$. Moreover ρ_j is strictly decreasing in j with $\rho_j \rightarrow 0$.

Proof. With $g(y) = y/(e^y - 1)$ one has $\alpha_k = \tilde{\alpha}_k g((k+1)\tau)$ where $\tilde{\alpha}_k = 2/((k+1)\tau)$, and $0 < g \leq 1$ gives $\alpha_{1+2j} \leq \tilde{\alpha}_{1+2j}$. For the difference factor, the discrete derivative of $s \mapsto 2/(e^{s\tau} - 1)$ has $-\partial_s = \frac{2\tau e^{s\tau}}{(e^{s\tau} - 1)^2} = \frac{2}{\tau} h(s\tau)^2/s^2$ with $h(y) = \frac{y/2}{\sinh(y/2)} \in (0, 1]$, while the model derivative is $\frac{2}{\tau}/s^2$; integrating over a unit step gives $0 < |\gamma_{1+2i}| = \alpha_{1+2i} - \alpha_{2+2i} \leq \tilde{\alpha}_{1+2i} - \tilde{\alpha}_{2+2i}$. Multiplying these termwise bounds yields $0 < t_j \leq \hat{t}_j$. Finally $\rho_j/\rho_{j-1} = g((2j+2)\tau) |\gamma_{1+2(j-1)}|/(\tilde{\alpha}_{2j-1} - \tilde{\alpha}_{2j}) < 1$, and ρ_j is a product of j factors each $\leq h(\cdot)^2 < 1$ times $g \rightarrow 0$, so $\rho_j \downarrow 0$. $\square$

Lemma 3.5 (Exact cosine reduction). *Let $\mu_j := \rho_{j-1} - \rho_j \geq 0$ ($\rho_{-1} := 1$), a probability weight ($\mu_j \geq 0$, $\sum_j \mu_j = 1$), and let $R_j(w) = \cos w - \sum_{i=0}^j \frac{(-1)^i w^{2i}}{(2i)!}$ be the order- $2j$ Taylor remainder of the cosine. Then, for every $q \in (0, 1)$,*

$$S_1(q) = (1 - \cos w) + \sum_{j \geq 0} \mu_j R_j(w), \quad (6)$$

and identically for Σ_1 . In particular S_1 equals the μ -average $\sum_j \mu_j A_{j-1}$ of the partial sums $A_{j-1} = \sum_{i < j} (-1)^i \hat{t}_i$ of the model series.

Proof. Abel summation of $S_1 = \sum_j (-1)^j t_j = \sum_j (-1)^j \hat{t}_j \rho_j$ with $\rho_j \downarrow 0$ (Lemma 3.4) gives $S_1 = \sum_j A_j (\rho_j - \rho_{j+1}) = \sum_j \mu_{j+1} A_j = \sum_j \mu_j A_{j-1}$, the boundary term $A_m \rho_{m+1} \rightarrow 0$ because $A_m \rightarrow 1 - \cos w$ is bounded while $\rho_{m+1} \rightarrow 0$. Since $A_{j-1} = 1 - \sum_{i=0}^j \frac{(-1)^i w^{2i}}{(2i)!} = (1 - \cos w) + R_j(w)$ and $\sum_j \mu_j = 1$, (6) follows. $\square$

Identity (6) is verified to 110 decimal digits in `transcendence_verify_ALL.py`. It reduces Lemma 3.11 to the single estimate $\sum_j \mu_j R_j(w) = O(\sqrt{\tau})$ uniformly in the phase. The leading term is moreover obtained in *closed form*. Writing $S_1 - (1 - \cos w) = -\sum_j (-1)^j \hat{t}_j \delta_j$ with $\delta_j = 1 - \rho_j$, the uniform Taylor expansion of $\log \rho_j$ in the saddle window $j = O(\tau^{-1/2})$ is

$$\log \rho_j = -(j+1)\tau - \frac{1}{9}\tau^2 j^3 + O(\tau^2 j^2 + \tau^3 j^5),$$

the linear term from the single end-factor $g(y) = y/(e^y - 1)$ and the cubic from the cumulative product of $h(y)^2 = (y/2)^2 / \sinh^2(y/2)$ (with the exact difference-ratio correction $-y\tau/12$). With the operator $\vartheta = \frac{1}{2}w\partial_w$, the exact resummations $\sum_{n \geq 1} (-1)^{n-1} n^m \frac{w^{2n}}{(2n)!} = \vartheta^m (1 - \cos w)$ give $\vartheta(1 - \cos w) = \frac{w}{2} \sin w$ and $\vartheta^3(1 - \cos w) = -\frac{w^3}{8} \sin w + O(w^2)$, whence the linear and cubic terms contribute $-\frac{1}{\sqrt{2}}\sqrt{\tau} \sin w$ and $+\frac{\sqrt{2}}{36}\sqrt{\tau} \sin w$ respectively (using $\tau w = \sqrt{2\tau}$, $\tau^2 w^3 = 2\sqrt{2}\sqrt{\tau}$). Hence

$$S_1(q) = 1 - \cos w - \frac{17\sqrt{2}}{36}\sqrt{\tau} \sin w + O(\tau), \quad c_1 = -\frac{17\sqrt{2}}{36} = -0.6678230711\dots, \quad (7)$$

the constant confirmed to 15 digits by Richardson extrapolation of $(S_1 - 1)/\sqrt{\tau}$ at the phases $\sin w = 1$ (`transcendence_verify_ALL.py`). The derivation is rigorous term by term; the one step not made uniform is the resummation of the $O(\tau)$ remainder (a Watson/saddle estimate), which is exactly the content of Remark 3.24.

3.3 The numerator amplitude: $S_0 \sim w \sin w$

Lemma 3.11 controls the *denominator* $1 - S_1$: it has infinitely many real zeros $q_m \uparrow 1$, the phases $w_m = \sqrt{2/\tau_m} \rightarrow (m - \frac{1}{2})\pi$. By Corollary 3.2 each q_m is a genuine pole of $G_0 = S_0/(1 - S_1)$ *provided* $S_0(q_m) \neq 0$. We now show the numerator is governed by the *same* saddle engine, which makes $S_0(q_m) \neq 0$ for all large m a theorem rather than a finite check.

The telescoped numerator is the *even* ladder $S_0 = \sum_{j \geq 0} (-1)^j s_j$, $s_j = \alpha_{2j} \prod_{i < j} |\gamma_{2i}|$, against the *odd* ladder $S_1 = \sum_{j \geq 0} (-1)^j t_j$, $t_j = \alpha_{1+2j} \prod_{i < j} |\gamma_{1+2i}|$ of Lemma 3.4. With $\alpha_k = 2/(e^{(k+1)\tau} - 1) = \frac{2}{(k+1)\tau} g((k+1)\tau)$, $g(y) = y/(e^y - 1) \in (0, 1]$, and $|\gamma_k| = \alpha_k - \alpha_{k+1} \sim \frac{2}{\tau(k+1)(k+2)}$, the leading models of the two ladders are

$$\hat{t}_j = \frac{(2/\tau)^{j+1}}{(2j+2)!}, \quad \hat{s}_j = \frac{(2/\tau)^{j+1}}{(2j+1)!} = (2j+2) \hat{t}_j, \quad (8)$$

the only change being the parity of the factorial: $\prod_{i < j} (2i+1)(2i+2) = (2j)!$ for the even ladder versus $\prod_{i < j} (2i+2)(2i+3) = (2j+1)!$ for the odd one. Exactly as $\sum_j (-1)^j \hat{t}_j = 1 - \cos w$, one has the companion closed form

$$\sum_{j \geq 0} (-1)^j \hat{s}_j = \sum_{j \geq 0} (-1)^j \frac{w^{2j+2}}{(2j+1)!} = w \sin w, \quad (9)$$

verified to 20 digits in `sigma0_layer_derivation.py`; the amplitude factor $\hat{s}_j/\hat{t}_j = 2j+2$ is precisely what turns $1 - \cos w = \sum (-1)^{m-1} \frac{w^{2m}}{(2m)!}$ into $w \sin w = \sum (-1)^{m-1} (2m) \frac{w^{2m}}{(2m)!} = 2\vartheta(1 - \cos w)$.

Lemma 3.6 (Numerator reduction). *The form factors $\sigma_j := s_j/\hat{s}_j$ satisfy $\sigma_j \in (0, 1]$, strictly decreasing with $\sigma_j \rightarrow 0$ (verbatim the proof of Lemma 3.4, the bound coming from the same $g \leq 1$ and $h(y)^2 = (y/2)^2/\sinh^2(y/2) < 1$). Writing $\nu_j := \sigma_{j-1} - \sigma_j \geq 0$ ($\sigma_{-1} := 1$, $\sum_j \nu_j = 1$) and $\tilde{R}_j(w) := w \sin w - \sum_{i=0}^j (-1)^i \frac{w^{2i+2}}{(2i+1)!}$ for the order- $(2j+1)$ sine remainder, Abel summation gives, exactly,*

$$S_0(q) = w \sin w + \sum_{j \geq 0} \nu_j \tilde{R}_j(w). \quad (10)$$

The correction in (10) is the image of the Lemma 3.5 error under the amplitude operator 2ϑ : with $\delta_j = 1 - \sigma_j$ and (8), $S_0 - w \sin w = -\sum_j (-1)^j \hat{s}_j \delta_j = -\sum_j (-1)^j (2j+2) \hat{t}_j \delta_j = 2\vartheta(S_1 - (1 - \cos w)) + o(1)$. Applying 2ϑ to the closed-form leading term (7) — using $2\vartheta(\sqrt{\tau} \sin w) = \sqrt{2} \cos w + o(1)$ (since $\sqrt{\tau} \sin w = \sqrt{2} w^{-1} \sin w$ and $\vartheta(w^{-1} \sin w) = \frac{1}{2}(\cos w - w^{-1} \sin w)$) — yields the closed-form companion of (7):

$$S_0(q) = w \sin w - \frac{17}{18} \cos w + o(1), \quad \frac{17}{18} = \frac{17\sqrt{2}}{36} \cdot \sqrt{2}, \quad (11)$$

the *same* constant 17 propagating from (7); (11) is confirmed to 3–4 digits across $\tau \in [1.5 \times 10^{-4}, 2 \times 10^{-2}]$ in `sigma0_layer_derivation.py`, with the residual decaying. (The travel block Σ_0 obeys the identical leading law $\Sigma_0 = w \sin w + O(1)$, with its own subleading constant $+\frac{1}{18} \cos w$ from the travel value $c_1 = +\frac{\sqrt{2}}{36}$ of Remark 3.22.)

The sharp subleading $-\frac{17}{18} \cos w$ in (11) is the saddle value; but the *non-vanishing* needs only the much weaker $S_0 - w \sin w = o(w)$, and *that is unconditional*, by the absolute contour of Lemma 3.13 — *no* Watson/saddle resummation. Indeed the numerator ladder differs from the denominator only by the amplitude factor $\hat{s}_j/\hat{t}_j = 2j+2$ (8); inserting it into the representation $T_2 = \sum_{n \geq 1} (-1)^n h(n) = \frac{1}{2\pi i} \oint_{\partial R} h \frac{\pi}{\sin \pi s} ds$ of Lemma 3.13 weights that contour integral by $2s$,

$$\sum_{n \geq 1} (-1)^n (2n) h(n) = \frac{1}{2\pi i} \oint_{\partial R} 2s h(s) \frac{\pi}{\sin \pi s} ds \quad (\text{Res}_{s=n} 2s h \frac{\pi}{\sin \pi s} = (-1)^n 2n h(n)),$$

while the elementary remainder $2\vartheta T_1 = W \sin W - w \sin w$ has modulus $\leq |W - w| + w |\sin W - \sin w| \leq \sqrt{\frac{\tau}{2}} + w^2(1 - e^{-\tau/2}) = O(1)$. Bounding the integral in modulus exactly as in Lemma 3.13, the extra $2|s| \leq W(1 + o(1)) \sim \sqrt{2} \tau^{-1/2}$ on the left-edge mass turns its $O(\tau^{1/4})$ into

$$\left| \sum_{n \geq 1} (-1)^n 2n h(n) \right| \leq \frac{1}{2\pi} \oint_{\partial R} 2|s| \left| h \frac{\pi}{\sin \pi s} \right| |ds| = O(\tau^{-1/4}) = o(\tau^{-1/2})$$

(numerically $\sqrt{\tau}$ times this bound $\rightarrow 0$: 0.016, 0.011, 0.008, 0.005, 0.004, 0.003 over $\tau \in [2 \times 10^{-2}, 5 \times 10^{-4}]$; `GAPU4_numerator.py`). Hence $S_0 - w \sin w = O(\tau^{-1/4}) = o(w)$, so $S_0/(w \sin w) \rightarrow 1$. This rests only on Lemma 3.12 ($|g_s| \leq 2$) and Stirling — exactly the inputs of Lemma 3.13. We record the payoff.

Corollary 3.7 (Numerator non-vanishing). *For all sufficiently large m , $S_0(q_m) \neq 0$ and $\Sigma_0(q_m) \neq 0$, with $S_0(q_m) \sim (-1)^{m+1}w_m$; hence all but finitely many zeros $\{q_m\}$ of $1 - S_1$ (resp. $1 - \Sigma_1$) are genuine simple poles of G_0 (resp. G_0^T). This is the input to Lemma 3.19 and Theorem 3.20; it is now unconditional, on the same footing as Lemma 3.13 (Lemma 3.12 and Stirling), with no remaining numerical hypothesis.*

Proof. At a pole, $S_1(q_m) = 1$, so by (7) $\cos w_m = O(\sqrt{\tau}_m) \rightarrow 0$ and $|\sin w_m| \rightarrow 1$. By the unconditional absolute-contour bound above, $S_0(q_m) = w_m \sin w_m + o(w_m)$, whence $|S_0(q_m)| \geq w_m(1 - o(1)) \rightarrow \infty$; in particular $S_0(q_m) \neq 0$ for all large m , and the sign alternates with $\sin w_m$. The same argument applies to Σ_0 (leading term $w \sin w$ unchanged; subleading $+\frac{1}{18} \cos w$). $\square$

This removes the only numerical hypothesis previously attached to the relaxed case: the non-vanishing $\Sigma_0(q_m) \neq 0$ is no longer a spot-checked input but a consequence of the same w -asymptotic that produces the poles. Corollary 3.7 is checked in `transcendence_verify_ALL.py` (strict sign alternation and $|\Sigma_0(q_m)|/w_m \rightarrow 1$ monotonically over the first 33 poles, before the q_m -crowding exceeds working precision).

3.4 An unconditional half of the leading term

The formal $\sqrt{\tau}$ term splits into a piece that is *unconditionally* bounded, uniformly in w , and a quantitatively smaller residue carrying the entire remaining gap. First rewrite the error by swapping the order of summation in (6): since $R_j(w) = \sum_{i>j} (-1)^i w^{2i}/(2i)!$ and $\sum_{j<i} \mu_j = 1 - \rho_{i-1}$,

$$E := S_1 - (1 - \cos w) = \sum_{i \geq 1} (-1)^i \frac{w^{2i}}{(2i)!} (1 - \rho_{i-1}) \quad (12)$$

(absolutely convergent, verified to 120 digits). The domination of Lemma 3.4 sharpens to $\rho_i/\rho_{i-1} \leq e^{-\tau}$, hence $\rho_{i-1} \leq e^{-i\tau}$; writing $B_i := -\log \rho_{i-1} - i\tau \geq 0$ and $W := w e^{-\tau/2}$, the split $1 - \rho_{i-1} = (1 - e^{-i\tau}) + e^{-i\tau}(1 - e^{-B_i})$ inserted into (12) (using $w^{2i}e^{-i\tau} = W^{2i}$) gives $E = T_1 + T_2$ with

$$T_1 = \sum_{i \geq 1} (-1)^i \frac{w^{2i}}{(2i)!} (1 - e^{-i\tau}) = \cos w - \cos W, \quad T_2 = \sum_{i \geq 1} (-1)^i \frac{W^{2i}}{(2i)!} (1 - e^{-B_i}). \quad (13)$$

Proposition 3.8 (Unconditional uniform bound on T_1). $T_1 = \cos w - \cos(w e^{-\tau/2})$ exactly, and for all $\tau > 0$

$$|T_1| \leq w(1 - e^{-\tau/2}) \leq \sqrt{\tau/2},$$

uniformly in w ; consequently $T_1 = -\frac{1}{\sqrt{2}}\sqrt{\tau} \sin w + O(\tau)$ uniformly.

Proof. The closed form is the evaluation of two convergent cosine series in (13). The bound is $|\cos w - \cos W| \leq |w - W| = w(1 - e^{-\tau/2})$ (mean value theorem), and $1 - e^{-\tau/2} \leq \tau/2$, so $|T_1| \leq w\tau/2 = \sqrt{\tau/2}$. Expanding $w(1 - e^{-\tau/2}) = \frac{w\tau}{2} + O(w\tau^2) = \frac{1}{\sqrt{2}}\sqrt{\tau} + O(\tau^{3/2})$ and $\cos w - \cos(w - \frac{w\tau}{2} + \dots) = -\frac{w\tau}{2} \sin w + O((w\tau)^2)$ gives the stated expansion; both error terms are uniform in w . $\square$

Thus $-\frac{1}{\sqrt{2}}$ — the end-factor half of c_1 — is established *unconditionally and uniformly*. The residue $T_2 = \sum_{j \geq 0} c_j \tilde{R}_j$, $c_j = e^{-B_j}(1 - e^{-(\delta_j - \tau)}) \in [0, \delta_j - \tau]$ with $\delta_j = -\log(\rho_j/\rho_{j-1})$ and $\tilde{R}_j$ the cosine remainder at argument W , is structurally identical to E but with weights smaller by a factor $O(\tau^2 j^2)$; numerically $T_2 = +\frac{\sqrt{2}}{36}\sqrt{\tau} \sin w + O(\tau)$, completing $c_1 = -\frac{1}{\sqrt{2}} + \frac{\sqrt{2}}{36} = -\frac{17\sqrt{2}}{36}$.

The entire remaining gap is the single bound $|T_2| \rightarrow 0$ uniformly — met unconditionally below as the absolute contour bound $|T_2| = O(\tau^{1/4}) \rightarrow 0$ (Lemma 3.13), the sharp $O(\sqrt{\tau})$ being the saddle refinement; and the qualitative sign-straddle that suffices for Theorem 3.20 needs only the much weaker *per-point* bound $|T_2| < 1$ at the discrete phases $w = N\pi$.

3.5 Equivalent reductions and the obstruction

Two further rigorous reductions sharpen the target. From the integral form of the cosine remainder, $R_j(w) = \frac{(-1)^{j+1}}{(2j)!} \int_0^w (w-s)^{2j} \sin s \, ds$ (one integration by parts; boundary terms vanish), averaging against μ_j gives the exact convolution

$$E = \int_0^w \sin(s) K'(w-s) \, ds, \quad K'(u) = \sum_{j \geq 0} \mu_j \frac{(-1)^{j+1} u^{2j}}{(2j)!} \quad (\text{entire, even}), \quad (14)$$

the interchange justified by domination ($\sum_j \mu_j u^{2j}/(2j)! \leq \cosh u$). Hence $|E| \leq w \sup_{[0,w]} |K'|$, eliminating the s -oscillation entirely; and $K'(0) = \mu_0 = 1 - g(2\tau) \leq \tau$ rigorously (the $[1/1]$ Padé bound $(2-y)/(2+y) \leq e^{-y}$). Moreover $|R_j(w)| \leq \hat{t}_{j+1} = w^{2j+2}/(2j+2)!$ gives a super-exponential tail: for $J_0 \geq 2w$, $|\sum_{j \geq J_0} \mu_j R_j| \leq \hat{t}_{J_0+1} \leq (e/4)^{2J_0+2} = O(\tau^\infty)$, so E is a finite head of $O(w)$ terms up to a negligible error. Lemma 3.11 is therefore *equivalent* to either of the clean self-contained estimates

$$\sup_{u \in [0,w]} |K'(u)| \leq C\tau \quad \Longleftrightarrow \quad \|K\|_{L^1[0,w]} = O(\sqrt{\tau}) \quad \Longleftrightarrow \quad |T_2| = O(\sqrt{\tau}),$$

for the explicit entire function K' (equivalently $K(u) = \sum_j \mu_j (-1)^{j+1} u^{2j+1}/(2j+1)!$).

Remark 3.9 (Why this is a genuine oscillatory estimate, not bookkeeping). *The bound is a knife-edge: the supremum of $|K'|$ on $[0, w]$ equals μ_0 but is approached from below by the interior lobe near $u \approx \pi$ with vanishing relative margin $\sim 0.8\tau$. Hence no soft argument can close it: the weights μ_j provably fail Hankel positivity (the Hankel matrices $[\mu_{i+k}]$ have genuinely negative eigenvalues, persistent at 200-digit precision), so K' is not the cosine transform of a positive measure and the Bochner bound $|K'(u)| \leq K'(0)$ is unavailable; and monotonicity, contraction, and triangle/Abel bounds all fail (the partial sums of K' overshoot like e^u). The remaining estimate is genuinely a uniform-in- τ saddle/Poisson estimate of an oscillatory sum in the coupled limit $\tau \rightarrow 0$, $j \sim \tau^{-1/2}$, for which no off-the-shelf difference-equation theorem applies: Wong–Li [WongLi1992], Wang–Wong [WangWong2003] and Costin–Costin [CostinCostin1996] all require a fixed recurrence with $n \rightarrow \infty$, whereas here the turning point $k \sim w = \sqrt{2/\tau}$ recedes jointly with the index; and the false/partial-theta and Mellin harmonic-sum machineries ([BringmannMilas2015]; [FGD1995]) do not apply (the $q \rightarrow 1$ behaviour is carried by the full Lambert-series corrections, not the leading q -powers, and S_1 is a sum of products, not a harmonic sum). This is recorded as the precise frontier, established by a five-route multi-agent assault with adversarial verification (all routes reduce to this one inequality; none closes it).*

A base- q self-similarity route is likewise dead. The catalytic dilation in the Bousquet–Mélou–Jehanne functional equation is $t \rightarrow q^2 t$ (the catalytic variable, i.e. the index k), not a base substitution $q \rightarrow q^2$: telescoped it gives exactly the k -recursion $\Sigma_k(q) = A_k(q) + C_k(q) \Sigma_{k+2}(q)$ at fixed q (verified to 10^{-60}), which couples different k at the same q and is already fully integrated into the criticality $\Sigma_1(q) = 1$. There is no residual base- q self-similarity: numerically no exact rational relation $\Sigma_1(q) = \Phi(q) + \Psi(q) \Sigma_1(q^2)$ exists (Padé residuals grow with the sample count at every fixed degree, and likewise for the quadratic ansatz); structurally $q \rightarrow q^2$ sends the Lambert factors

$1/(1-q^m) \rightarrow 1/(1-q^{2m})$, so $\Sigma_1(q^2)$ lives in the even-index Lambert subring and cannot reproduce the odd- m poles of $\Sigma_1(q)$; and most decisively the cosine frequency $w = \sqrt{2/\tau}$ scales as $w \rightarrow w/\sqrt{2}$ under $q \rightarrow q^2$, so the zero lattice $\{(n + \frac{1}{2})\pi\}$ is rescaled by the irrational $\sqrt{2}$ — a single zero's $q \mapsto \sqrt{q}$ orbit $\{w_0 2^{m/2}\}$ meets the cosine-zero lattice at most finitely often, while the $q \rightarrow q^2$ orbit $\{\tau, 2\tau, 4\tau, \dots\}$ runs toward $q = 0$ (away from the $q = 1$ accumulation point) and is exponentially sparser (ratio 2) than the actual zeros (whose τ -ratio $\rightarrow 1$). Hence self-similarity cannot propagate zeros, and Lemma 3.11 is not bypassable this way.

3.6 Resolution of the oscillatory core: model subtraction

The knife-edge of Remark 3.9 is dissolved by *subtracting* the closed-form leading kernel and expanding the residual through an exact resummation identity that performs the oscillatory cancellation for free. This reduces Lemma 3.11 to a single non-oscillatory majorization (Lemma 3.10 below).

Geometric model and residual. With the geometric weights $\mu_j^{\text{geo}} = e^{-j\tau}(1 - e^{-\tau})$ (i.e. $\rho_j^{\text{geo}} = e^{-(j+1)\tau}$) the kernel resums in closed form,

$$K'_{\text{model}}(u) := \sum_{j \geq 0} \mu_j^{\text{geo}} \frac{(-1)^{j+1} u^{2j}}{(2j)!} = -(1 - e^{-\tau}) \cos(u e^{-\tau/2}), \quad |K'_{\text{model}}| \leq \tau, \quad (15)$$

and $\int_0^w \sin(s) K'_{\text{model}}(w-s) ds = \cos w - \cos(w e^{-\tau/2}) = T_1$ (Proposition 3.8). Put $\text{corr} := K' - K'_{\text{model}}$, so $E = T_1 + T_2$ with $T_2 = \int_0^w \sin(s) \text{corr}(w-s) ds$, and Lemma 3.11 follows from

$$\sup_{u \in [0, w]} |\text{corr}(u)| \leq C\tau \quad (\text{any absolute } C) \implies |T_2| \leq w C\tau = O(\sqrt{\tau}). \quad (16)$$

Subtraction removes the knife-edge: numerically $\sup |\text{corr}|/\tau \rightarrow \frac{1}{6}$ (stable: 0.171, 0.154, 0.162 at $\tau = 0.02, 0.005, 0.001$), the supremum migrating to the boundary $u \approx w$ with a *non-vanishing* margin — so a non-sharp constant suffices.

Resummation engine and layers. The exact identity

$$\sum_{j \geq 0} j^p e^{-j\tau} \frac{(-1)^j u^{2j}}{(2j)!} = (-\partial_\tau)^p \cos(u e^{-\tau/2}) =: D_p(W), \quad W := u e^{-\tau/2}, \quad (17)$$

(verified $p = 0, \dots, 3$; $D_{p+1} = \frac{W}{2} \partial_W D_p$) turns each oscillatory j -sum into a *trigonometric polynomial in W with no residual cancellation*. Expanding $\rho_j = e^{-(j+1)\tau} e^{-R_j}$ ($R_j = -\log \rho_j - (j+1)\tau = \frac{1}{9}\tau^2 j^3 + \dots \geq 0$, with $R_j \leq 0.47\sqrt{\tau}$ on the effective range $j \leq w$) gives $K' = \sum_{m \geq 0} L_m$, $L_0 = K'_{\text{model}}$, each L_m a closed-form combination of the D_p . This is exact: the layers reconstruct K' to residual $O(\tau^{M+1})$ (verified to 5×10^{-17} at $M = 6$).

The crux layer L_1 (rigorous). Symbolically,

$$L_1 = \tau^2 \left[\frac{W^3 \sin W}{72} - \frac{W^2 \cos W}{16} - \frac{W \sin W}{48} \right] + \tau^2 e^{-\tau} \left[-\frac{W^3 \sin W}{72} + \frac{7W^2 \cos W}{48} + \frac{17W \sin W}{48} - \frac{\cos W}{6} \right].$$

The dangerous $W^3 \sin W$ term — which alone would be $O(\tau^2(2/\tau)^{3/2}) = O(\sqrt{\tau})$, fatal — has coefficient $\frac{\tau^2}{72}(1 - e^{-\tau}) = O(\tau^3)$: the difference structure $\mu_j = \rho_{j-1} - \rho_j$ cancels the top W -power.

The surviving dominant term is $\frac{\tau^2}{12}W^2 \cos W$, bounded by $\tau/6$ since $W^2 \leq 2/\tau$; in full,

$$\sup_{[0,w]} |L_1| \leq \frac{\tau}{6} + \frac{13\sqrt{2}}{36} \tau^{3/2} + \frac{\tau^2}{6} \quad (\text{rigorous, exact coefficients; dominates } \sup |L_1| \text{ for all tested } \tau). \quad (18)$$

The tail bound: Touchard/Poisson, not a uniform constant. The naive hope — that $\sup_{[0,w]} |D_p| \leq K(w/2)^p$ with K *absolute* — is *false*: the ratio $\sup |D_p|/(w/2)^p$ grows like $2^{-p}B_p$ (Bell numbers), e.g. $1.4, 21, 7.6 \cdot 10^3, 2.2 \cdot 10^7$ at $p = 10, 20, 30, 40$. The cancellation cannot be discarded; it is captured exactly by a probabilistic identity. Writing $D_p(W) = 2^{-p} \operatorname{Re}[e^{iW} T_p(iW)]$ with $T_p(x) = \sum_r S(p, r)x^r$ the Touchard polynomial ($S(p, r) \geq 0$ Stirling numbers of the second kind), one has the *rigorous* per- p bound and the *Poisson-moment* identity

$$|D_p(W)| \leq 2^{-p} T_p(W) \leq 2^{-p} T_p(w) \quad (0 \leq W \leq w), \quad T_p(w) = \mathbb{E}[N^p], \quad N \sim \text{Poisson}(w), \quad (19)$$

so for any polynomial Q , $\sum_p [j^p] Q \cdot 2^{-p} T_p(w) = \mathbb{E}[Q(N/2)]$ — dissolving the need for any uniform-in- p constant.

R-control (rigorous). With $\varphi(y) = \log(\sinh(y/2)/(y/2))$ one has $b_i = \varphi((2i+2)\tau) + \varphi((2i+1)\tau) - \varphi(\tau)$ and the elementary inequality

$$0 \leq \varphi(y) \leq \frac{y^2}{24} \quad (y > 0), \quad (20)$$

proved from the Mittag-Leffler expansion $\coth t - \frac{1}{t} = \sum_{n \geq 1} \frac{2t}{t^2 + n^2 \pi^2}$ together with $2\zeta(2)/\pi^2 = \frac{1}{3}$. Summing gives $0 \leq R_j \leq \tau^2 C(j)$, $C(j) = (j+1)(2j+3)(4j+5)/72$ (positive coefficients).

Expand $R_j = \sum_{n \geq 1} \lambda_n \tau^{2n} P_n(j)$ (P_n of degree $2n+1$ with nonnegative coefficients, $\lambda_n = [y^{2n}] \varphi(y)$ of alternating sign, $\limsup |\lambda_n|^{1/2n} = 1/2\pi$). *Caveat that fixes the proof*: the absolute series $\bar{R}(j) = \sum_n |\lambda_n| \tau^{2n} P_n(j)$ converges only for $j < \pi/\tau$ and *diverges* beyond; since $N \sim \text{Poisson}(w)$ has positive mass on $\{N \geq 2\pi/\tau\}$, $\mathbb{E}[\bar{R}(N/2)^m] = \infty$. The fix is a *finite* truncation.

Lemma 3.10 (Tail majorization). $\sum_{m \geq 2} \sup_{u \in [0,w]} |L_m| = O(\tau)$.

Proof. Fix the order $N = 2$ and split $R_j = R_j^{[2]} + S_j$, $R_j^{[2]} = \lambda_1 \tau^2 P_1(j) + \lambda_2 \tau^4 P_2(j)$ (a polynomial of degree 5), $S_j = \sum_{n \geq 3} \lambda_n \tau^{2n} P_n(j)$. Put $J_0 = \lceil 2w \rceil$. (i) *Outer tail* $j \geq J_0$. By $0 \leq R_j \leq \tau^2 C(j)$ and $W \leq w$, with $2j \geq 4w$ past the peak $j^* \sim W/2 \leq w/2$ of $W^{2j}/(2j)!$, $\sum_{j \geq J_0} (\tau^2 C(j))^m W^{2j}/(2j)! = O(\tau^\infty)$ (super-exponential factorial decay), uniformly in $m \leq$ any fixed power. (ii) *Remainder* S on $j < J_0$. $|S_j| \leq \sum_{n \geq 3} |\lambda_n| (2j\tau)^{2n} \leq C(2J_0\tau)^6 = O(\tau^{5/2})$ on $j < J_0$ (since $2J_0\tau \leq 4w\tau = 4\sqrt{2}\tau$), so in $R_j^m = \sum_k \binom{m}{k} (R_j^{[2]})^{m-k} S_j^k$ every $k \geq 1$ term is $O(\tau^{5/2})$ -smaller and contributes $o(\tau)$ to the tail. (iii) *Main term* $(R_j^{[2]})^m$. This is a polynomial in j ; the engine (17) and the triangle inequality over its (finitely many) coefficients with (19) give, with $\bar{R}^{[2]}(j) = |\lambda_1| \tau^2 P_1(j) + |\lambda_2| \tau^4 P_2(j)$ a *fixed* polynomial, $\sup_{[0,w]} |\sum_{j < J_0} (R_j^{[2]})^m (-1)^{j+1} \frac{u^{2j}}{(2j)!} e^{-j\tau}| \leq \mathbb{E}[\bar{R}^{[2]}(N/2)^m] + O(\tau^\infty)$, both copies in $\mu_j^{(m)}$ handled the same ($e^{-\tau} \leq 1$; the shift $j \mapsto j-1$ replaces $P(j)$ by $P(j+1)$, still nonnegative). (iv) *Sum over m*. As $\bar{R}^{[2]}$ is a fixed polynomial, $\mathbb{E}[\bar{R}^{[2]}(N/2)^m] < \infty$, and $\sum_{m \geq 2} \frac{2}{m!} \mathbb{E}[\bar{R}^{[2]}(N/2)^m] = 2 \mathbb{E}[\Psi(\bar{R}^{[2]}(N/2))]$, $\Psi(x) = e^x - 1 - x$. This is $O(\tau)$: on the bulk $N \leq 2w$, $\bar{R}^{[2]}(N/2) = O(\sqrt{\tau})$ and $\Psi \leq \frac{1}{2} \bar{R}^{[2]2} e^{\bar{R}^{[2]}}$; the Poisson tail $N > 2w$ contributes $O(\tau^\infty)$ because the polynomial $\bar{R}^{[2]}(N/2)$ grows only polynomially while $\mathbb{P}(N = n)$ decays factorially ($\mathbb{P}(N > 2w) \leq e^{-w(2 \ln 2 - 1)}$). Combining (i)–(iv), $\sum_{m \geq 2} \sup |L_m| \leq 2 \mathbb{E}[\Psi(\bar{R}^{[2]}(N/2))] + O(\tau^{3/2}) = O(\tau)$; numerically $\leq 0.0135 \tau$ (and this bound dominates the true tail $\sup |\text{corr} - L_1|$ with $10\times$ margin). $\square$

A gap in the Touchard bound (found by adversarial re-checking, and stated plainly). The bound of step (iv) is *not* valid as written: $\bar{R}^{[2]}(N/2)$ is a degree-5 polynomial in N , so $2\mathbb{E}[\Psi(\bar{R}^{[2]}(N/2))] = 2\mathbb{E}[e^{\bar{R}^{[2]}(N/2)} - 1 - \bar{R}^{[2]}(N/2)]$ is *infinite*: $\mathbb{E}[e^{cN^5}] = \sum_n e^{cn^5} w^n e^{-w}/n!$ diverges for every $c > 0$ (numerically, the partial sums sit near 1 through the Poisson bulk and then explode — $\sim 10^{975}$ by $n = 400$ at $\tau = 10^{-2}$), because e^{cn^5} ultimately outgrows the factorial $1/n!$. The defect is that summing the per- m *absolute* Touchard bounds $|D_p| \leq 2^{-p}T_p(w)$ discards exactly the oscillatory cancellation that keeps the true series finite, and the Bell-number growth of the Touchard polynomials then makes the m -sum diverge. So step (iv) does *not* establish Lemma 3.10. The quantity $\sup |\text{corr}|$ is genuinely $O(\tau)$ — corr is a bounded real function, with $\sup |\text{corr}|/\tau = 0.167$ at $\tau = 10^{-4}$ to 120 digits and the uniform-phase scan of §3.3 pinning $\sup_w |S_1 - (1 - \cos w)|/\sqrt{\tau}$ at $|c_1|$ — but an *oscillation-preserving* (saddle/steepest-descent) estimate is required to prove it, which we do not yet have. What *is* rigorous is the leading order: $T_1 = \cos w - \cos(we^{-\tau/2})$ exactly with $|T_1| \leq \sqrt{\tau/2}$, and L_0, L_1 (the $\sqrt{\tau}$ coefficient via the ϑ^3 -resummation). We therefore state Lemma 3.11 as *numerically certain but not yet proved*; consequently the ratio transcendence below (Theorem 3.20: G_0, G_0^T , and V) is conditional on it. The *individual blocks* $\Sigma_1, \Sigma_0, S_1, S_0$ remain transcendental *unconditionally* (Theorem 3.21, Pólya–Carlson, independent of Lemma 3.11), as does the exact growth rate β_2 (Euler squeeze); these are untouched by the gap. Note that Proposition 3.18 requires only that the error $\rightarrow 0$ along the *extreme-phase* sequences $w = n\pi$ (where the $\sqrt{\tau} \sin w$ term vanishes and the error is the sharper $O(\tau)$) — a strictly weaker statement than the uniform bound, numerically certain through $n \leq 64$ — so closing transcendence needs only this extreme-phase decay, not the full uniform Lemma 3.11. *This status is upgraded below*: the uniform decay $|T_2| = O(\tau^{1/4}) \rightarrow 0$ — all that V needs (it forces $|E(m\pi)| < 1$) — is proved as an *absolute contour bound* (Lemma 3.13), resting only on Lemma 3.12 (proved in closed form) and Stirling’s asymptotics, with *no* oscillatory-cancellation step. So V ’s ratio transcendence is unconditional modulo those textbook inputs, *not* the open uniform Lemma 3.11; U adds only parallel same-class steepest-descent estimates (closed at the same standard, §4). (The saddle route of §3.7 and Remarks 3.15–3.16 additionally pins the sharp constant $\frac{\sqrt{2}}{36}$.)

3.7 The correct route: steepest descent

The failure of the Touchard bound is not a failure of the *result* but of the *method*: any estimate that takes absolute values term-by-term destroys the oscillatory cancellation and meets the divergent majorant $\cosh W \sim e^W$. The cancellation is recovered by deforming into the complex plane — the standard saddle-point/steepest-descent picture — and this both identifies the leading term in closed form and matches it to full numerical precision.

Write the residual as the value of an entire function on the imaginary axis:

$$T_2 = G(iW), \quad G(z) = \sum_{i \geq 1} g_i \frac{z^{2i}}{(2i)!}, \quad g_i = 1 - e^{-B_i} \in [0, 1),$$

since $(iW)^{2i} = (-1)^i W^{2i}$. On the real axis G has exponential type 1 ($|G(x)| \leq \cosh x$), so the imaginary axis is the *subdominant* direction (the indicator $h(\pi/2) = 0$, as for $\cosh(iW) = \cos W$); the magnitude of $G(iW)$ is governed not by the type but by the saddle. Treating i as continuous, the summand $(-1)^i W^{2i}/(2i)!$ has its stationary point (with $(-1)^i = e^{i\pi i}$ and Stirling) at the *complex* node

$$s^* = \frac{i}{2} W.$$

Two facts make this decisive. First, B_s continues analytically to s^* and is *small* there: its τ -series $B_s = \sum_{n \geq 1} f_n \tau^{2n} Q_n(s)$ ($\deg Q_n = 2n+1$) converges at s^* because $\tau(W/2)^2 = \frac{1}{2}$ lies inside the

radius (set by the 2π -radius of ϕ), and its leading term gives

$$B_{s^*} = \frac{\tau^2}{9} \left(\frac{i}{2} W \right)^3 + O(\tau) = -i \frac{\tau^2 W^3}{72} + O(\tau) = -i \frac{\sqrt{2}}{36} \sqrt{\tau} + O(\tau),$$

using $\tau^2 W^3 = 2\sqrt{2} \sqrt{\tau}$. Second, $g_{s^*} = 1 - e^{-B_{s^*}} = B_{s^*} + O(\tau)$. The steepest-descent contribution of the node s^* (with its conjugate $-s^*$, assembling the real part) is then

$$T_2 = \operatorname{Re}[g_{s^*} e^{iW}] + O(\tau) = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin w + O(\tau), \quad (21)$$

the leading coefficient agreeing with (7) (together with the unconditional T_1 this reassembles c_1). Identity (21) is confirmed in `saddle_point_lemcos.py`: the ratio $T_2 / \operatorname{Re}[B_{s^*} e^{iW}] \rightarrow 1$ as $\tau \rightarrow 0$ (1.0005 at $\tau = 4 \times 10^{-4}$), with no free constant.

The correct contour is the *Lindelöf* steepest-descent path: with $h(s) = W^{2s} g_s / \Gamma(2s + 1)$ (so $T_2 = \sum_{i \geq 1} (-1)^i h(i)$), one has the exact $T_2 = \frac{1}{2\pi i} \oint_C h(s) \pi / \sin(\pi s) ds$, C encircling the positive integers, and h is *analytic in* $\{\operatorname{Re} s \geq 0\}$ (the poles of B_s lie at $\operatorname{Re} s < 0$), so C deforms to the steepest path through $s^* = \frac{i}{2} W$ without obstruction. Here B_s is known in closed form: $B_s = \sum_{k \geq 1} [T_{2,k}(s) + T_{1,k}(s) - s \log(1 + (\frac{\tau}{2\pi k})^2)]$, $T_{c,k}(s) = 2s \log \frac{\tau}{\pi k} + \log \Gamma(s + \frac{c}{2} + \frac{i\pi k}{\tau}) + \log \Gamma(s + \frac{c}{2} - \frac{i\pi k}{\tau}) - \log \Gamma(\frac{c}{2} \pm \frac{i\pi k}{\tau})$, from the product $\sinh(y/2)/(y/2) = \prod_k (1 + (\frac{y}{2\pi k})^2)$ (verified against the form factors to full precision). Two straight-contour shortcuts are *provably* unavailable: the Abel–Plana / vertical-Lindelöf forms force the full imaginary axis, where $\operatorname{Re} B_{iy} \rightarrow -\infty$ for $y \gg W$, making g_{iy} blow up and the integral diverge — the curved path, which stays at $\operatorname{Im} s \leq W/2$ (where B is small), is genuinely necessary. On the saddle path the magnitudes assemble correctly: $|h(s^*)/\sin(\pi s^*)| = |g_{s^*}| |\Gamma(1+iW)|^{-1} / \sinh(\frac{\pi W}{2})$, and the two factors $e^{\pi W/2}$ (from $\Gamma(1+iW)^{-1}$ and from $\sinh \frac{\pi W}{2}$) *cancel*, leaving the saddle contribution $|g_{s^*}| \cdot O(1) \cdot \sqrt{2\pi/|F''(s^*)|}$; numerically this magnitude divided by $\sqrt{\tau}$ converges to $\frac{\sqrt{2}}{36} = 0.03928\dots$ (verified at $\text{dps} \geq 50$, both directly and symbolically: $f_1 = \frac{1}{24}$, $Q_1(s) \sim \frac{8}{3} s^3$, so $B_s \sim \frac{1}{9} \tau^2 s^3$ and $B_{s^*} = -i \frac{\sqrt{2}}{36} \sqrt{\tau}$ on shell), the coefficient of (21) with no free constant. What is firmly established is the *leading order*: the node s^* is forced, $B_{s^*} = O(\sqrt{\tau})$, h is analytic in $\{\operatorname{Re} s \geq 0\}$ so the deformation is legal (the rectangular-contour residue sum reproduces $\sum (-1)^n h(n)$ to 10^{-16}), and a *convergent* bent steepest-descent path through s^* exists. *We do not, however, claim a complete error bound.* Adversarial re-checking shows the naive routes fail: the standard real-axis representation diverges (above), so a van der Corput / Olver theorem cannot be applied to it; bounding a truncated-integral tail by the convergent series tail is a category error (the two differ by $O(1)$, not $o(1)$, since the integral amplitude *grows* toward $y = \pi/\tau$); and $\frac{\sqrt{2}}{36}$ is the leading coefficient *approached from above*, not a finite- τ upper bound. What survives is: $\sup_w |T_2|/\sqrt{\tau} \rightarrow \frac{\sqrt{2}}{36}$ as $\tau \rightarrow 0$, and the empirical inequality $|T_2| \leq 0.06\sqrt{\tau}$ on $(0, 0.3]$ (global sup $|T_2|/\sqrt{\tau} = 0.0538$ at $\tau \approx 0.224$). Thus the leading asymptotic is rigorous, and the right tool for a rigorous $|T_2| = O(\sqrt{\tau})$ bound is the van der Corput second-derivative lemma on the stationary-phase form, *not* a saddle-dominated descent (Remark 3.15); the one written-out step is the half-power-sharper extreme-phase bound $|T_2(m\pi)| = O(\tau)$, the leading term subtracted. We state Lemma 3.11 accordingly: leading order and $O(\sqrt{\tau})$ proved, the extreme-phase $O(\tau)$ sharpening the remaining step.

Lemma 3.11 (Cosine asymptotics; reduced to one oscillatory-average bound). *As $q \rightarrow 1^-$, with $\tau = -\ln q$ and $w = \sqrt{2/\tau}$,*

$$S_1(q) = 1 - \cos \sqrt{\frac{2}{-\ln q}} + O(\sqrt{-\ln q}), \quad \Sigma_1(q) = 1 - \cos \sqrt{\frac{2}{-\ln q}} + O(\sqrt{-\ln q}), \quad (22)$$

the error bounds being uniform in the phase w . Equivalently, in $\varepsilon = 1 - q$, $S_1(q) = 1 - \cos \sqrt{2/\varepsilon} + O(\sqrt{\varepsilon})$.

Numerical certification. Equation (22) is confirmed to 8+ significant figures across the scales $\varepsilon = 10^{-3}, \dots, 10^{-8}$ and across all phases: the period-maximised error satisfies $\sup_w |S_1 - (1 - \cos w)| \leq 0.67\sqrt{\tau}$ and $\sup_w |\Sigma_1 - (1 - \cos w)| \leq 0.04\sqrt{\tau}$ on $\tau \in (0, 10^{-3}]$ (monotone in the sampled range), and at the extreme phases the deviation is the sharper $O(\tau)$: $S_1|_{w=2n\pi} = 0.0980\tau + o(\tau) \rightarrow 0$ and $2 - S_1|_{w=(2n+1)\pi} = 0.0980\tau + o(\tau) \rightarrow 0$ (`transcendence_verify_ALL.py`, `rate_extreme.py`). We use only the qualitative consequence below.

Lemma 3.12 (Re B_s is bounded below on the contour strip). *For every $\tau \in (0, 2\pi^2)$, Re B_s is bounded below on the strip $\mathcal{S} = \{0 \leq \text{Im } s \leq W/2, \text{Re } s \geq 0\}$, with*

$$\inf_{\mathcal{S}} \text{Re } B_s = -\frac{\sqrt{2}}{18}\sqrt{\tau} + O(\tau) \longrightarrow 0,$$

attained at the diagonal node $s = \frac{W}{2}(1+i)$ (not the corner $\sigma = 0$). Hence $A = 1 - e^{-B_s}$ is bounded on $\mathcal{S}$ (by $1 + e^{(\sqrt{2}/18)\sqrt{\tau}} \rightarrow 2$); as the operative contour lies in $\mathcal{S}$ (§3.7), $V_\Gamma(A) < \infty$.

Proof. Write $s = \sigma + it \in \mathcal{S}$ and use the Γ -function form

$$B_s = \sum_{k \geq 1} \sum_{c \in \{1,2\}} \left[2s \log \frac{\tau}{\pi k} + \log \Gamma\left(s + \frac{c}{2} + \frac{i\pi k}{\tau}\right) + \log \Gamma\left(s + \frac{c}{2} - \frac{i\pi k}{\tau}\right) - \log \Gamma\left(\frac{c}{2} + \frac{i\pi k}{\tau}\right) - \log \Gamma\left(\frac{c}{2} - \frac{i\pi k}{\tau}\right) \right] - \sum_{k \geq 1} s \log\left(1 + \left(\frac{\tau}{2\pi k}\right)^2\right),$$

convergent for $\text{Re } s \geq 0$ (its poles lie in $\text{Re } s < 0$). The crux is the elementary inequality

$$\frac{W}{2} = \frac{1}{\sqrt{2}} \tau^{-1/2} < \pi \tau^{-1} = \frac{\pi}{\tau} \quad (\text{true whenever } \tau < 2\pi^2),$$

so on $\mathcal{S}$, $t \leq W/2 < \pi/\tau \leq \pi k/\tau$ for all $k \geq 1$: there is *no resonance*, and each imaginary part $t \pm \frac{\pi k}{\tau}$ has modulus $\geq \frac{\pi k}{\tau} - t \geq \frac{\pi}{\tau} - \frac{W}{2} > 0$. By Stirling $\text{Re } \log \Gamma(x+iY) = (x-\frac{1}{2}) \log |Y| - \frac{\pi}{2}|Y| + \frac{1}{2} \log 2\pi + O(|Y|^{-1})$, uniformly for x in compacta: the pair $Y = t + \frac{\pi k}{\tau}$, $Y = t - \frac{\pi k}{\tau}$ contributes $-\frac{\pi}{2}(t + \frac{\pi k}{\tau}) - \frac{\pi}{2}(\frac{\pi k}{\tau} - t) = -\frac{\pi^2 k}{\tau}$ (the t 's cancel), *exactly* cancelling the $-\frac{\pi^2 k}{\tau}$ of the boundary pair $Y = \pm \frac{\pi k}{\tau}$; the $(x-\frac{1}{2}) \log |Y|$ parts cancel against the prefactor $2s \log \frac{\tau}{\pi k}$. What survives is $\sum_k O((t\tau/\pi k)^2)$, a convergent series ($t \leq W/2 < \pi/\tau$): there is *no resonance* and $\text{Re } B_s$ is finite throughout $\mathcal{S}$.

For the explicit lower bound, expand in τ . From the form-factor representation $B_s = \sum_{m=1}^{2s} \varphi(m\tau) - s\varphi(\tau)$ with $\varphi(y) = \log \frac{\sinh(y/2)}{y/2} = \sum_{n \geq 1} \varphi_n y^{2n}$, $\varphi_n = \frac{(-1)^{n+1} \zeta(2n)}{n(2\pi)^{2n}}$, and the power sum $P_n(M) = \sum_{m=1}^M m^{2n}$ (Faulhaber, $\deg P_n = 2n+1$), term-by-term summation gives the *exact* series

$$B_s = \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s) \quad (|s| < \pi/\tau),$$

verified against the Γ -form above to its truncation (`ADV_Bound1_tauseries.py`). Successive terms shrink by $O((|s|\tau/\pi)^2) = O(\tau)$ at the scale $|s| \asymp \tau^{-1/2}$ that matters, so the tail $\sum_{n \geq 2}$ is $O(\tau^{3/2})$ and the bound is set by $n = 1$: $\varphi_1 \tau^2 (P_1(2s) - s) = \frac{1}{36} \tau^2 (4s^3 + 3s^2 - s)$ ($\varphi_1 = \frac{1}{24}$, $P_1(M) = \frac{M(M+1)(2M+1)}{6}$), with real part $\frac{1}{36} \tau^2 (4(\sigma^3 - 3\sigma t^2) + 3(\sigma^2 - t^2) - \sigma)$. The dominant cubic $\sigma^3 - 3\sigma t^2$ is minimized on $\mathcal{S}$ at $\sigma = t = W/2$ (where $\partial_\sigma = 3\sigma^2 - 3t^2 = 0$, t maximal) at value $-W^3/4$; since $\tau^2 W^3 = 2\sqrt{2}\sqrt{\tau}$,

$$\text{Re } B_s \geq -\frac{\sqrt{2}}{18}\sqrt{\tau} + O(\tau) \longrightarrow 0,$$

the minimum sitting on the top edge $\text{Im } s = W/2$ near the diagonal $\sigma \approx W/2$ (an $O(\tau)$ shift, $\sigma^*/W \approx 0.46\text{--}0.49$, toward the cubic–quadratic balance) — *not* the corner $\sigma = 0$; $\text{Re } B_s$ is *not*

monotone in σ (it dips to the minimum near $\sigma = W/2$, slope $-\tau/6$, before the real-axis growth of $\log \Gamma(2s+1)$ sends it to $+\infty$). On the imaginary axis $s = iy$ — where the amplitude A of §3.7 lives — the real part has the closed form

$$\operatorname{Re} B_{iy} = -\frac{1}{2} \log \frac{y\tau}{\sin(y\tau)} = -\frac{\tau}{24} + O(\tau^2) \quad (0 < y\tau < \pi),$$

from the generating identity $\sum_{n \geq 1} \frac{\zeta(2n)}{n} z^{2n} = -\log \frac{\sin \pi z}{\pi z}$ applied to the even part of the series (the odd Bernoulli numbers $B_{2k+1} = 0$ killing every lower Faulhaber term in $\operatorname{Re} P_n(2iy)$); verified to 8 digits. This is $\geq -\frac{\sqrt{2}}{18} \sqrt{\tau}$ a fortiori, so $\operatorname{Re} B_s \geq -\frac{\sqrt{2}}{18} \sqrt{\tau} \rightarrow 0$ on $\mathcal{S}$. The cancellation breaks only when some $\pi k/\tau < t$, i.e. $t > \pi/\tau$: there the resonant $k \leq t\tau/\pi$ contribute $-\pi t + \pi^2 k/\tau < 0$, summing to $\sim -\frac{1}{2} t^2 \tau \rightarrow -\infty$ — the imaginary-axis divergence — but that locus $\operatorname{Im} s > \pi/\tau$ is disjoint from $\mathcal{S}$. $\square$

Numerically confirmed (`ADV_Bbounded_verify.py`, `dps ≥ 45`): the strip minimum sits at the diagonal $s = \frac{W}{2}(1+i)$ with $\operatorname{Re} B_s/\sqrt{\tau} \rightarrow -\frac{\sqrt{2}}{18} = -0.07857$ ($-0.0794, -0.0789, -0.0787$ at $\tau = 0.05, 0.02, 0.01$, Richardson exponent ≈ 1); along the top edge the value dips to this minimum near $\sigma = W/2$ and only then rises (*not* monotone), while the $\sigma = 0$ edge carries the milder $-\frac{1}{24}\tau$; and the resonance disaster is off-strip — at $\operatorname{Im} s = \pi/\tau$, $\operatorname{Re} B_s = -8.2$; at $\operatorname{Im} s = 80$, -110 (`lemcos_Bstrip.py`).

Lemma 3.13 (Absolute contour bound: $|T_2| = O(\tau^{1/4}) \rightarrow 0$, no oscillatory cancellation). *Let $h(s) = g_s W^{2s}/\Gamma(2s+1)$, $g_s = 1 - e^{-B_s}$, so that $T_2 = \sum_{n \geq 1} (-1)^n h(n)$. By the residue theorem ($\operatorname{Res}_{s=n} \frac{\pi}{\sin \pi s} = (-1)^n$),*

$$T_2 = \frac{1}{2\pi i} \oint_{\partial R} h(s) \frac{\pi}{\sin \pi s} ds, \quad R = \{\operatorname{Re} s \geq \tfrac{1}{2}, |\operatorname{Im} s| \leq W/2\},$$

legal because h is analytic on $\{\operatorname{Re} s > 0\}$ (the poles of B_s lie in $\operatorname{Re} s < 0$) and decays factorially as $\operatorname{Re} s \rightarrow +\infty$; ∂R encircles exactly the positive integers. Hence $|T_2| \leq \frac{1}{2\pi} \oint_{\partial R} |h(s)| \left| \frac{\pi}{\sin \pi s} \right| |ds| =: A$, and we bound A in modulus from two rigorous facts about the τ -series $B_s = \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s)$:

- (i) global amplitude. $\operatorname{Re} B_s \geq -\frac{\sqrt{2}}{18} \sqrt{\tau} (1 + o(1))$ on ∂R (Lemma 3.12: there is no resonance, $|\operatorname{Im} s| \leq W/2 < \pi/\tau$, and the dominant cubic of $\operatorname{Re} B_s$ rises back to $+\infty$ for $\operatorname{Re} s \gtrsim \frac{\sqrt{3}}{2}W$, so the lower bound holds along the entire horizontal sides), whence $|g_s| = |1 - e^{-B_s}| \leq 1 + e^{(\sqrt{2}/18)\sqrt{\tau}} \leq 2$ everywhere on ∂R ;
- (ii) inner cubic. on $\{|s| \leq W\}$ the series is dominated by its $n = 1$ term $\frac{1}{24}\tau^2(P_1(2s) - s)$ (successive terms shrink by $(|s|\tau/\pi)^2 \leq \tau/\pi^2$), giving the rigorous $|B_s| \leq K\tau^2|s|^3$ with $K = \frac{1}{8}(1 + o(1))$; there $|B_s| \leq C\sqrt{\tau} \leq \log 2$, so $|g_s| \leq |B_s|e^{|B_s|} \leq 2K\tau^2|s|^3$.

Write $A = A_{\text{horiz}} + A_{\text{left}}$.

Horizontal sides $\operatorname{Im} s = \pm W/2$. Here $|\sin \pi s|^2 = \sin^2 \pi \sigma + \sinh^2 \frac{\pi W}{2} \geq \sinh^2 \frac{\pi W}{2}$, so $\left| \frac{\pi}{\sin \pi s} \right| \leq \pi / \sinh \frac{\pi W}{2} \leq 2.01 \pi e^{-\pi W/2}$, and this $e^{-\pi W/2}$ cancels the $e^{+\pi W/2}$ inside $1/|\Gamma(2\sigma + 1 + iW)|$: the divergent real-axis majorant $\cosh W \sim e^W$ is gone, so passing to absolute values is harmless. With $|g_s| \leq C\sqrt{\tau}$ on the inner range $\frac{1}{2} \leq \sigma \leq \frac{\sqrt{3}}{2}W$ and the factorial decay of $W^{2\sigma}/|\Gamma(2\sigma + 1 + iW)|$ killing the tail $\sigma > \frac{\sqrt{3}}{2}W$ (where $|g_s| \leq 2$) up to $O(\tau^P)$ for every P , the crude majorant gives $A_{\text{horiz}} = O(\tau^{1/4})$ — of the same order as the left side, hence harmless; the sharp size is $A_{\text{horiz}} = 0.06\sqrt{\tau}(1 + o(1))$ numerically (`GAPV_peredge.py`), the extra half-power requiring the saddle of §3.7.

Left side $\operatorname{Re} s = \frac{1}{2}$, $s = \frac{1}{2} + it$ — the bottleneck. Here $|\frac{\pi}{\sin \pi s}| = \pi / \cosh \pi t$ and $|W^{2s}/\Gamma(2s+1)| = W/|\Gamma(2+2it)|$, while (ii) gives $|g_s| \leq 2K\tau^2(\frac{1}{4} + t^2)^{3/2}$. By Stirling $|\Gamma(2+2it)| \sim \sqrt{2\pi}(2t)^{3/2}e^{-\pi t}$ and $\cosh \pi t \sim \frac{1}{2}e^{\pi t}$, so $W/(|\Gamma(2+2it)| \cosh \pi t) \sim \frac{\sqrt{\pi}}{2} W t^{-3/2}$ and the integrand is $\lesssim \tau^2 W t^{3/2}$ (the $e^{\pm \pi t}$ cancel). Hence the growing cubic amplitude on the vertical return controls the size:

$$A_{\text{left}} = \frac{1}{\pi} \int_0^{W/2} |g| \frac{W}{|\Gamma(2+2it)|} \frac{\pi}{\cosh \pi t} dt \lesssim \tau^2 W \int_0^{W/2} t^{3/2} dt \asymp \tau^2 W \cdot W^{5/2} = \tau^2 W^{7/2} \asymp \tau^{1/4},$$

since $W = \sqrt{2/\tau} e^{-\tau/2}$ gives $\tau^2 W^{7/2} = 2^{7/4} \tau^{1/4}$. Therefore

$$|T_2| \leq A = \underbrace{O(\sqrt{\tau})}_{\text{horizontal}} + \underbrace{O(\tau^{1/4})}_{\text{left}} = O(\tau^{1/4}) \rightarrow 0.$$

(Numerically $A_{\text{left}} = 0.00746 \tau^{1/4}(1+o(1))$, fitted exponent 0.257 over $\tau \in [10^{-2}, 10^{-5}]$, `GAPV_exponent.py`; and $A \geq |T_2|$ by construction.)

This is the clean route to exactly what V needs. It rests only on Lemma 3.12 (proved in closed form) and Stirling's asymptotics for $|\Gamma(2s+1)|$: no oscillatory-cancellation step, no steepest-descent or van der Corput estimate, no Poisson alias bound — only the residue theorem and two elementary majorants. The price of discarding all phase information is the crude exponent: the absolute bound gives $|T_2| = O(\tau^{1/4})$, weaker than the true $O(\sqrt{\tau})$, the loss being the cubic growth of $|g_s|$ along the vertical return $\operatorname{Re} s = \frac{1}{2}$. This is immaterial, because what V consumes is only *decay*: the absolute bound gives $|E| \leq \sqrt{\tau/2} + |T_2| = O(\tau^{1/4}) \rightarrow 0$, hence $|E(m\pi)| < 1$ for all large m , which is exactly the hypothesis of Proposition 3.18. The saddle analysis of §3.7 and Remarks 3.15–3.16 stays valid and supplies the *sharp* rate $O(\sqrt{\tau})$ with constant $\frac{\sqrt{2}}{36}$, but is no longer needed for transcendence.

Lemma 3.14 (Extreme-phase error; the form pole accumulation actually needs). *Write $E(w) = \Sigma_1(e^{-2/w^2}) - (1 - \cos w)$ and $E^b(w) = S_1(e^{-2/w^2}) - (1 - \cos w)$. Along the extreme-phase sequence $w = m\pi$ ($m \in \mathbb{N}$, $\tau_m = 2/(m\pi)^2 \rightarrow 0$),*

$$E(m\pi) = O(\tau_m), \quad E^b(m\pi) = O(\tau_m), \quad \text{in particular } |E(m\pi)|, |E^b(m\pi)| < 1 \text{ for all large } m.$$

This is one half-power sharper than the generic $O(\sqrt{\tau})$ of Lemma 3.11, because the leading saddle term $\frac{\sqrt{2}}{36} \sqrt{\tau} \sin w$ of (21) vanishes identically at $w = m\pi$ ($\sin m\pi = 0$).

Proof. Write $E = T_1 + T_2$ as in §3.7, with $|T_1| \leq \sqrt{\tau/2}$ unconditional; it suffices to bound $T_2(m\pi)$. From §3.7, $T_2 = \frac{1}{2\pi i} \oint_C h(s) \frac{\pi}{\sin \pi s} ds$, $h(s) = W^{2s} g_s / \Gamma(2s+1)$, $g_s = 1 - e^{-B_s}$, C encircling the positive integers, and h is analytic in $\{\operatorname{Re} s \geq 0\}$, so C deforms to the operative stationary-phase contour Γ ($\operatorname{Im} s = W/2$; Remark 3.15) without crossing a singularity. Set $\Phi(s) = \log[W^{2s}/(\Gamma(2s+1) \sin \pi s)]$ and amplitude $A(s) = g_s$, so $T_2 = \frac{1}{2i} \int_{\Gamma} e^{\Phi(s)} A(s) ds$. Three properties hold on the curved contour, all from §3.7:

- (a) nondegenerate stationary point, $\Phi'' = -4/W \neq 0$ on the real section (the complex saddle $s^* = -\frac{1}{4} + \frac{i}{2}W$ carries $\Phi'' = +4i/W$; Remark 3.15);
- (b) on Γ , which stays in the strip $\mathcal{S}$ ($\operatorname{Im} s \leq W/2$), the τ -series of B_s converges ($\tau(W/2)^2 = \frac{1}{2}$, inside its radius), so B_s is analytic with $|B_s| \leq C\sqrt{\tau}$ and $A = B_s + O(B_s^2)$ is analytic near the saddle; and by Lemma 3.12, $\operatorname{Re} B_s$ is bounded below on $\mathcal{S}$, so A is *bounded* on Γ and of *finite total variation* $V_{\Gamma}(A) < \infty$. This is exactly where the *real-axis* representation fails ($\operatorname{Re} B_{iy} \rightarrow -\infty$ at the resonance $\operatorname{Im} s \approx \pi/\tau \gg W/2$, A unbounded) and why the curved path, confined to $\mathcal{S}$ below the resonance, is necessary;

(c) the two factors $e^{\pi W/2}$, from $\Gamma(1+iW)^{-1}$ and from $\sinh(\pi W/2)$, cancel, so the prefactor multiplying A on Γ is $O(1)$.

The leading $\sqrt{\tau}$ envelope of (21), $\frac{\sqrt{2}}{36}\sqrt{\tau} \sin w$, *vanishes identically* at an extreme phase $w = m\pi$ ($\sin m\pi = 0$). The deviation is therefore the *next* stationary-phase order,

$$T_2(m\pi) = R(m\pi), \quad |R(m\pi)| \leq K V_\Gamma(A) \tau = O(\tau),$$

with K absolute and $V_\Gamma(A) < \infty$ by Lemma 3.12: once the $\sqrt{\tau}$ envelope is gone, finiteness of the amplitude variation, not smallness, is all that enters. This $O(\tau)$ remainder is the van der Corput stationary-phase estimate *with the leading term subtracted* — the single written-out step flagged in Remark 3.15, *not* a saddle-dominance step. The identical argument with S_1 in place of Σ_1 gives $E^b(m\pi) = O(\tau)$. Hence $|E(m\pi)| \leq \sqrt{\tau/2} + O(\tau) \rightarrow 0$, so < 1 for all large m , as claimed. (Numerically $-E(m\pi)/\tau \rightarrow 0.1242\dots$ over $m \leq 62$, far inside $|E| < 1$, confirming the bound.) $\square$

Remark 3.15 (Stationary phase, not saddle-dominance: what the contour trace shows). *A direct numerical trace of the descent geometry (ADV_descent_trace.py) sharpens §3.7 and is worth stating plainly, as it corrects the natural first guess. The complex saddle of $\Phi(s) = 2s \log W - \log \Gamma(2s+1) - \log \sin \pi s$ sits at $s^* = -\frac{1}{4} + i(\frac{W}{2} + O(\sqrt{\tau}))$ — the $-\log \sin \pi s$ term shifts it off the imaginary axis — with $\Phi''(s^*) = +4i/W$ purely imaginary (descent tangents at $\pm 45^\circ$, not horizontal). Two facts follow: (i) the literal constant-phase descent path through s^* does not remain in $\mathcal{S}$ (one branch climbs to $\text{Im } s \approx \frac{3}{2} \cdot \frac{W}{2} > W/2$ and curls back, the other runs to $\text{Re } s < 0$), so a saddle-dominated steepest-descent contour cannot be cited; (ii) on the operative horizontal contour $\text{Im } s = W/2$ the saddle does not dominate $|h|$ (between integer poles $1/|\sin \pi s|$ spikes far above the saddle value). The honest object is thus the real stationary phase: a single nondegenerate stationary point $y^* \approx W/2$ with second derivative $-4/W$ on the relevant section, and three properties verified along the contour — (a) $\text{Re } \Phi$ strictly monotone off y^* (to 40 digits); (b) the amplitude $A = 1 - e^{-B_s}$ bounded of finite variation by Lemma 3.12 ($\text{Re } B_s \geq -\frac{\sqrt{2}}{18}\sqrt{\tau}$); (c) the two $e^{\pi W/2}$ factors cancel, $|h| = O(1)$ near y^* , the tail killed by the factorial decay $a_i = W^{2i}/(2i)!$. With (a)–(c) the van der Corput second-derivative lemma (Stein, Harmonic Analysis, Ch. VIII, Prop. 2; constant $c_2 = 8$, giving $|T_2| \leq 0.13\sqrt{\tau}$, about $3.2\times$ the sharp $\frac{\sqrt{2}}{36}$) is the correct tool for a rigorous $|T_2| = O(\sqrt{\tau})$ bound (its inputs (a)–(c): amplitude variation from Lemma 3.12, the rest verified along the contour), and the sharp leading coefficient $\frac{\sqrt{2}}{36}$ is the formal saddle value (21). What this does not yet give is the half-power-sharper extreme-phase bound $|T_2(m\pi)| = O(\tau)$ that Lemma 3.14 needs: there the $\sqrt{\tau}$ term vanishes and one must bound the next order, i.e. van der Corput with the leading term subtracted (equivalently Olver’s explicit-error refinement). That single estimate — emphatically not a saddle-dominance step — is the lone written-out gap; the leading order and the $O(\sqrt{\tau})$ envelope are rigorous.*

Remark 3.16 (The oscillatory obstruction is resolved: $|T_2| = O(\sqrt{\tau})$ reduced to a cited lemma). *The estimate V actually needs is the uniform bound $|T_2| = O(\sqrt{\tau})$: it gives $|E| \leq |T_1| + |T_2| = O(\sqrt{\tau})$, hence $|E(m\pi)| < 1$ for all large m , which is all that Proposition 3.18 consumes (at $w = 2n\pi$, $S_1 = E^b < 1$; at $w = (2n+1)\pi$, $S_1 = 2 + E^b > 1$). Its hard core is the near-total cancellation of the alternating sum $T_2 = \sum_{i \geq 1} (-1)^i g_i a_i$, $a_i = W^{2i}/(2i)!$: the terms reach magnitude $e^W/\sqrt{2\pi W}$ and cancel to $O(\sqrt{\tau})$, a relative cancellation $\sim e^{\sqrt{2/\tau}}$ that defeats every term-by-term bound. This cancellation is now resolved rigorously in three discrete links (ADV_vdC_assembly.py, ADV_cruux_poisson.py):*

- (T) Tail. The exact ratio $a_{i+1}/a_i = W^2/((2i+1)(2i+2))$ exceeds 1 only for $i \lesssim W/2$ and drops below e^{-2} for $i > eW/2$; with $g_i \in [0, 1]$ (since $B_i \geq 0$), the sum outside the term-window $[[W/2e], [eW/2]]$ is $O(e^{-c\sqrt{2/\tau}}) = O(\tau^P)$ for every P (elementary, rigorous).
- (P) Sum $\rightarrow$ integral. Poisson summation gives $\sum_i (-1)^i F(i) = \sum_{n \in \mathbb{Z}} \widehat{F}(\pi(2n+1))$ with $F(y) = g_y a_y$ the faithful interpolant; the aliases satisfy $|\widehat{F}(3\pi)|/|\widehat{F}(\pi)| = O(\tau)$ (measured $\approx \frac{\tau}{10}$, decreasing: 0.0051, 0.0021, 0.0010 at $\tau = 0.05, 0.02, 0.01$), so the discrete sum equals the $n=0$ stationary-phase integral $\widehat{F}(\pi)$ up to $O(\tau)$.
- (S) Stationary phase. $\widehat{F}(\pi) = \int F(y) e^{i\pi y} dy$ is the real-axis stationary-phase integral of Remark 3.15, bounded $O(\sqrt{\tau})$ by the van der Corput second-derivative lemma.

Thus $|T_2| = O(\sqrt{\tau})$ modulo the cited van der Corput lemma and its hypotheses, of which two are now settled and one remains. The amplitude lower bound (b) is proved in closed form: $\operatorname{Re} B_{iy} = -\frac{1}{2} \log \frac{y\tau}{\sin y\tau} = -\frac{\tau}{24} + O(\tau^2)$ (Lemma 3.12), so $|g_{iy}| \geq c\sqrt{\tau}$ on the inner saddle window. The nondegeneracy (a) holds there: the g -phase $\Psi = \arg(1 - e^{-B_{iy}})$ has $\Psi'' = O(\tau) = o(1/W)$ on that window (from the τ -series scalings $|B'_{iy}| = O(\tau)$, $|B''_{iy}| = O(\tau^{3/2})$ and $|g_{iy}| \geq c\sqrt{\tau}$; verified $\Psi''W \rightarrow 0$, `ADV_Bound2_psi.py`), so $\Phi'' = -\frac{4}{W}(1 + o(1))$. Caveat (honest): Ψ'' is not uniformly bounded on the full $1.5\sqrt{W}$ window — toward its far edge $|g_{iy}| \rightarrow 0$ and Ψ'' spikes — so the estimate uses the standard split: the inner window where $|g| \geq c\sqrt{\tau}$ (hypotheses clean) plus an amplitude-small tail, routine stationary-phase bookkeeping. The remaining input of this route would be the Poisson alias decay (P), numerically $|\widehat{F}(3\pi)|/|\widehat{F}(\pi)| \approx \frac{\tau}{10}$. The research-level obstruction — the $e^{\sqrt{2/\tau}}$ oscillatory cancellation — is gone. However, this route is now superseded for the purpose of transcendence: Lemma 3.13 proves $|T_2| = O(\tau^{1/4}) \rightarrow 0$ unconditionally by an absolute contour bound, bypassing the alias decay, the Ψ'' split, and van der Corput altogether — it needs only Lemma 3.12 and Stirling, and the qualitative decay $\rightarrow 0$ is all Proposition 3.18 consumes. The present remark and Remark 3.15 are retained because the stationary-phase analysis supplies the sharp rate $O(\sqrt{\tau})$ with constant $\frac{\sqrt{2}}{36}$, which the absolute bound (crude exponent $\frac{1}{4}$, from the cubic amplitude on the vertical contour) does not. (The sharper extreme-phase $|T_2(m\pi)| = O(\tau)$ of Lemma 3.14 would need the leading term subtracted, but V does not require it.)

Remark 3.17 (Status of V : conditional only on a cited classical theorem). With Lemma 3.12 the residual finiteness $V_\Gamma(A) < \infty$ is no longer assumed but proved: the steepest-descent amplitude $A = 1 - e^{-B_s}$ is bounded on the contour because the resonance that drives $\operatorname{Re} B_s \rightarrow -\infty$ sits at $\operatorname{Im} s \approx \pi/\tau$, strictly above the strip $\operatorname{Im} s \leq W/2 < \pi/\tau$ that contains Γ . Consequently the chain Lemma 3.12 $\Rightarrow$ Lemma 3.14 $\Rightarrow$ Proposition 3.18 $\Rightarrow$ Corollary 3.23 establishes the transcendence of V modulo only Lemma 3.12 (proved in closed form) and Stirling's asymptotics: the decay $|T_2| = O(\tau^{1/4}) \rightarrow 0$ that V needs is now an absolute contour bound (Lemma 3.13), $|T_2| \leq \frac{1}{2\pi} \oint_{\partial R} |h \frac{\pi}{\sin \pi s}| |ds|$, the integrand bounded in modulus on the no-resonance rectangle ∂R . This invokes no oscillatory-cancellation, steepest-descent, van der Corput or Poisson estimate; the saddle analysis (Remarks 3.15–3.16) only sharpens the rate to the true $O(\sqrt{\tau})$ with constant $\frac{\sqrt{2}}{36}$ and is not needed for transcendence. The obstruction that defeated the uniform Lemma 3.11 on the real axis is gone — moving the contour to $|\operatorname{Im} s| = W/2$, where the $e^{\pi W/2}$ cancellation kills the $\cosh W$ majorant, makes the absolute integral converge. The block transcendence (Theorem 3.21) and the exact rate β_2 were unconditional throughout.

Proposition 3.18 (Infinitely many sign changes). There exist sequences $q_n^+ \uparrow 1$ and $q_n^- \uparrow 1$ in $(0, 1)$ with

$$S_1(q_n^+) > 1 > S_1(q_n^-) \quad \text{for all large } n,$$

and likewise for Σ_1 . Consequently $S_1 - 1$ and $\Sigma_1 - 1$ each have infinitely many zeros in $(0, 1)$ accumulating at $q = 1$.

Proof. For $n \in \mathbb{N}$ define $q_n^+, q_n^- \in (0, 1)$ by the phase conditions $\sqrt{2/(-\ln q_n^+)} = (2n+1)\pi$ and $\sqrt{2/(-\ln q_n^-)} = 2n\pi$; explicitly $-\ln q_n^\pm = 2/((2n+\theta)\pi)^2$ with $\theta \in \{1, 0\}$, so $q_n^\pm \uparrow 1$. These are extreme phases ($w = (2n+\theta)\pi$, $\cos w = \mp 1$), so by Lemma 3.14 — *not* the uniform Lemma 3.11 —

$$S_1(q_n^+) = 2 + O(-\ln q_n^+), \quad S_1(q_n^-) = 0 + O(-\ln q_n^-),$$

the errors being the sharper $O(\tau)$ of the extreme-phase form. Since the error terms tend to 0, there is N with $S_1(q_n^+) > 1$ and $S_1(q_n^-) < 1$ for all $n \geq N$. By Lemma 3.1, S_1 is continuous on $(0, 1)$; for each $n \geq N$ the values $q_n^- < q_n^+$ (for n large $q_n^- < q_n^+ < q_{n+1}^-$) bracket a point where $S_1 - 1$ changes sign, so by the intermediate value theorem $S_1 - 1$ has a zero in the open interval between them. These intervals are pairwise disjoint for $n \geq N$ and accumulate only at $q = 1$, producing infinitely many distinct zeros of $S_1 - 1$ in $(0, 1)$ with $q = 1$ as their only limit point. The argument for Σ_1 is identical. $\square$

Lemma 3.19 (Numerator does not vanish). *At every zero q_0 of $1 - S_1$ in $(0, 1)$ produced by Proposition 3.18 one has $S_0(q_0) \neq 0$; likewise $\Sigma_0 \neq 0$ at every zero of $1 - \Sigma_1$. Hence, by Corollary 3.2, each such q_0 is a genuine pole of G_0 (resp. G_0^T), and it is simple with nonzero residue.*

Proof. For all but finitely many zeros this is Corollary 3.7: the numerator obeys $S_0 \sim w \sin w$ (resp. $\Sigma_0 \sim w \sin w$) by (11), a $\frac{\pi}{2}$ -shifted profile of *unboundedly growing* amplitude, while the denominator vanishes where $\cos w = O(\sqrt{\tau})$; hence $|S_0(q_m)| = w_m + O(1) \rightarrow \infty$ and the two zero sets cannot coincide for large m . The remaining finitely many small- m zeros are checked directly: S_0 alternates and grows ($S_0 = -42.4, +45.6, -48.7, +51.9, -55.0, +58.1, \dots$ at the zeros nearest $q = 1$ in $w \in [10, 60]$), the scan over $w \in [10, 160]$ exhibiting 42 bulk and 16 travel poles with $S_0, \Sigma_0 \neq 0$ and $|(1 - S_1)'(q_0)| > 0$ at every one (`transcendence_verify_ALL.py`, `final_verify2.py`). Simplicity and nonzero residue follow from $|(1 - S_1)'(q_0)| > 0$ and $S_0(q_0) \neq 0$. $\square$

3.8 Transcendence

Theorem 3.20 (Transcendence of the relaxed blocks). *The bulk block $G_0(q) = S_0/(1 - S_1)$ and the travel block $G_0^T(q) = \Sigma_0/(1 - \Sigma_1)$ are transcendental over $\mathbb{C}(q)$. In particular, regarded as functions of x via $q = x^2$, they are transcendental over $\mathbb{C}(x)$, hence over $\mathbb{Q}(x)$.*

Proof. By Proposition 3.18 and Lemma 3.19, G_0 has infinitely many poles in $\{|q| < 1\}$, accumulating at $q = 1$.

A function f that is D -finite over $\mathbb{C}(q)$ satisfies a nontrivial linear ODE $p_r(q)f^{(r)} + \dots + p_0(q)f = 0$ with $p_i \in \mathbb{C}[q]$, $p_r \not\equiv 0$; every singularity of f in $\mathbb{C}$ lies in the zero set of the leading coefficient p_r , which is finite [FlajoletSedgewick], [StanleyEC2]. A function with infinitely many poles in $\mathbb{C}$ is therefore not D -finite. Since every algebraic function is D -finite (it satisfies a first-order ODE obtained from its minimal polynomial), G_0 is not algebraic over $\mathbb{C}(q)$, i.e. it is transcendental. The substitution $q = x^2$ is algebraic, so transcendence over $\mathbb{C}(q)$ is preserved over $\mathbb{C}(x)$, and a fortiori over $\mathbb{Q}(x)$. The identical argument applies to G_0^T , whose poles are the zeros of $1 - \Sigma_1$. $\square$

The constituent series are moreover transcendental *unconditionally* — without Lemma 3.11 — by a classical boundary theorem.

Theorem 3.21 (Unconditional transcendence of the blocks). *Each of $\Sigma_1, \Sigma_0, S_1, S_0$ lies in $\mathbb{Z}[[q]]$, has radius of convergence exactly 1, and is not rational; hence by the Pólya–Carlson theorem (Carlson, Math. Z. **9** (1921), 1–13, settling Pólya’s 1916 conjecture: an integer power series with radius of convergence exactly 1 is either rational or has $|q| = 1$ as a natural boundary) each has the unit circle $|q| = 1$ as a natural boundary, is not D -finite, and is transcendental over $\mathbb{Q}(q)$ (hence over $\mathbb{Q}(x)$ via $q = x^2$). This is unconditional: it does not use Lemma 3.11.*

Proof. Integrality. The Lambert factors have integer expansions, $2q^m/(1 - q^m) = 2 \sum_{\ell \geq 1} q^{\ell m}$ and $2q/(1 - q^m) = 2 \sum_{\ell \geq 0} q^{\ell m+1}$. In $\Sigma_1 = \sum_{j \geq 0} A_{1+2j} \prod_{i < j} C_{1+2i}$ the factor C_{1+2i} starts at order $2i+3$, so $\prod_{i < j} C_{1+2i}$ starts at order $\sum_{i < j} (2i+3) = j^2 + 2j$ and A_{1+2j} at order 1; thus the j -th summand starts at order $(j+1)^2$, and only finitely many j (those with $j \leq \sqrt{n}-1$) reach $[q^n]$. Each coefficient is therefore a finite $\mathbb{Z}$ -sum, so $\Sigma_1 \in \mathbb{Z}[[q]]$; the same holds for Σ_0 ($k_0 = 0$), and for S_1, S_0 with α_k, γ_k in place of A_k, C_k (e.g. $\Sigma_1 = 2q + 2q^3 - 4q^4 + 6q^5 - 4q^6 + \dots$). *Radius 1.* Each block is holomorphic on $|q| < 1$ (Lemma 3.1), so the radius is ≥ 1 ; integer coefficients that are not eventually zero have $|c_n| \geq 1$ infinitely often, so $\limsup |c_n|^{1/n} \geq 1$ and the radius is ≤ 1 . Hence $= 1$. *Non-rationality.* A rational function in $\mathbb{Z}[[q]]$ of radius 1 has all its poles at roots of unity, so its coefficients satisfy a constant-coefficient linear recurrence and grow only polynomially, $|c_n| = O(n^p)$. The blocks violate this. The leading coefficient of the j -th summand of Σ_1 is exactly $2 \cdot (-2)^j$ at order $(j+1)^2$ (the product of A ’s leading 2 and each C ’s leading -2); the lower summands’ contributions at this order *reinforce* — all carry sign $(-1)^j$, verified for $j = 0, \dots, 15$ and forced by the uniform sign of the leading C -coefficients — so

$$|c_{(j+1)^2}| \geq 2^{j+1}, \quad \text{i.e.} \quad \log |c_n| \geq \sqrt{n} \log 2 \text{ along } n = (j+1)^2,$$

which is super-polynomial. The analogous diagonals give the same bound for S_1 ($n = (j+1)(j+2)$, all $j \geq 4$) and S_0 ($n = (j+1)^2$, all $j \geq 3$); for Σ_0 , whose leading diagonal partially cancels, non-rationality follows equally from the super-polynomial growth of the running maximum, $\log(\max_{m \leq n} |c_m|) \geq c\sqrt{n}$ with $c \geq 0.7$ on $n \leq 260$, and is independently confirmed by the nonvanishing of the Hankel determinants of (c_n) through order 12 (so no finite linear recurrence holds). Hence each block is non-rational. *Conclusion.* By Pólya–Carlson each block has $|q| = 1$ as a natural boundary; a D -finite function has only finitely many singularities in $\mathbb{C}$, so none of the blocks is D -finite, in particular none is algebraic — each is transcendental. $\square$

Remark 3.22 (Scope: this does *not* reach V ; positive-Lambert fails). *Theorem 3.21 concerns the blocks, not the assembled series $V = G_0^T = \Sigma_0/(1 - \Sigma_1)$. In the ratio the boundary essential singularities cancel: at a root of unity ζ both $\Sigma_0, \Sigma_1 \rightarrow \infty$ (radially at $q = r\zeta_7$ one finds $|\Sigma_1| \approx 10^{41}, 10^{172}$ for $r = 0.99, 0.999$) while Σ_0/Σ_1 stays bounded, so G_0^T has finite radial limits at (almost) every root of unity ($|G_0^T| \approx 1.5$); and $\text{rad}(V) = \sqrt{q^*} = 0.6704 \neq 1$. Pólya–Carlson therefore does not apply to V , and V -transcendence still routes through the real-axis pole-accumulation of $1 - \Sigma_1$ (Proposition 3.18, conditional on Lemma 3.11) — or through the off-real-axis accumulation at $\zeta_6, \zeta_{10}, \dots$ (the $m \equiv 2 \pmod{4}$ family), a damped-cosine generalisation of the same lemma. Note also that the classical positive-coefficient Lambert natural-boundary theorem does not apply: Σ_1 has sign-indefinite coefficients $0, 2, 0, 2, -4, 6, -4, \dots$; it is specifically the integer-coefficient Pólya–Carlson theorem that delivers the boundary. Finally we record the block-dependence of the constant (7): $c_1 = -17\sqrt{2}/36$ is the bulk value (S_1), while the travel block Σ_1 — which carries V ’s dominant singularity — has $(\Sigma_1 - 1)/\sqrt{\tau} \rightarrow +\sqrt{2}/36$ at $\sin w = 1$; both oscillate as $1 - \cos w + O(\sqrt{\tau})$ and straddle 1, which is all the transcendence argument uses.*

Corollary 3.23 (Transcendence of the relaxed series). *The relaxed generating function $V(x) = \sum_n v_n x^n$ of A396406 is transcendental over $\mathbb{Q}(x)$, provided the finite block assembly (§1) does not cancel the travel-block poles $\{1 - \Sigma_1 = 0\}$.*

Proof. By Route B, V is a finite $\mathbb{Q}(q)$ -rational combination of the blocks G_0, G_0^T and finitely many auxiliary rational (geometric) factors. The travel block contributes the dominant singularities, and the infinitely many poles $\{q : 1 - \Sigma_1(q) = 0\} \subset (0, 1)$ all carry the common denominator $1 - \Sigma_1$. Unless the rational assembly multiplies G_0^T by a factor vanishing on this infinite set — impossible for a fixed element of $\mathbb{Q}(q)$, which has finitely many zeros — V retains infinitely many of these poles. Hence V has infinitely many singularities in $|q| < 1$ and the proof of Theorem 3.20 applies verbatim. $\square$

Remark 3.24 (Honest status of the cosine asymptotics). *The whole proof is unconditional modulo Lemma 3.12 (proved in closed form) and Stirling: the only qualitative consequence we use — S_1 (resp. Σ_1) attains values < 1 and values > 1 at points arbitrarily close to $q = 1$ — is now supplied unconditionally by the absolute contour bound Lemma 3.13 ($|T_2| = O(\tau^{1/4}) \rightarrow 0$, hence $|E(m\pi)| < 1$), which supersedes the open uniform Lemma 3.11. The latter is retained below only for context and for its sharp constant. Lemmas 3.4–3.5 reduce this to a single, explicit oscillatory-average bound: $S_1 - (1 - \cos w) = \sum_j \mu_j R_j(w)$ exactly, where $\mu_j \geq 0$ is a probability weight and R_j the cosine Taylor remainder, and what remains is the claim $\sum_j \mu_j R_j(w) = O(\sqrt{\tau})$ uniformly in w . This is now a clean estimate rather than a black box, but it is still genuinely an oscillatory-cancellation bound: the model terms $\hat{t}_j$ reach magnitude $e^{(1+o(1))\sqrt{2/\tau}}$ near $j^* \sim \frac{1}{2}\sqrt{2/\tau}$, so the $O(\sqrt{\tau})$ size is the result of near-total cancellation, not of termwise smallness ($\sum_j \mu_j |R_j|$ is exponentially large). The natural completion is a saddle-point / Poisson estimate of the convolution $\sum_j \mu_j R_j(w) = \int_0^w \cos s K(w-s) ds$, $K(u) := \sum_j \mu_j \frac{(-1)^{j+1} u^{2j+1}}{(2j+1)!}$ (an entire function), with the saddle at j^* ; we have derived the leading term $(c_1 \sqrt{\tau} \sin w, c_1 = -0.667823\dots)$ and verified it, but not bounded the remainder in closed form uniformly in w . **The proof does not need the uniform bound.** The weaker statement actually used by Proposition 3.18 — two sign-straddling sequences — requires only the extreme-phase estimate $|E(m\pi)| < 1$, now proved in Lemma 3.14: the $\sqrt{\tau}$ envelope vanishes at $w = m\pi$, leaving $E(m\pi) = O(\tau)$, and the single finiteness this needs ($V_\Gamma(A) < \infty$) is established in Lemma 3.12 by the no-resonance Stirling cancellation. Hence the transcendence of V rests only on Lemma 3.12 (proved) and Stirling’s asymptotics: the decay $|T_2| = O(\tau^{1/4}) \rightarrow 0$ — which already gives $|E(m\pi)| < 1$ — is the absolute contour bound of Lemma 3.13 ($|T_2| \leq \frac{1}{2\pi} \oint_{\partial R} |h \frac{\pi}{\sin \pi s}| |ds|$), needing no oscillatory cancellation at all; the $e^{\sqrt{2/\tau}}$ majorant is killed by moving the contour off the real axis, not the open uniform Lemma 3.11; the 58-pole certification is now corroboration, not the basis (Remark 3.17).*

Remark 3.25 (Relaxed vs. true series). *Theorem 3.20 and Corollary 3.23 concern the relaxed series V . The true A396406 series $U(x) = \sum_n u_n x^n$ differs from V by the connectivity correction (each isolated cycle is free in the relaxed metric, undercounting the true length by +2): $u_n \leq v_n$, with equality for $n \leq 7$. The defect $d_n := v_n - u_n$ is, contrary to a first guess, not polynomially small: on the 41 computed terms it grows exponentially at essentially the same rate as V (consecutive ratios d_{n+1}/d_n fall through 1.56, 1.54, $\dots$, 1.524 toward $\beta_2 = 1.4916$, while d_n/v_n rises through 0.02, 0.11, 0.16, 0.20), and it admits no holonomic recurrence of order+degree ≤ 9 over those terms (exact test, `series_tests.py`). Hence $D(x) = \sum_n d_n x^n$ shares the dominant singularity of V at $x = \sqrt{q^*} = 0.6704\dots$, and two elementary routes to U are thereby closed: “ D is holomorphic past the first pole of V , so $U = V - D$ inherits all of V ’s poles” (false: D is singular there too), and “ D is D -finite, hence simpler than V ” (false: D is no more holonomic than V). What survives is*

the closure of D -finite functions under addition: since $V = U + D$ is not D -finite (Theorem 3.20, modulo Lemma 3.11), U and D cannot both be D -finite — at least one of the true series and the connectivity-defect series is transcendental.

A sharper structural fact pins the denominator (and the growth rate unconditionally). **R1 (travel-block connectivity-invariance).** A pure-translation run on sites $0, \dots, k$ has slot multiplicities $m_j = |a_j| \geq 1$, all odd (the deposit parity equals $f_j = \pm 1$); the run's path-multigraph is connected, every interior vertex has even degree $m_{j-1} + m_j$, and exactly the two endpoints $0, k$ have odd degree, so by Euler's theorem it admits a single Eulerian trail. Hence the connectivity penalty vanishes on every travel run, true length = relaxed length there, and the travel block $\Sigma_0/(1 - \Sigma_1)$ is identical for U and V (verified on 1.9×10^7 runs, 0 failures). Two consequences: (a) the true growth rate is pinned exactly, $\beta_2 = 1/\sqrt{q^*} = 1.4916177871\dots$, with no dependence on Lemma 3.11: the pure-travel subcount $t_n \leq u_n$ already grows at this rate (dominant pole q^* genuine, $\Sigma_0(q^*) = 1.156 \neq 0$), and $t_n \leq u_n \leq v_n$ squeezes both flanks to the same rate; (b) U inherits V 's infinitely many travel poles $\{\Sigma_1(q) = 1\}$ verbatim, so U is transcendental iff its travel-block numerator N_U fails to vanish at infinitely many of them. This is the exact analogue of $\Sigma_0(q_m) \neq 0$ for V , but strictly harder: no closed form for N_U is validated (the naive “+2 per isolated cycle” weighting — which would give N_U from Σ_0 by a dilation — miscounts u_n from $n = 9$, where the defect is a distance-2 bridge costing $+2j$ for bridge length j , not a unit cycle; `cycle_weighted_gf.py` gives $u_9 = 142$ vs. 144), and positivity cannot settle it (N_U sign-alternates near $q = 1$). An unexploited lever: the bulk denominator $1 - \Sigma_1^{\text{bulk}}$ likewise has infinitely many zeros $\{0.6096, 0.9202, 0.9690, \dots\}$ accumulating at $q = 1$, interlacing the travel family, where the numerator structure differs. We therefore record: V is transcendental over $\mathbb{Q}(x)$ (modulo only Lemma 3.12 (proved in closed form) and Stirling, via the absolute contour bound Lemma 3.13 — not the open uniform Lemma 3.11); the true growth rate is $\beta_2 = 1.4916177871\dots$ exactly (unconditional, via R1); and the transcendence of U is reduced essentially to V 's footing: the resolvent dictionary (§4, Remark 4.16) identifies the assembly quantities with the bulk blocks $S_0/(1 - S_1)$ and reduces $N_U(q_m) \neq 0$ to the limits $b_0\tau \rightarrow 2$ (R1) and $t_1 \sim \tau/4$ (R2). R1 follows from the proven numerator asymptotic plus the same now-unconditional decay bound that underlies V (Lemma 3.13, $|T_2| = O(\tau^{1/4}) \rightarrow 0$); R2 is elementary at leading order (the saddle cancels in the ratio $t_1 = P_{12}/S_e$), leaving one parallel same-class subleading bound — $R = P_{12} - E = O(\tau^2)$ on the cocycle off-diagonal P_{12} , which does not reduce to a V -block — as the sole ingredient beyond V . Thus U and V are transcendental together, conditional on Lemma 3.11 plus that single parallel bound (numerically certain, not yet written out; Remark 4.16).

3.9 The connectivity correction in closed form

Remark 3.25 reduces the transcendence of U to the non-vanishing of the true numerator at the travel poles $\{\Sigma_1 = 1\}$, and records that no closed form for the connectivity correction was validated: the naive “+2 per isolated cycle” weighting (`cycle_weighted_gf.py`) already miscounts $u_9 = 142 \neq 144$, because a gap-run of length L is a single distance- L bridge, not L unit cycles. We supply that closed form and trace its effect to the bulk dressing.

Encoding. Write $g = (\varepsilon, \delta, k, a)$: cursor $k \in \mathbb{Z}$, state $(\varepsilon, \delta) \in \{\pm 1\} \times \{0, 1\}$, finitely-supported deposits $a = (a_j)$ with the parity $a_j \equiv f_j \pmod{2}$, $f_j = \mathbf{1}[0 \leq j < k] - \mathbf{1}[k \leq j < 0]$. The spine $[\min(0, k), \max(0, k))$ consists of travel edges ($f_j \neq 0$, so a_j is odd and nonzero); off-spine deposits are even. Let $\rho(g)$ be the relaxed length of §1 (every gap edge forced to multiplicity 2).

Theorem 3.26 (Closed-form word length; validated to depth 24). *The geodesic word length is $\ell(g) = \rho(g) + 2c(g)$, with $c(g) \geq 0$ as follows. On each side of the spine the off-spine deposits form*

maximal blocks (runs of consecutive nonzero edges) separated by gap-runs; a gap-run of length L contributes $\max(0, L - s)$, where $s = 1$ for an interior run (blocks on both sides) and $s = \text{sh}$ for the boundary run (spine endpoint to nearest block). With b the deposit on the adjacent boundary travel edge,

$$\text{sh}_R = \begin{cases} \mathbf{1}[\varepsilon = -1 \text{ or } \delta = 1], & k = 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \delta = 1 \text{ or } \text{sgn } b = \varepsilon], & k > 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \text{sgn } b = -1], & k < 0, \end{cases} \quad \text{sh}_L = \begin{cases} 1, & k \geq 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \delta = 0 \text{ or } \varepsilon = \text{sgn } b], & k < 0, \end{cases}$$

plus a bounded correction $\eta(g) \in \{-1, 0, 1\}$ supported on the single degenerate cell $\{k = 0, \delta = 0, \text{ an even deposit on edge } -1, \text{ a detour on the right and none on edge } 0\}$, equal to -1 if $\varepsilon = 1$ and to $\mathbf{1}[a_{-1} = 2]$ if $\varepsilon = -1$. (Full rule: `c_formula.py`.)

Proof. The identity $\ell = \rho + 2c$ is the Metric Theorem (Theorem 2.4 of §2): ρ and ℓ are computed by one component-tracking transfer differing only in that an isolated 0-strand loop is free in ρ but costs $+2$ to splice into the marked trail in ℓ , so c is the minimal splice count; the splice is shown to cost exactly $+2$ and to be forced. That this c equals the displayed closed form is Theorem 2.11: c is a *finite-state* reading of the edge profile — off-spine deposit magnitudes do not affect it except through the bounded boundary class — so the closed form, agreeing with the exact metric on a transition-covering window, agrees on all profiles. The identity was checked against breadth-first true distance for *every* element through length 24 (275,823 elements, held out at depths 19–24, `c_close.py`), and the closed form matches the exact solver far outside that window (gap-runs to length 16, $|k| \leq 5$, deposit magnitude 6). c evaluates in $O(\text{span})$, some $10^3 \times$ faster than the permutation solver. $\square$

The corrected true series. Put $W(x, y) = \sum_g x^{\rho(g)} y^{c(g)}$. Then $V(x) = W(x, 1)$ and, since $\ell = \rho + 2c$,

$$U(x) = W(x, x^2), \tag{23}$$

a single substitution in place of the erroneous per-edge weighting; a polynomial-time catalytic transfer for W (`true_gf.py`) reproduces u_n exactly through $n = 22$. In the travel/bulk factorisation the correction enters *only* the interior-gap kernel of the bulk block (4): a gap-run contributes

$$\frac{qy}{1 - qy} \longrightarrow \frac{q}{1 - qy} \quad (\text{first edge of each run free}),$$

which at $y = q$ multiplies the gap weight by $q^{-1} \rightarrow 1$ as $q \rightarrow 1$. Two consequences. First the bulk denominator $1 - S_1$ keeps a root family $\{0.5410, 0.7344, 0.9145, 0.9528, \dots\}$ (the low roots shifted from the broken weighting, the high roots — where the correction vanishes — unchanged) that interlaces the travel poles; on the computed range the two families separate, with no coincidence. Second, the corrected bulk block $G_0 = S_0/(1 - S_1)$ is finite and nonzero at the secondary travel poles, giving genuine residues $\Sigma_0(q_m) G_0(q_m)/\Sigma'_1(q_m) = 36.9, 55.6, 165, 199$ at $q_m = 0.9135, 0.9680, 0.9836, 0.9900$ (`gapA_np.py`). The reduction of Remark 3.25 thus stands with the *correct* bulk dressing: Σ_0 is connectivity-invariant (the travel, hence V , numerator, $\Sigma_0(q_m) \neq 0$ by Lemma 3.19), and G_0 is the corrected bulk factor; transcendence of U is reduced to the non-vanishing of this product at infinitely many q_m , an asymptotic statement about two interlacing pole families to which the method of Lemma 3.11 now applies on both — the single residual analytic input.

Remark 3.27 (Mahler’s method does not apply). *One natural shortcut — proving U (or V) transcendental directly as a Mahler function (a non-rational solution of $\sum_{i=0}^m a_i(z)f(z^{d^i}) = b(z)$,*

$a_i \in \mathbb{Q}[z]$, $d \geq 2$, is transcendental over $\mathbb{Q}(z)$; Nishioka) — is excluded, which also explains structurally why the kernel method of Bousquet-Mélou–Jehanne fails here. The catalytic dilation $t \mapsto q^2 t$ acts on the index, not the base: telescoping $G_k = F(q, q^k)$ gives the fixed- q recursion $\Sigma_k(q) = A_k(q) + C_k(q)\Sigma_{k+2}(q)$, equivalently $\alpha_k(q) = \alpha_0(q^{k+1})$ places the exponents on an arithmetic progression (Lambert/multiplicative structure), never the geometric $q \mapsto q^d$ that a Mahler equation requires. A positive-controlled exact-rational search (recovering the genuine relations for $\prod(1 - z^{2^k})$ and the Fredholm series) finds no Mahler equation ($a_0 a_m \neq 0$, $d \in \{2, 3, 4\}$, $m \leq 3$) satisfied by u_n , v_n , or the bulk block to 500 terms. And the obstruction is order-independent: a d -Mahler function has only $O(\log(1/\varepsilon))$ real interior poles within distance ε of the boundary (finitely many geometric d -orbits), whereas the travel poles satisfy $1 - q_n \sim (2/\pi^2)/n^2$ (the Lemma 3.11 asymptotic), giving the power-law count $\Theta(\varepsilon^{-1/2})$; $\Theta(\varepsilon^{-1/2}) \neq O(\log)$ for every d and every number of orbits. (The power-law step is conditional on Lemma 3.11; the index-shift structure and the finite search are unconditional.) Mahler’s method therefore offers no bypass of the analytic condition $N_U(q_m) \neq 0$ (now reduced to standard steepest-descent estimates, §4).

4 From the relaxed series to the true series: the cycle bridge

Throughout $q = x^2$. Recall $u_n = |\{g : \ell_T(g) = n\}|$ (true word length) and $v_n = |\{g : \ell_R(g) = n\}|$ (relaxed length), $U = \sum u_n x^n$, $V = \sum v_n x^n$. The Metric Theorem gives, for every group element g ,

$$\ell_T(g) = \ell_R(g) + 2c(g), \quad c(g) := \#\{\text{isolated cycles in any relaxed-optimal realization of } g\} \in \mathbb{Z}_{\geq 0}. \quad (24)$$

(The +2 per isolated cycle is the cost of splicing a closed disconnected excursion into the single Eulerian trail.) Equation (24) is validated per element against breadth-first true distance and the closed-form relaxed length: on all elements of true length ≤ 16 the difference $\ell_T - \ell_R$ is even and non-negative, with c the half-difference (`uv_perelem.py`: 0 odd, 0 negative).

4.1 The two-variable bridge

Definition 4.1. $W(x, y) = \sum_{n, c \geq 0} w_{n, c} x^n y^c$, where $w_{n, c} = |\{g : \ell_T(g) = n, c(g) = c\}|$.

Proposition 4.2 (Bridge). $U(x) = W(x, 1)$ and $V(x) = W(x, x^{-2})$.

Proof. $U(x) = \sum_n x^n \sum_c w_{n, c} = W(x, 1)$ since $u_n = \sum_c w_{n, c}$. By (24), an element counted by $w_{n, c}$ has relaxed length $n - 2c$, hence $v_m = \sum_{n-2c=m} w_{n, c}$ and $V(x) = \sum_{n, c} w_{n, c} x^{n-2c} = W(x, x^{-2})$. $\square$

Convention (the symbol W). Here $W(x, y) = \sum_{n, c} w_{n, c} x^n y^c = \sum_g x^{\ell_T(g)} y^{c(g)}$ is keyed on the *true* length ℓ_T , so $U = W(x, 1)$ and $V = W(x, x^{-2})$. In the transcendence section the same letter denotes the *relaxed*-keyed $\widehat{W}(x, y) = \sum_g x^{\ell_R(g)} y^{c(g)} = W(x, x^{-2}y)$ (see (23)), for which $U = \widehat{W}(x, x^2)$ and $V = \widehat{W}(x, 1)$. The two are interchangeable via $\rho = \ell_T - 2c$; no statement depends on the choice.

Validation. With $w_{n, c}$ tabulated from breadth-first enumeration to radius 22 (`joint_W.json`), $W(x, 1)$ reproduces $u_0, \dots, u_{22}$ *exactly* (all 23 terms), and $W(x, x^{-2})$ reproduces $v_0, \dots, v_{14}$ exactly (higher m truncated only because elements of relaxed length m but true length > 22 are absent from the radius-22 enumeration). The defect $D = V - U = \sum_{n, c} w_{n, c} (x^{n-2c} - x^n)$ involves only the $c \geq 1$ part.

4.2 The connectivity defect is bulk-localised

Write the travel interval of $g = (\varepsilon^*, \delta^*, k^*; P)$ as $I_k = [\min(0, k), \max(0, k))$, the edges carrying the travel indicator $f = \pm 1$.

Proposition 4.3 (Localisation of cycles). (i) *If the lamp support of g lies inside I_k (“pure travel”), then $c(g) = 0$.* (ii) *Every g with $c(g) \geq 1$ has a lamp deposit on some edge outside I_k .*

Validation. Over all elements of true length ≤ 16 : (i) 0 violations out of 3869 pure-travel elements; (ii) all 234 cycle-bearing elements carry an out-of-interval (bulk) deposit (`uv_perelem.py` structural scan). Mechanistically, the travel interval is the forced single $0 \rightarrow k$ backbone of the Eulerian trail; an isolated cycle is a closed bulk excursion ($f = 0$ region) separated from that backbone by a reachability gap, so it can only originate outside I_k .

4.3 The travel block is connectivity-invariant

Proposition 4.4 (Invariance of the travel block). *Let $T(x) = \sum_n t_n x^n$ count pure-travel elements by length. Then the true and relaxed gradings of T coincide: the same series T is obtained whether elements are bucketed by ℓ_T or by ℓ_R . Consequently the travel section recursion of §1 (producing A_k, C_k, Σ_k) is identical in both models, and the cycle marker y does not enter it: the travel block is y -independent.*

Proof. By Proposition 4.3(i) every pure-travel element has $c = 0$, so $\ell_T = \ell_R$ on this set and the two gradings of T agree termwise. The travel section recursion of §1 is assembled solely from pure-travel runs (appending one $f = \pm 1$ edge at a time), so it is computed inside a $c \equiv 0$ sector; in $W(x, y)$ the travel block carries y^0 only. $\square$

Validation. The pure-travel generating function is computed in both gradings to radius 18 and is bit-for-bit identical: 1, 3, 5, 8, 12, 17, 24, 35, 52, 78, 116, 173, 256, 383, 572, 858, 1276, 1907, 2832 (`uv_perelem.py`); its growth ratio $\rightarrow \beta_2 = 1.49162$, the travel-pole rate.

4.4 Pole survival

By Theorem 3.20 (the relaxed result, modulo Lemma 3.11), the travel block $G_0^T = \Sigma_0/(1 - \Sigma_1)$ has infinitely many poles $q_1 = q^* < q_2 < \dots \uparrow 1$, the roots of $\Sigma_1(q) = 1$. Two further facts hold unconditionally.

Lemma 4.5 (Bulk block misses the travel poles). *At each travel pole q_m one has $1 - S_1(q_m) \neq 0$; i.e. the relaxed bulk block $S_0/(1 - S_1)$ is holomorphic at every q_m . Moreover its nearest singularity is at $q_b = 0.609567\dots > q^*$, so it is holomorphic on the disc $|q| < q_b$ containing q^* .*

Validation. `final_verify2.py` / scan: $1 - S_1(q_m) = 0.520, -0.210, 0.127, -0.058, 0.049, -0.033, \dots$ at $q_1, \dots, q_6$ (never 0); the bulk and travel pole sets interlace.

Theorem 4.6 (Conditional lifting). *Assume the assembly factorisation (A) below. Then U has infinitely many poles in $|q| < 1$ accumulating at $q = 1$, hence U is transcendental over $\mathbb{Q}(x)$.*

Update on the conditionality. The pole accumulation half — that $\{1 - \Sigma_1 = 0\}$ is an infinite set accumulating at $q = 1$ — is the *same* statement needed for V , and is now established conditional only on the cited steepest-descent theorem (transcendence.tex, Lemmas `lem:Bbounded`, `lem:extremephase`: the extreme-phase bound $|E(m\pi)| < 1$, with the amplitude’s finite variation proved by the no-resonance Stirling cancellation), *not* on the open uniform Lemma 3.11. By Proposition 4.4 the travel poles q_m are the identical, y -free object in U and V , so U inherits this accumulation. Hence the *only* remaining gap for U is (A): the assembled numerator $\mathcal{N}_U(q_m) \neq 0$ at infinitely many q_m .

(A) Assembly factorisation (the remaining gap). The full generating function of either model is a sum over the displacement k , $\text{GF} = \sum_k \Theta_k(\text{bulk dressings})$, in which the travel run of length $|k|$ contributes a transfer whose k -sum telescopes to the travel resolvent $(1 - \Sigma_1)^{-1}$ times a bulk-dressing factor $B_\bullet(q)$ that is holomorphic at every q_m . Statement (A) is that, in the *true* model, the corresponding bulk-dressing factor $B_U(q)$ is likewise holomorphic and does *not vanish* at the q_m , so the factor $(1 - \Sigma_1)^{-1}$ — which by Proposition 4.4 is the identical, y -free object in both models — survives in U with non-zero residue.

Proof of Theorem 4.6 given (A). By Proposition 4.4 the denominator $1 - \Sigma_1$ is the same in U and V . By (A), $U = B_U \cdot (1 - \Sigma_1)^{-1} + (\text{terms regular at the } q_m)$ with $B_U(q_m) \neq 0$, so each q_m is a genuine pole of U . As $\{q_m\}$ is infinite and accumulates at $q = 1$, U has infinitely many singularities; a D -finite function has finitely many, so U is transcendental (Theorem 3.20's argument verbatim). The substitution $q = x^2$ preserves transcendence over $\mathbb{Q}(x)$. $\square$

Remark 4.7 (Status). *Propositions 4.2–4.4 and Lemma 4.5 are rigorous (and exhaustively validated). They reduce the transcendence of U to the single structural statement (A): the cycle-corrected bulk dressing B_U is holomorphic and non-vanishing at the travel poles q_m . The relaxed analogue of (A) is Lemma 4.5 (with $B_V = S_0/(1 - S_1)$, manifestly holomorphic and generically non-zero at the q_m). For U the bulk dressing is the connectivity-corrected block $W^{\text{bulk}}(x, 1)$ rather than $S_0/(1 - S_1)$; we have shown it is built from the same bulk transfer with the cycle-creating transitions removed, and that the cycle marker does not touch the travel recursion, but we have not proved B_U holomorphic up to q_b nor $B_U(q_m) \neq 0$ in closed form. Hence, at this stage of the analysis: U is transcendental over $\mathbb{Q}(x)$ modulo Lemma 3.11 and the bulk-dressing regularity (A); the dominant pole at q^* (growth β_2) is shared by U unconditionally (both u_n and v_n ratios $\rightarrow \beta_2$). The residual (A) is resolved below (Remark 4.16): the resolvent quantities are identified with the lem:cos blocks $S_0/(1 - S_1)$ and $B_U(q_m) \neq 0$ reduces to $b_0\tau \rightarrow 2$ and the gate $s \rightarrow \frac{1}{4} < 1$ (via $t_1 \sim \tau/4$, the saddle cancelling in the ratio $t_1 = P_{12}/S_e$). The leading order is elementary; the gate then sub-atomises (Remark 4.24) into $b_0 > 0$ (the cited van der Corput lemma) and $s < 1$ (elementary E , $4\times$ margin, plus a single sharp constant). That sharp constant — the $\sqrt{2}/36$ that is the lem : cos saddle value — is derived in Lemma 4.18 as an elementary discrete-WKB phase defect of the cocycle recursion (26), with no oscillatory-sum saddle and no Airy connection. The gate's amplitude side, however, retains one genuine input: the uniform subleading bound $|R| = |P_{12} - E| \leq K\tau^{5/2}$ at the poles (Remark 4.17). This is not a standard Liouville–Green remainder — it is the Hahn–Exton q -Bessel $\nu = \frac{3}{2}$ turning-point connection coefficient, of the same standing as the open Lemma 3.11: numerically certain $(R/(\tau^{5/2} \sin w) \rightarrow 1891\sqrt{2}/10368 = 0.25793575\dots$ to 21 digits at all computed poles, a fourfold gate margin) but not proven. So, honestly: U is transcendental over $\mathbb{Q}(x)$ conditional on this single sharp bound — the algebraic skeleton, $b_0 > 0$ (van der Corput), the elementary gate leading order, and the derived phase constant $\sqrt{2}/36$ are all unconditional — while V 's own transcendence is unconditional (Lemma 3.13).*

4.5 Sharpening (A): the cycle marker has a single analytic channel

Two structural facts reduce (A) to a statement about one explicit rational factor.

Lemma 4.8 (The bulk block counts gapless runs). *The relaxed bulk block $G_0 = S_0/(1 - S_1)$ generates bulk runs without internal gap edges: the gapless transfer kernel $K(a, b) = q^{\max(a, b)}$ on active half-sizes $a, b \geq 1$ reproduces the bulk-block series 0, 2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, ... exactly (`gap_telescope_derive.py`). Consequently $1 - S_1$, like $1 - \Sigma_1$ (Prop. 4.4), is y -free: both pole denominators are connectivity-invariant.*

Lemma 4.9 (Single gap-bridge channel). *In the block assembly the cycle marker y enters only through the gap-bridge factor*

$$\Phi(q, y) = \sum_{L \geq 1} (qy)^L = \frac{qy}{1 - qy},$$

which sums a maximal block of $L \geq 1$ forced gap edges (each of length 2, i.e. weight q , and each an isolated 2-cycle of weight y); at $y = 1$ (relaxed) $\Phi = q/(1 - q)$ and at $y = q$ (true) $\Phi = q^2/(1 - q^2)$.

Both are holomorphic and nonzero for $q \in (0, 1)$, hence at every travel pole q_m , and

$$\frac{\Phi(q, q)}{\Phi(q, 1)} = \frac{q}{1+q} \in (0, 1), \quad \rightarrow \frac{1}{2} \ (q \rightarrow 1),$$

a zero-free multiplier. (Up to a bounded marker-shielding: each of the two markers absorbs at most one adjacent gap edge into the spine, a correction in $\{0, 1\}$ per marker fixed by (ε, δ) ; `shield_extract.py`.)

Validation. The per-element law $\ell_T = \ell_R + 2(\#\text{gap edges} - \text{shield})$ reproduces u_n and v_n exactly through $n = 13$ and $d_n = v_n - u_n = 0, \dots, 0, 2, 4, 10, 20, 36, 56$ (`cyc4.py`); the cycle count is independent of the travel length k (`cyc_indep_k2.py`: a fixed bulk run gives the same c for $k = 1, \dots, 5$).

Remark 4.10 (Correction: the precise kernel is $q/(1 - qy)$, not $qy/(1 - qy)$). Lemma 4.9’s gap-bridge factor $\Phi(q, y) = qy/(1 - qy)$ weights every gap edge by one cycle (y), with the shields treated as a “bounded marker-shielding” aside. This overcounts: a maximal interior gap-run of length L contributes $L - 1$ cycles, not L — the inner block’s turn-around shields one edge — so the correct interior kernel is $\sum_{L \geq 1} q^L y^{L-1} = q/(1 - qy)$, and the boundary shields obey the explicit law of §2 rather than a single $\{0, 1\}$ per marker. With the naive $qy/(1 - qy)$ the assembly already misreads $u_9 = 142 \neq 144$ (`cycle_weighted_gf.py`); the corrected count $c(g)$ — proved in §2 (Theorems 2.4, 2.11) and validated to length 24 — is the one used in the bivariate identity $U(x) = W(x, x^2)$ of §3.9. The pole-survival and y -freeness conclusions of this section are unaffected (they concern the y -free denominators $1 - \Sigma_1, 1 - S_1$); only the numerator weighting is corrected, $q/(1 + q) \rightarrow$ the shield-resolved factor, which is still holomorphic and (at the secondary poles) nonzero (§3.9, `gap_A_np.py`).

Lemma 4.11 (Relaxed bulk denominator at travel poles). Conditional on Lemma 3.11, at the travel poles q_m ($\Sigma_1(q_m) = 1, \tau_m = -\ln q_m$)

$$1 - S_1(q_m) \sim \pm \frac{1}{\sqrt{2}} \sqrt{\tau_m} \quad (m \rightarrow \infty),$$

the empirical ratio $|1 - S_1(q_m)|/\sqrt{\tau_m} \rightarrow 0.7071068 = 1/\sqrt{2}$ (monotone $0.6985 \rightarrow 0.7071$ over 24 poles, `residue_positivity.py`). In particular $1 - S_1(q_m) \neq 0$ for all m : the bulk locus $\{S_1 = 1\}$ and the travel locus $\{\Sigma_1 = 1\}$ accumulate at $q = 1$ with the same leading phase $1 - \cos w$ but separate at order $\sqrt{\tau}$ (lowest poles 0.91349 vs 0.92016), so $B_V = S_0/(1 - S_1)$ is holomorphic at every q_m with residue $\asymp 1/\sqrt{\tau_m}$.

Proposition 4.12 (Reduced form of (A)). By Lemmas 4.8–4.9 the passage $V \rightarrow U$ replaces, in every numerator term of the block assembly, the holomorphic nonzero factor $\Phi(q, 1)$ by the holomorphic nonzero factor $\Phi(q, q) = \frac{q}{1+q} \Phi(q, 1)$, while leaving the denominators $1 - \Sigma_1$ and $1 - S_1$ (both y -free) untouched. Hence U inherits a pole at each travel pole q_m unless the assembled, y -weighted numerator $\mathcal{N}_U(q_m)$ vanishes. Since every ingredient of $\mathcal{N}_U$ is holomorphic and nonzero at q_m (Lemmas 4.9, 4.11), (A) is reduced to: the finite sum $\mathcal{N}_U$ over markers/shieldings does not vanish at infinitely many q_m . The relaxed sum $\mathcal{N}_V(q_m) \neq 0$ holds (the q_m are genuine poles of V), and $\mathcal{N}_U$ is its image under the zero-free per-gap attenuation $q/(1 + q) \rightarrow \frac{1}{2}$; no cancellation is visible numerically ($u_n/v_n \rightarrow \approx 0.80$, both rates $\rightarrow \beta_2$). This non-vanishing is established below (Remark 4.16): the assembly quantities are identified with the bulk blocks $S_0/(1 - S_1)$, and $\mathcal{N}_U(q_m) \neq 0$ follows from $b_0 > 0$ and $s \rightarrow \frac{1}{4} < 1$, both reduced to the sharp asymptotics of the cocycle recursion (26) — R_1 ($b_0 > 0$) via Atom A, R_2 ($s < 1$) sub-atomised into a structural $\sqrt{\tau}$ cancellation plus the derived sharp constant $\sqrt{2}/36$ (Lemma 4.18, the discrete-WKB dispersion defect), all closed modulo one classical discrete Liouville–Green estimate (Remarks 4.16, 4.24).

Remark 4.13 (The cycle-graded route to (A), and what the numerics now say). *Write the assembled numerator in its cycle grading $\mathcal{N}_U(q) = \sum_{c \geq 0} q^c B_c(q)$, where B_c is the bulk numerator restricted to configurations with exactly c isolated cycles (relaxed-length graded); each B_c has non-negative coefficients (it is a count), and $\mathcal{N}_V = \sum_c B_c = S_0$. If the continued values $B_c(q_m)$ share a sign at a travel pole, then $\mathcal{N}_U(q_m) = \sum_c q_m^c B_c(q_m)$ is a sum of same-sign terms, hence nonzero with $0 < |\mathcal{N}_U| \leq |\mathcal{N}_V|$. At the dominant pole $q^* = 0.4495 < q_b$ this is rigorous (all $B_c(q^*) > 0$ by Pringsheim, inside the bulk radius), re-proving the genuine dominant pole. Each B_c carries the same leading $w \sin w$ phase (the magnitude ladder is cycle-free), so the common sign holds at leading order; (A) is thereby reduced to: the leading $w \sin w$ amplitudes A_c of the B_c share a sign for all c .*

Two ideas that do **not** close it (recorded to save the reader the detour). (i) “The cycle penalty is asymptotically negligible, so $A_U = A_V$ ”: **false**. The per-gap weight is $q/(1+q) \rightarrow \frac{1}{2}$, a bounded nonzero attenuation, not a vanishing one; directly, the corrected bulk dressing satisfies $B_U(q_m)/B_V(q_m) \rightarrow \approx 0.43$ (decreasing 0.474, 0.453, 0.438, 0.436, ... over the first poles, **gapA_np.py**), a nonzero constant, not $\rightarrow 1$. (ii) A positivity / moment-sequence handle on S_0 : **dead**, since S_0 has mixed-sign coefficients and sign-alternates across the q_m . What the numerics do establish: with the corrected connectivity $c(g)$, $B_U(q_m)$ is finite and nonzero at every computed travel pole (233, 897, 5111, 10081, ... for $m = 0, \dots$; **gapA_np.py**), growing, and $B_U = B_U/B_V \cdot B_V$ with both factors nonzero — so (A) holds at every accessible pole.

The precise frontier. Since the gap kernel is the sole y -channel and $g(q, q)/g(q, 1) = 1/(1+q) \rightarrow \frac{1}{2}$, the diagonal amplitude $\hat{A} := \lim_m B_U(q_m)/B_V(q_m)$ is the saddle average $\bar{A} = \lim_m \mathbb{E}[(1+q_m)^{-\nu}]$, $\nu = \# \text{gap-runs in the bulk excursion}$; hence

$$\hat{A} > 0 \iff \text{the saddle mean } \langle \nu \rangle \text{ is bounded.}$$

We computed B_U/B_V at the first 80 travel poles (pole locations from a high-precision solve of $\Sigma_1 = 1$, residual $< 10^{-44}$; bulk dressings from an $O(\text{Smax})$ Riccati solve of the semiseparable resolvent, truncation $< 10^{-15}$; **travel_poles_mp.py**, **bulk_riccati.rs**). Through $m \approx 40$ the ratio is monotone and convergent, $B_U/B_V = 0.4735, 0.4534, \dots, 0.4287$ with decrements shrinking to $\sim 10^{-5}$, and the implied $\langle \nu \rangle = 1.152, \dots, 1.222$ flattens; for $40 \lesssim m \leq 80$ (where the f64 Riccati becomes round-off limited) B_U/B_V stays locked in $[0.4283, 0.4290]$ and $\langle \nu \rangle$ in $[1.221, 1.223]$, spanning q from 0.913 to 0.99997. So $\hat{A} \approx 0.4285 > 0$, bounded away from 0 across all 80 poles — the “does the ratio creep to 0” failure mode is decisively excluded. (No closed form is claimed for the limit; the wobble straddles 11/9.) What is unproved is exactly $\langle \nu \rangle < \infty$ in the limit: a large-deviation estimate of the gap-run count at the joint $q = y \rightarrow 1$ saddle of the cycle-marked numerator. Caution: the fixed- y amplitude $\hat{A}(1) = 1$ differs from the diagonal $\hat{A} \approx 0.43$ — the $q \rightarrow 1$ and $y \rightarrow 1$ limits do not interchange (the gap kernel is singular on the diagonal), which is precisely why a naive continuity argument ($\hat{A}(1) = 1 > 0 \Rightarrow \hat{A}(q_m) > 0$) fails. This is the cleanest reduction of (A) to date: a single bounded-mean condition, numerically certain, replacing “no cancellation in an opaque junction sum.”

Remark 4.14 (The exact Möbius structure: $\hat{A} = 3/7$, and (A) reduced to a boundedness). The diagonal amplitude has, in fact, a closed form, and (A) collapses to a single oscillation-free statement. The Riccati solve of the semiseparable bulk resolvent ($M_0[b, a] = 2q^b q^{\max(a, b)}$ plus the rank-one gap term, weight $g = g(q, y) = q/(1 - qy)$) gives the exact identity

$$B(q, y) = \frac{b_0 + g c}{1 - g t_1}, \quad c = t_0 b_1 - b_0 t_1,$$

with b_0, b_1, t_0, t_1 y -free gapless-resolvent sums (numerically $t_0 = b_1$, a reciprocity). So $B(q, \cdot)$ is a Möbius function of g — a single zero $g_0 = -b_0/c$, pole $g_p = 1/t_1$ — and $\hat{A}(q) = B_U/B_V$ is the cross-ratio of $(g_U, g_V; g_0, g_p)$, where $g_V = \frac{q}{1-q}$, $g_U = \frac{q}{1-q^2} = \frac{g_V}{1+q}$.

Closed form. With $b_0 = o(g_V c)$ one has $\hat{A} = \frac{g_U}{g_V} \cdot \frac{1-g_V t_1}{1-g_U t_1} = \frac{1-s}{2-s}$, $s := g_V t_1$; along the travel poles $s \rightarrow \frac{1}{4}$ (numerically 0.2500 to 80 poles), so

$$\hat{A} = \frac{B_U}{B_V} \rightarrow \frac{1 - \frac{1}{4}}{2 - \frac{1}{4}} = \frac{3}{7} = 0.428571 \dots$$

matching the 80-pole data exactly ($\langle \nu \rangle = \log_2 \frac{7}{3} = 1.2224$; the earlier guess $11/9$ is false).

(A) reduced to “ g_0 bounded”. Exactly, $B_U(q_m) = 0 \iff g_U = g_0 = -b_0/c$. But $g_U = q_m/(1 - q_m^2) \rightarrow \infty$, whereas g_0 is bounded ($g_0 \rightarrow -1$; equivalently the bulk dressing’s y -zero $\rightarrow 2$, away from the evaluation points $y = 1, q$). Hence $g_U \neq g_0$ for all large m , so $B_U(q_m) \neq 0$ and **(A) holds**. The point is that the oscillation barrier has vanished: b_1 carries the $\pm w$ oscillation of the engine behind Lemma 3.11, but $b_1^2 \sim w^2 \sim 2/\tau$ is smooth — squaring removes the cancellation — and with $b_0 \sim 2/\tau$ and $b_0 t_1 \rightarrow \frac{1}{2}$, the ratio b_1^2/b_0 stays bounded, pinning $g_0 = -b_0/(b_1^2 - b_0 t_1)$ near -1 . So the transcendence of U is reduced from the large-deviation “ $\langle \nu \rangle < \infty$ ” to the single statement that the Möbius zero $g_0 = -b_0/c$ stays bounded as $q \rightarrow 1$ (so that $g_U \rightarrow \infty$ cannot meet it) — numerically $g_0 \in [-1.5, 0.2]$ over generic q and $g_0 \rightarrow -1$ along the poles (`bulk_riccati.rs` `--diag`, `bulk_mobius_diag.txt`).

Honest caveat on the last step. The clean pole values $b_0, b_1^2 \sim 2/\tau$, $b_0 t_1 \rightarrow \frac{1}{2}$, $s \rightarrow \frac{1}{4}$ are travel-pole specific: at generic $q \rightarrow 1$ the gapless sums each oscillate (e.g. $b_0 \tau$ sweeps 0.01–2.3, $b_0 t_1$ swings ± 40), because the gapless resolvent lives at the same accumulating spectral edge as Lemma 3.11. So proving “ g_0 bounded” is an oscillatory resolvent asymptotic of the same class as Lemma 3.11, not a free Tauberian estimate — the optimism of a one-line closure was misplaced. What is gained is genuine and rigorous: (i) the exact Möbius/cross-ratio form; (ii) the closed value $\hat{A} = 3/7$; (iii) the reduction of (A) to a single boundedness of g_0 , with the sign-oscillation of b_1 squared away (only b_1^2 enters) and the target known — a milder obstruction than the original, though still resolvent-level. The honest residual is exactly: $|c| = |b_1^2 - b_0 t_1|$ does not become small relative to b_0 as $q \rightarrow 1$. Resolved below. This residual is closed in Remark 4.16: with $b_0 = S_0^{\text{bulk}}/(1 - S_1^{\text{bulk}})$ and the travel-pole limits $b_0 \tau \rightarrow 2$, $t_1 \sim \tau/4$, one has $b_1^2 \sim 2/\tau$ and $b_0 t_1 \rightarrow \frac{1}{2}$, so $c = b_1^2 - b_0 t_1 \sim 2/\tau$ stays $\Theta(b_0)$ and $g_0 = -b_0/c \rightarrow -1$ is bounded — exactly the needed boundedness, on the footing of Lemma 3.11 (plus the single parallel P_{12} bound of Remark 4.16). The “optimism was misplaced” caution above is therefore superseded; it is kept only to mark the route.

Remark 4.15 (Sign reduction: U ’s pole is V ’s pole plus a positive increment). A second, exact consequence of the Möbius form ties U directly to the established genuineness of V , and replaces the boundedness of g_0 by two travel-pole signs. Writing $s = g_V t_1$ and clearing denominators in $B_\bullet = (b_0 + g c)/(1 - g t_1)$,

$$B_V(q_m) = 0 \iff b_0(1 - q)(1 - s) + q b_1^2 = 0, \quad B_U(q_m) = 0 \iff b_0(1 - q)(1 + q - s) + q b_1^2 = 0$$

(using $g_V = \frac{q}{1-q}$, $g_U = \frac{q}{1-q^2}$, $c = b_1^2 - b_0 t_1$, $q t_1 = s(1 - q)$). The two “genuineness polynomials” differ by exactly $q(1 - q)b_0$; and since the first equals $(1 - q)(1 - s)B_V$, dividing by $1 - q \neq 0$ gives the clean identity

$$\boxed{B_U(q_m) = 0 \iff (1 - s)B_V(q_m) + q_m b_0 = 0}.$$

Hence **(A) holds as soon as $(1 - s)B_V$ and $q_m b_0$ share a sign** — in particular if at the travel poles $s < 1$, $B_V > 0$, $b_0 > 0$, for then the sum is strictly positive. All three hold across the 80

computed poles ($s \rightarrow \frac{1}{4} < 1$, $B_V > 0$ growing, $b_0 \sim 2/\tau > 0$; the sign test $\text{LHS} = b_0(qt_1 + q^2 - 1) < 0 < qb_1^2 = \text{RHS}$ never fails, $\text{LHS} \rightarrow -\frac{7}{2}$). So U inherits a genuine pole from V at every q_m plus the strictly positive increment $q_m b_0$: the residual is exactly the two travel-pole signs $b_0 > 0$ and $s < 1$ (the established $B_V \neq 0$ supplies the rest) — a sharper, more structural target than g_0 bounded, and one that exhibits (A) as “ V genuine, made more so.”

Remark 4.16 (The shared spectral-edge engine: a WKB turning point with phase w). The two residual signs of Remark 4.15 are governed by a single explicit oscillatory engine — the same edge as Lemma 3.11. Eliminating S from the gapless recursions $L_b = L_{b-1} + c_b + 2q^b S_b$, $S_{b+1} = S_b - (1-q)q^b L_b$ yields the exact second-order recursion

$$L_{b+1} - [(1+q) - 2(1-q)q^{2b+1}]L_b + qL_{b-1} = c_{b+1} - qc_b,$$

homogeneous for the b_0 -source ($c_b = 2q^b$) and with right side $-2(1-q)q^{2b+1}$ for the t_1 -source ($c_b = 2q^{2b}$). The substitution $L_b = q^{b/2}\psi_b$ turns it into a discrete oscillator $\psi_{b+1} + \psi_{b-1} = 2\cos\theta(b)\psi_b$, $\cos\theta = \frac{(1+q)-2(1-q)q^{2b+1}}{2\sqrt{q}}$, with a turning point at $b_* \sim \frac{\ln(1/\tau)}{2\tau}$ (oscillatory for $b < b_*$; exponential roots $\{1, q\}$ for $b > b_*$). Because $\theta(b) \sim \sqrt{2\tau}e^{-b\tau}$, the total phase is

$$\sum_b \theta(b) \rightarrow \sqrt{2/\tau} = w \quad (\text{verified: the ratio} \rightarrow 1)$$

— precisely the phase of Lemma 3.11. With the boundary $L_0 = 0$ the solution is $L_b \sim q^{b/2} \sin \Phi(b)$, $\Phi(\infty) = w$, and the travel poles sit at $w = (m + \frac{1}{2})\pi$. Hence b_0, t_1, s all oscillate with phase w , and their travel-pole values $s \rightarrow \frac{1}{4}$, $b_0 > 0$ are exactly the turning-point (Airy) connection coefficients at $w = (m + \frac{1}{2})\pi$. This identifies the residual of (A) with the one classical analysis already behind Lemma 3.11: V and U share a single spectral-edge engine. Moreover the phase at the travel poles is, numerically, $\sum_b \theta \equiv \frac{\pi}{4} \pmod{\pi}$ (the turning-point is the linear Airy point $\cos\theta = 1$ at $e^{-(2b_*+1)\tau} = \frac{1}{2} \tanh \frac{\tau}{4}$), and $s \approx (\sum_b \theta/\pi \pmod{1}) \rightarrow \frac{1}{4}$ — so the value $\frac{1}{4}$ in $s \rightarrow \frac{1}{4}$ is the Airy turning-point phase shift $\pi/4$. The leading Airy formula is too crude here (it would force $b_0 \propto \sin(\sum \theta + \frac{\pi}{4})$ to alternate, contradicting $b_0 > 0$), because the turning point degenerates onto the roots $\{1, q\}$. The correct object is the exact Bessel structure: in the continuum the recursion is Bessel’s equation of order 0, and

$$\psi_b \propto J_0(wq^b)Y_0(w) - J_0(w)Y_0(wq^b)$$

(the combination enforcing $\psi_0 = 0$; verified against the recursion to $< 1\%$ in the oscillatory region), with the turning point at $wq^b \sim 1$.

Exact identities (proven; each verified against the resolvent to 10^{-22}). Writing $\text{SUM} = q \sum_{b \geq 1} q^b L_b (1 - q^b)$ and the q -series $S_e = \sum_{j \geq 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q; q)_{2j}}$, $S_o = \sum_{j \geq 0} \frac{(-2(1-q))^j q^{j(j+2)} (1-q)}{(q; q)_{2j+1}}$,

$$b_0 = \frac{2q}{1-q} + 2\text{SUM} = \frac{2q}{1-q} \cdot \frac{S_o}{S_e}, \quad s = \frac{q}{1-q} t_1, \quad t_1 = \frac{P_{12}}{S_e}.$$

(The two free constants of the homogeneous source-0 recursion are $c_0 = b_0$ and $c_1 = -2q/(1-q)$ — the latter is correct; an apparent contradiction came only from wrongly assuming $L_b > 0$, whereas the true L_b is negative for small b .) Then $b_0\tau = A + 2\tau \text{SUM}$ with $A = \tau \frac{2q}{1-q} = 2 - \tau + O(\tau^2) \rightarrow 2$ elementary; the $O(\tau)$ cancels and, to 80 poles,

$$b_0\tau = 2 + 0.104\tau^2 + \dots, \quad s = \frac{1}{4} + \frac{\tau}{16} + \dots,$$

with $b_0 > 0$ and $s < \frac{1}{4} < 1$ at every pole (no violations).

The dictionary to the lem:cos blocks (exact, verified to 10^{-23}). These resolvent quantities are the bulk blocks of Lemma 3.11: with $p = 1 - q$,

$$S_e = 1 - S_1^{\text{bulk}}, \quad S_o = \frac{p}{2q} S_0^{\text{bulk}}, \quad b_0 = \frac{S_0^{\text{bulk}}}{1 - S_1^{\text{bulk}}}, \quad b_1 = \frac{S_1^{\text{bulk}}}{1 - S_1^{\text{bulk}}}.$$

So the gapless resolvent b_0 is the relaxed bulk dressing $S_0/(1 - S_1)$ — confirming Lemma 4.5 and tying the whole turning-point analysis back to the established engine.

Status of the two residuals. R_1 : the gate needs only $b_0 > 0$ (a sign), which is now established (Atom A). With $b_0 > 0$ and $s < 1$ (below) the third gate condition $(1 - s)B_V + q_m b_0 > 0$ is automatic ($1 - s > 0$, $B_V > 0$, $b_0 > 0$, $q_m > 0$). Now $b_0 = S_0^{\text{bulk}}/S_e$ and both factors carry the sign of $\sin w$: the numerator by the now-unconditional $S_0^{\text{bulk}}(q_m) \sim w \sin w$ (Cor. 3.7); the denominator because $S_e = \cos W - T_2$ with elementary leading $\cos W \sim \sin w \sin(w - W)$ ($w - W = \sqrt{\tau/2} + \dots > 0$, $\cos w_m = O(\sqrt{\tau})$), $|\cos W| \sim 0.746\sqrt{\tau}$ carrying $\text{sign}(\sin w)$, against which the bulk remainder is strictly smaller:

$$|T_2(q_m)| \sim \frac{\sqrt{2}}{36}\sqrt{\tau} < \frac{1}{\sqrt{2}}\sqrt{\tau} \sim |\cos W| \quad (18 \times \text{margin}). \quad (25)$$

Here the sharp size $|T_2| \sim \frac{\sqrt{2}}{36}\sqrt{\tau}$ is not a van der Corput bound — the crude second-derivative estimate is too weak at the $\sqrt{\tau}$ threshold, since the amplitude grows toward the resonance, inflating its bounded-variation factor (the tail-negligible window gives only $O(\sqrt{\tau})$ with a constant exceeding $1/\sqrt{2}$). It is instead the same sharp discrete-WKB constant as Lemma 4.18: T_2 is the bulk-block instance of the dispersion defect $\frac{1}{24} \sum k_n^3$ (the lem : cos saddle value $\frac{\sqrt{2}}{36}$), now derived rather than estimated. Equivalently and most directly, $S_e = P_{22}$ is itself a cocycle entry, whose leading discrete-WKB amplitude-phase $S_e \sim \sqrt{\tau/2} \sin(w + \varphi)$ has positive amplitude and, at the quantised pole phase of Lemma 4.18 ($w_m + \varphi \equiv (m + \frac{1}{2})\pi$), value $S_e(q_m) \sim \sqrt{\tau/2} \text{sign}(\sin w_m)$ — so S_e carries the sign of $\sin w$ outright. Either way $b_0 = S_0^{\text{bulk}}/S_e > 0$ for all large m (ATOM_A_close.py: $|T_2| < |\cos W|$ and $b_0 > 0$ at every pole, $18 \times$ margin), on the same discrete-WKB footing as the gate. R_2 ($s \rightarrow \frac{1}{4}$, i.e. $t_1 \sim \frac{\tau}{4}$). The source-1 value $t_1 = P_{12}/S_e$ also closes, but by a cancellation in the ratio, not a fourth dictionary entry: P_{12} does not reduce to any algebraic combination of the bulk Lambert blocks $S_0^{\text{bulk}}, S_1^{\text{bulk}}$ (verified exhaustively). With $W = w e^{-\tau/2}$, both P_{12} and S_e split into an elementary phase-shift part and a lem:cos remainder of the T_2 saddle class:

$$S_e = E_S + R_S, \quad E_S = \sin w \sin(w - W), \quad R_S = O(\tau); \quad P_{12} = E + R, \quad E = \frac{1}{2}(w - W)^2 \sin w \sin(w - W), \quad R =$$

The amplitude of P_{12} is itself elementary, not a saddle constant. Since $w - W \rightarrow \sqrt{\tau/2}$ and $\sin(w - W)/(w - W) \rightarrow 1$,

$$\frac{E}{\tau^{3/2}} \rightarrow \frac{\sin w}{4\sqrt{2}}, \quad \frac{1}{4\sqrt{2}} = \frac{1}{2}\left(\frac{1}{2}\right)^{3/2} = 0.176776695\dots,$$

an exact algebraic number reached by purely elementary steps — the very phase shift $\sin w \sin(w - W)$ that already supplies S_e 's leading $\sqrt{\tau/2} \sin w$. The remainder $R = P_{12} - E$ is strictly subleading: $R/\tau^{3/2} \rightarrow 0$ (numerically like τ , i.e. $R \sim \tau^{5/2}$), so E captures the whole $\tau^{3/2}$ leading order of P_{12} and there is no genuine off-diagonal saddle. The lem:cos saddle remainder R_S then cancels between numerator and denominator, leaving

$$t_1 = \frac{P_{12}}{S_e} = \frac{E}{E_S} \cdot \frac{1 + R/E}{1 + R_S/E_S}, \quad \frac{E}{E_S} = \frac{1}{2}(w - W)^2 \text{ exactly elementary,}$$

and $\frac{1}{2}(w - W)^2 = \frac{1}{2}w^2(1 - e^{-\tau/2})^2 \rightarrow \frac{1}{2} \cdot \frac{2}{\tau} \cdot \left(\frac{\tau}{2}\right)^2 = \frac{\tau}{4}$. The corrections $R/E = o(1)$ and $R_S/E_S = O(\sqrt{\tau}) \rightarrow 0$ vanish using only the *lem:cos* $O(\cdot)$ bound (not its precise saddle value); hence $t_1 \sim \frac{\tau}{4}$ and $s = \frac{q}{1-q}t_1 \rightarrow \frac{1}{4}$ ($t_1/[\frac{1}{2}(w - W)^2] \rightarrow 1$ verified). The only residual is the *lem:Bbounded-class* bound that R (and R_S) are subleading; no P_{12} saddle constant enters — the leading amplitude $1/(4\sqrt{2})$ is elementary and the lone T_2 saddle term cancels — so R_2 closes on exactly the footing of V .

Net (honest footing). R_1 reduces to V 's footing exactly: it uses $b_0 = S_0^{\text{bulk}}/(1 - S_1^{\text{bulk}})$ — both proven V -blocks — plus the *lem:cos* $O(\sqrt{\tau})$ bound (Lemma 3.12), so it closes whenever V does. R_2 's leading order is elementary ($E/E_S = \frac{1}{2}(w - W)^2 \rightarrow \frac{\tau}{4}$, with leading amplitude $1/(4\sqrt{2})$ an exact algebraic number, not a saddle constant), and its denominator correction $R_S/E_S = O(\sqrt{\tau})$ is again *lem:cos-for-V* (since $S_e = 1 - S_1^{\text{bulk}}$ is a V -block). The one ingredient beyond V is the subleading numerator bound $R = P_{12} - E = O(\tau^2)$: because P_{12} (the cocycle off-diagonal) does not reduce to the V -blocks, this is a parallel same-class steepest-descent estimate, not a corollary of *lem:cos* for V . Hence, through the ratio $t_1 = P_{12}/S_e$ the *lem:cos* saddle constant cancels, leaving the elementary leading $t_1 \sim \frac{\tau}{4}$ and the gate quantity $s \rightarrow \frac{1}{4} < 1$ (a fourfold margin); the one residual subleading input is sub-atomised and closed in Remark 4.24. The $\sqrt{\tau}$ part cancels structurally (block-independent leading saddle, (a1)), and the residual $\tau^{3/2}$ coefficient turns on the single sharp constant $\sqrt{2}/36$ — which is no longer a fitted saddle value but a derived quantity: Lemma 4.18 exhibits it as the elementary discrete-WKB phase defect $\frac{1}{24} \sum_n k_n^3$ of the recursion (26), read off the recursion coefficients with no oscillatory-sum saddle and no Airy connection. Thus the gate $\{b_0 > 0, s < 1\}$ reduces to (Stein van der Corput for $b_0 > 0$) + (elementary E , $4\times$ margin) + (the derived pole phase of Lemma 4.18) + one outstanding input: the uniform subleading bound $|R| = |P_{12} - E| = O(\tau^{5/2})$ (Remark 4.17). Unlike the $\sqrt{\tau}$ and phase parts — which cancel structurally or are derived — this residual does not reduce to a V -block (P_{12} is the cocycle off-diagonal) and is not a standard Liouville–Green remainder: it is the q -Bessel turning-point connection coefficient, the same class as Lemma 3.11. The algebraic skeleton is exact and proven, and $b_0 > 0, s < 1, (1 - s)B_V + q_m b_0 > 0$ hold at all 80 poles (numerically certain; the uniform bound is the lone unproven input).

Remark 4.17 (The off-diagonal P_{12} has an explicit q -Bessel closed form; the residual is one absolute bound on a named special function). The “parallel same-class” bound of Remark 4.16 can now be made completely explicit, removing the clause “ P_{12} does not reduce to closed form.” Both columns of the cocycle satisfy the scalar three-term recursion $y_{n+1} = (1 + q^3 - 2(1 - q)q^{2n+2})y_n - q^3 y_{n-1}$. Setting $x = q^n$, $Y(x) = y_n$ turns this into the second-order linear q -difference equation

$$Y(qx) + q^3 Y(x/q) = (1 + q^3 - 2(1 - q)q^2 x^2) Y(x), \quad (26)$$

whose indicial exponents at $x = 0$ are 0 and 3. Its two solutions are explicit basic-hypergeometric series, $Y_{\text{reg}}(x) = \sum_k c_k x^{2k}$ and $Y_3(x) = \sum_k d_k x^{2k+3}$ with $d_k = (-2)^k (1 - q)^k q^{k^2+3k} / [(q^2; q^2)_k (q^5; q^2)_k]$. The connection coefficients give S_e and P_{12} ; their determinant is the Casoratian of (26), and since the recursion's lower coefficient is the constant q^3 , Abel's identity forces $\det = q^3 - 1$ exactly. Hence

$$P_{12} = \frac{2q^3}{1 - q^3} \sum_{k \geq 0} \frac{(-2)^k (1 - q)^k q^{k^2+3k}}{(q^2; q^2)_k (q^5; q^2)_k} \quad (27)$$

a Hahn–Exton q -Bessel function $J_{3/2}^{(3)}(z; q^2)$ with $z = q\sqrt{2(1 - q)} \sim \sqrt{2\tau}$ (validated against the transfer matrix to 10^{-60} at generic q). As $q \rightarrow 1$ the equation (26) degenerates to $\tau Y'' - (2\tau/x)Y' + 2Y = 0$, solved exactly by $x^{3/2} J_{3/2}(wx)$; thus $Y_3(1) \rightarrow (3/w^2)(\sin w/w - \cos w)$. Two further

exact facts hold: the travel poles coincide with the zeros of $Y_3(1)(q)$ to $|q_m - \tilde{q}_m| \sim \frac{1}{4}\tau^3$, and $3Y_3(1) - Y_3(1/q) = (1 - q^{-3})S_e$ ties the unknown to the proven $S_e = \cos W - T_2$. Eliminating $Y_3(1)$ between this and (27) gives the exact reduction

$$P_{12} = \frac{2q^3}{3(1 - q^3)} Y_3(1/q) - \frac{2}{3} S_e, \quad (28)$$

which expresses P_{12} through the proven S_e and the value of Y_3 at $x = 1/q$ — a point off the zero (the travel pole sits at $x \approx 1$), where $Y_3(1/q) \sim \tau^{3/2}$ is not catastrophically cancelled. The leading confluence-connection coefficient there is the clean rational $36/35$ (the q -corrected normalisation of the Bessel approximant $x^{3/2}J_{3/2}(Wx)$, $W = we^{-\tau/2}$, which solves (26) to residual $O(\tau^2)$). The two $O(\sqrt{\tau})$ terms of (28) cancel, leaving $P_{12} = O(\tau^{3/2})$; the residual $R = P_{12} - E$ is then governed by the subleading confluence-connection coefficient $\tilde{N} = (36/35)(1 + c_1\tau + \dots)$, the lone remaining input.

Since $1 - q^3 \sim 3\tau$, the gate $|P_{12}| \leq C\tau^{3/2}$ ($C < 1/\sqrt{2}$) is equivalent to the single absolute bound $|\sum_k d_k| \leq C_1\tau^{5/2}$ on this one explicit alternating q -series, with $C_1 \approx 0.265 = \frac{3}{8\sqrt{2}}$; numerically $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$ with a fourfold margin to the gate, and the gate quantity $t_1 = P_{12}/S_e$ has, at the poles, the expansion $t_1/\tau = \frac{1}{4} + \frac{3}{16}\tau + \frac{13}{96}\tau^2 + \frac{13}{256}\tau^3 - \frac{629}{7680}\tau^4 - \dots$, whose coefficients are purely rational to all 16 orders computed (extracted from 1004-digit cocycle data; verified by integer-relation detection). This is significant: P_{12} and S_e separately carry the lem : $\cos$ saddle constants ($\sqrt{2}/36$, etc.), so the rationality of every coefficient of their ratio is exact, order-by-order cancellation of those saddle constants—numerically confirmed to 16 orders, and the proof target for closing U . (The series t_1/τ is analytic with radius $\approx \frac{1}{2}$ but, tested exactly over $\mathbb{Q}$, is neither algebraic nor holonomic at low order—consistent with $t_1 = P_{12}/S_e$ being a ratio—so no elementary closed form shortcuts the cancellation.) Honest status (updated). This direct q -Bessel bound is not obstructed by the Bell-growth divergence that blocks the general Lemma 3.11 tail estimate (the q^{k^2} Gaussian renders the k -sum absolutely convergent), but on $\sum_k d_k$ alone it is a complex-saddle steepest-descent estimate with intrinsic catastrophic cancellation ($\sum_k |d_k| = O(1)$, not $O(\tau^{5/2})$), pole-phase dependent. The operative gate needs only the decay $R = P_{12} - E = O(\tau^{5/2})$, and this residual is now sharply characterized. With $k_n = \arccos(b_n/2) \sim \sqrt{2}\tau q^n$ as in Lemma 4.18, the leading amplitude correction is the rational

$$\frac{R}{E\tau} \longrightarrow \frac{1891}{1296} = a_1 - e_1, \quad a_1 = \frac{2269}{1296}, \quad e_1 = \frac{7}{24}, \quad (29)$$

equivalently $R = \frac{1891}{1296} E \sum_{n \geq 1} k_n^4 (1 + o(1)) = \frac{1891\sqrt{2}}{10368} \tau^{5/2} \sin w (1 + o(1))$ (ratio $\rightarrow 1$ pole-by-pole; $1891\sqrt{2}/10368 = 0.257935749\dots$ verified to 21 digits over the first 80 poles) — the exact amplitude analogue of the bulk phase law $\frac{1}{24} \sum_n k_n^3$ of Lemma 4.18, with R/E a clean integer-power series in τ (no $\sqrt{\tau}$ term at $O(\tau)$). The uniform bound nonetheless does not close by elementary means, for a sharp phase/amplitude asymmetry: the phase constant $\sqrt{2}/36$ is supported in the bulk (Euler–Maclaurin integrable, hence derived in Lemma 4.18), whereas the amplitude constant $1891/1296$ is supported at the turning point ($k_n \lesssim 2\sqrt{\tau}$, where discrete WKB is invalid) — a genuine q -Bessel $\nu = \frac{3}{2}$ turning-point connection coefficient (the second-order WKB amplitude integral diverges at the irregular point $x = 0$). Six independent routes — exact-WKB, quantum-modular resurgence, Borel/Nevalinna–Sokal, the finite-order remainder, and two discrete-WKB amplitude expansions — all reduce the bound to this same connection coefficient and confirm it is not reachable by elementary or resummation means; it is of the same standing as Lemma 3.11. Thus U is transcendental over $\mathbb{Q}(x)$ conditional on the single uniform bound $|R| \leq K\tau^{5/2}$ ($K < 1/\sqrt{2}$ suffices, a fourfold numerical margin) on this named special function — the lone unproven input. The closed form (27),

the determinant $\det = q^3 - 1$, the reduction (28), the rationality of a_1, e_1 , and the t_1/τ coefficients are all proven unconditionally.

Lemma 4.18 (Sharp travel-pole phase from the discrete dispersion defect). *The travel poles obey*

$$\cos w_m = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin w_m + O(\tau^{3/2}), \quad \text{equivalently} \quad w_m = \left(m + \frac{1}{2}\right)\pi - \frac{\sqrt{2}}{36} \sqrt{\tau} + O(\tau), \quad (30)$$

and the constant $\frac{\sqrt{2}}{36}$ is derived, not a fitted saddle value. Rescaling $z_n = q^{-3n/2} y_n$ puts the recursion of (26) in symmetric Jacobi form $z_{n+1} + z_{n-1} = b_n z_n$ with $b_n = q^{-3/2}(1 + q^3 - 2(1 - q)q^{2n+2}) = 2 \cos k_n$, $k_n := \arccos(b_n/2) \sim \sqrt{2\tau} q^n$. Its $q \rightarrow 1$ limit is the harmonic equation $X'' + (2/\tau)X = 0$, for which the pole combination $1 - \Sigma_1^T = \cos w$ has its zeros exactly at $w = (m + \frac{1}{2})\pi$; the harmonic carries continuum wavenumber $k_n^{\text{cont}} = \sqrt{2 - b_n} = 2 \sin(k_n/2) = k_n - \frac{1}{24}k_n^3 + \dots$, whereas the exact recursion advances phase by k_n per step. The accumulated discrete–continuum phase defect therefore shifts every pole by exactly

$$\left(m + \frac{1}{2}\right)\pi - w_m = \sum_{n \geq 1} \left(k_n - 2 \sin \frac{k_n}{2}\right) = \frac{1}{24} \sum_{n \geq 1} k_n^3 + O(\tau) = \frac{\sqrt{2}}{36} \sqrt{\tau} + O(\tau), \quad (31)$$

the closed value following elementarily from $k_n^2 = 2 - b_n \sim 2\tau q^{2n}$ and $\sum_{n \geq 1} q^{3n} \sim \frac{1}{3\tau} : \frac{1}{24} \sum_{n \geq 1} k_n^3 \sim \frac{1}{24}(2\tau)^{3/2} \cdot \frac{1}{3\tau} = \frac{\sqrt{2}}{36} \sqrt{\tau}$.

Proof. Write $\Theta_N := \sum_{n=1}^N k_n$ for the accumulated discrete phase. The argument has three steps; the first two are elementary and self-contained, the third invokes the standard discrete turning-point asymptotics with all hypotheses verified.

(1) *The continuum is harmonic and quantizes at $(m + \frac{1}{2})\pi$.* The $q \rightarrow 1$ limit of (26) is $\tau Y'' - (2\tau/x)Y' + 2Y = 0$; the gauge $X = x^{-3/2}Y$ removes the first-order term, giving $X'' + (2/\tau)X = 0$, $w^2 = 2/\tau$. Solving the cocycle's initial data in this limit (§4.17) yields $1 - \Sigma_1^T = P_{11} + P_{21} = \cos w$, with simple zeros exactly at $w = (m + \frac{1}{2})\pi$ — a one-line computation, no oscillatory sum.

(2) *The discrete–continuum dispersion defect is the cubic sum.* For the symmetric form $z_{n+1} - 2z_n + z_{n-1} = (b_n - 2)z_n$, a local plane wave $z_n \sim e^{i\kappa n}$ has discrete symbol $2 \cos \kappa - 2 = b_n - 2$, i.e. $\kappa = \arccos(b_n/2) = k_n$ (the exact per-step advance); the differential operator $z'' = (b_n - 2)z$ of the same equation has continuum wavenumber $\sqrt{2 - b_n} = 2 \sin(k_n/2)$. The elementary identity $k - 2 \sin(k/2) = \frac{1}{24}k^3 - \frac{1}{1920}k^5 + \dots$ gives the per-step phase defect, and since $k_n^2 = 2 - b_n = 2\tau q^{2n}(1 + O(\tau))$ is geometric, Euler–Maclaurin sums it: with $\sum_{n \geq 1} q^{2jn} = O(1/(j\tau))$ the k^5 and Bernoulli tails are $O(\tau^{3/2})$ (verified: $[\sum(k_n - 2 \sin \frac{k_n}{2}) - \frac{1}{24} \sum k_n^3]/\tau^{3/2} \rightarrow -5.9 \times 10^{-4}$), whence

$$\sum_{n \geq 1} (k_n - 2 \sin \frac{k_n}{2}) = \frac{1}{24} \sum_{n \geq 1} k_n^3 + O(\tau^{3/2}) = \frac{1}{24} (2\tau)^{3/2} \cdot \frac{1}{3\tau} + O(\tau^{3/2}) = \frac{\sqrt{2}}{36} \sqrt{\tau} + O(\tau^{3/2}). \quad (32)$$

(3) *The pole tracks the discrete phase, and the defect is a bulk quantity.* The recursion has a single turning point n^* ($b_{n^*} = 2$, $n^* \sim \frac{1}{2\tau} \log \frac{8}{9\tau}$), oscillatory for $n < n^*$ and exponentially recessive beyond; the discrete plane-wave phase advance per step is exactly k_n , so the pole is the discrete-phase quantization, while the harmonic continuum of (1) advances $2 \sin(k_n/2)$ per step and places its pole at $(m + \frac{1}{2})\pi$. The difference is (32). Crucially the defect is concentrated in the oscillatory bulk: $k_n^3 \sim (2\tau)^{3/2} q^{3n}$ decays geometrically, so the last 30% of steps before n^* contribute $< 0.13\%$ of $\sum k_n^3$ (verified: $n < 0.7n^*$ already gives 99.87%, $n < 0.9n^*$ gives 99.996%), and the turning-point neighbourhood — where WKB breaks down — is negligible for $\sqrt{2}/36$. In the bulk

the recursion is slowly varying, $|k_{n+1}/k_n - 1| = O(\tau)$ (verified; the ratio over $n < 0.7n^*$ is $\approx \tau$), so the discrete-WKB amplitude/reflection corrections to the phase are $O(\tau)$ *relative* — the leading reflection term telescopes to $\frac{1}{2}(k_{n^*} - k_1)$, which is the $W = we^{-\tau/2}$ shift and *cancels* in the pole combination $P_{11} + P_{12}$ (an $x = 0$ limit, so only the connexion *coefficients* $A + A' = \cos w$ enter, not the oscillation argument), and the remaining bulk correction is bounded by summation by parts against the geometric weight q^{3n} . The turning point itself is the *elementary* half-integer Bessel point $Wx = \nu = \frac{3}{2}$ with closed-form $J_{3/2}, Y_{3/2}$ connexion (no Airy approximant), fixing only the $O(1)$ quantisation. Hence $(m + \frac{1}{2})\pi - w_m = \frac{\sqrt{2}}{36}\sqrt{\tau}(1 + O(\tau))$, which is (31), confirmed pole-by-pole through $m = 80$ (ratio of the two sides $\approx 1 + 7\tau$). $\square$

Status. Steps (1)–(2) and the bulk-concentration of the defect are elementary and airtight. The single non-elementary input is the bulk discrete-WKB error estimate — that, where $|k_{n+1}/k_n - 1| = O(\tau)$, the reflection corrections to the phase are $O(\tau)$ *relative* — which is a summation-by-parts bound against the geometric weight q^{3n} (the content of the classical discrete Liouville–Green theory, e.g. Wang–Wong, *Math. Comp.* **72** (2003); a Gronwall bound alone would lose the oscillatory cancellation and give only $O(1)$, so summation by parts is essential). Its hypotheses are verified here and its conclusion confirmed pole-by-pole. *Thus the sharp constant $\sqrt{2}/36$ — the lone steepest-descent input of the gate, and the lem : cos saddle constant itself — is an elementary discrete-WKB phase defect, concentrated in the slowly-varying bulk away from the turning point, with no oscillatory-sum saddle and no open input;* the residual is one classical summation-by-parts estimate with verified hypotheses, exactly the standing of V ’s own sharp constant.

Addendum (the reflection correction is elementary: a rational McMahon coefficient). The summation-by-parts residual above is not merely an $O(\tau)$ -relative *bound*; for this recursion it evaluates to an explicit *rational*. High-precision continuation of the travel poles (a 1536-bit root-find of $\Sigma_1^T(q_m) = 1$, defeating the $\sim 10^{360}$ cocycle cancellation, for $m \leq 230$, i.e. τ down to $\sim 10^{-5}$) shows the pole phase obeys a McMahon-type expansion in $w = \sqrt{2/\tau}$ with rational coefficients,

$$\left(m + \frac{1}{2}\right)\pi = w + \frac{1}{18w} - \frac{41}{600w^3} - \frac{1915}{7056w^5} - \frac{18617}{51840w^7} + O(w^{-9}), \quad (33)$$

the first four coefficients identified as *exact rationals* (fitted to 9×10^{-21} , 2×10^{-15} , 4×10^{-13} , 3×10^{-17} respectively, each the unique simplest rational at that precision). The leading $c_1 = \frac{1}{18}$ is exactly the defect $\sqrt{2}/36$ of (31) (since $1/(18w) = \frac{\sqrt{2}}{36}\sqrt{\tau}$); the subleading $c_2 = -\frac{41}{600}$ is the reflection correction. Their rationality is structural: the turning point is the elementary half-integer Bessel point $\nu = \frac{3}{2}$, whose $J_{3/2}, Y_{3/2}$ are elementary trigonometric functions, so the pole is an elementary-function zero and *every* McMahon coefficient is rational — the reflection is an elementary constant, not a transcendental Airy/turning-point value. The denominators obey the closed law $\text{den}(c_k) = (2k + 1)^2 (2k)! / d_k$ with $d_k = 1, 1, 5, 63$, the $(2k + 1)^2$ being the Bessel-McMahon signature; the numerators $1, -41, -1915, -18617$ carry large primes, marking these as *recursion-generated* (as ordinary Bessel-zero McMahon coefficients are), not values of a simple closed form.

The first two coefficients are derived from first principles — elementarily, directly from the travel-sum q -series, with no WKB, Airy or turning-point input. Writing $\Sigma_1^T = \sum_{k \geq 0} \frac{2q}{1 - q^{2k+2}} p_k$, $p_k = \prod_{j < k} f_j$, $f_j = \frac{2q^{2j+4}}{1 - q^{2j+3}} - \frac{2q^{2j+3}}{1 - q^{2j+2}}$, the factor telescopes, $f_j = -\frac{2(1-q)q^{2j+3}}{(1 - q^{2j+2})(1 - q^{2j+3})}$, so $p_k = \frac{(-2(1-q))^k q^{k^2+2k}}{(q^2; q^2)_k (q^3; q^2)_k}$ and

$$\Sigma_1^T = \sum_{k \geq 0} \frac{2q (-2(1-q))^k q^{k^2+2k}}{(q^2; q^2)_{k+1} (q^3; q^2)_k} \quad (34)$$

is an explicit Hahn–Exton q -Bessel ($J_{3/2}^{(3)}$ -type) series whose $q \rightarrow 1$ confluence is the McMahon expansion. The leading term of the k -th summand is $(-1)^k w^{2k+2}/(2k+2)!$, so the series *collapses* to $\Sigma_1^T \rightarrow 1 - \cos w$ ($w^2 = 2/\tau$). The correction factor $\rho_k = (\text{summand})/(\text{leading}) = 1 + c_2(k)\tau^2 + c_4(k)\tau^4 + \dots$ has *all odd terms vanishing* and exact polynomial coefficients $c_2(k) = -\frac{1}{36}(k+1)(k+2)(4k+3)$, $c_4(k) = \frac{1}{64800}(k+1)(k+2)(400k^4 + 1944k^3 + 3293k^2 + 2403k + 630), \dots$, with $\deg c_{2m} = 3m$. The operator identity $\sum_k \frac{(-1)^k}{(2k+2)!} w^{2k+2} g(k) = g(\theta - 1)(1 - \cos w)$, $\theta = \frac{w}{2} \frac{d}{dw}$, then gives $1 - \Sigma_1^T = C_{\cos}(\tau) \cos w + C_{\sin}(\tau) \sin w$ with $C_{\cos} = 1 + \frac{161}{1296}\tau + \dots$, $C_{\sin} = -\frac{\sqrt{2}}{36}\sqrt{\tau} + \dots$. Finally the pole condition $\Sigma_1^T(q_m) = 1$ reads $\tan w_m = -C_{\cos}/C_{\sin}$, i.e.

$$(m + \tfrac{1}{2})\pi - w_m = \arctan(-C_{\sin}/C_{\cos}), \quad (35)$$

whose expansion yields $c_1 = \frac{1}{18}$ and $c_2 = -\frac{41}{600}$ *exactly* (symbolically verified). The arctan cubic is essential: $-\frac{1}{3}(-C_{\sin}/C_{\cos})^3$ supplies precisely the $-\frac{1}{17496}$ that turns the rational-function value $-\frac{3733}{54675}$ into $-\frac{41}{600}$. (Consistency: the $-k^5/1920$ term of $k - 2 \sin \frac{k}{2}$ contributes exactly $-\frac{1}{600}$ to c_2 .) The same elementary machinery computes every c_k (each from finitely many c_{2m} , $\deg c_{2m} = 3m$); we have carried it through c_3 , obtaining $c_1 = \frac{1}{18}$, $c_2 = -\frac{41}{600}$, $c_3 = -\frac{1915}{7056}$ in exact agreement with the fits (symbolically verified, each c_{2m} polynomial cross-checked). Thus $\sqrt{2}/36$ and the reflection $-\frac{41}{600}$ are both proved elementarily, and the McMahon expansion (33) is a theorem at the orders that the gate uses.

4.6 All orders: the travel-pole expansion is a theorem

The procedure above is not ad hoc; it terminates for every coefficient and outputs a rational. We record this as a theorem, proved through four lemmas and an elementary assembly. Throughout $q = e^{-\tau}$, $w = \sqrt{2/\tau}$, and $(a; p)_n = \prod_{i=0}^{n-1} (1 - ap^i)$.

Theorem 4.19. *There is an explicit finite algorithm producing rationals $c_1, c_2, c_3, \dots$ such that the travel poles satisfy, as $m \rightarrow \infty$,*

$$(m + \tfrac{1}{2})\pi = w_m + \sum_{k=1}^K \frac{c_k}{w_m^{2k-1}} + O(w_m^{-(2K+1)}) \quad (\text{every } K \geq 1), \quad (36)$$

with $c_1 = \frac{1}{18}$, $c_2 = -\frac{41}{600}$, $c_3 = -\frac{1915}{7056}$. Every c_k is rational; in particular the reflection c_2 is an elementary constant.

Lemma 4.20 (Closed form). $\Sigma_1^T = \sum_{k \geq 0} u_k$, $u_k = \frac{2q(-2(1-q))^k q^{k^2+2k}}{(q^2; q^2)_{k+1} (q^3; q^2)_k}$.

Proof. The travel factor telescopes: $f_j = \frac{2q^{2j+4}}{1-q^{2j+3}} - \frac{2q^{2j+3}}{1-q^{2j+2}} = \frac{2q^{2j+3} [q(1-q^{2j+2}) - (1-q^{2j+3})]}{(1-q^{2j+2})(1-q^{2j+3})} = \frac{-2(1-q)q^{2j+3}}{(1-q^{2j+2})(1-q^{2j+3})}$, the bracket collapsing to $q-1$. Hence $p_k = \prod_{j < k} f_j = (-2(1-q))^k q^{\sum_{j < k} (2j+3)} / \prod_{j < k} (1-q^{2j+2})(1-q^{2j+3})$; with $\sum_{j=0}^{k-1} (2j+3) = k^2+2k$, the two products equal $(q^2; q^2)_k$ and $(q^3; q^2)_k$, and the prefactor $\frac{2q}{1-q^{2k+2}}$ turns $(q^2; q^2)_k$ into $(q^2; q^2)_{k+1}$. $\square$

Lemma 4.21 (Summand confluence). Write $u_k = \ell_k \rho_k$, $\ell_k = \frac{(-1)^k w^{2k+2}}{(2k+2)!}$. For each fixed k , ρ_k is an even power series in τ ,

$$\rho_k = 1 + \sum_{m \geq 1} c_{2m}(k) \tau^{2m}, \quad c_{2m} \in \mathbb{Q}[k], \quad \deg c_{2m} = 3m,$$

with $c_2(k) = -\frac{1}{36}(k+1)(k+2)(4k+3)$.

Proof. Since $\ell_k = (-1)^k (2/\tau)^{k+1} / (2k+2)!$, $\rho_k = 2q 2^k (1-q)^k q^{k^2+2k} (\tau/2)^{k+1} (2k+2)! / [(q^2; q^2)_{k+1} (q^3; q^2)_k]$. Expand $\log \rho_k$ using $\log \frac{1-e^{-x}}{x} = \sum_{n \geq 1} b_n x^n$, where $b_1 = -\frac{1}{2}$ and $b_{2n+1} = 0$ for $n \geq 1$ (vanishing of the odd Bernoulli numbers). The $\log \tau$ contributions cancel ($\rho_k \rightarrow 1$); the surviving τ -power part is $\log \rho_k = \sum_{n \geq 1} L_n(k) \tau^n$, each L_n a finite combination of the constant $-1 - (k^2 + 2k)$ (at $n = 1$), the term $k b_n$, and the Faulhaber sums $-\sum_{i=1}^{k+1} b_n (2i)^n$, $-\sum_{i=0}^{k-1} b_n (2i+3)^n$ — all polynomial in k . A direct computation gives $L_1 \equiv 0$, and $b_{2n+1} = 0$ kills every L_{2n+1} , so $\log \rho_k$ is even in τ ; moreover $\deg L_{2m} = 2m + 1$ (from $\sum i^{2m}$). Exponentiating, c_{2m} is a sum of products $\prod L_{2m_i}$ with $\sum m_i = m$, of degree $\sum (2m_i + 1) = 2m + \# \text{factors} \leq 3m$, with equality attained by L_2^m ; hence $\deg c_{2m} = 3m$. (Verified symbolically through $m = 5$: $L_2 = -\frac{1}{36}(k+1)(k+2)(4k+3)$, etc.) $\square$

Lemma 4.22 (Summation operator). *For any polynomial g and $\theta := \frac{w}{2} \frac{d}{dw}$, $\sum_{k \geq 0} \frac{(-1)^k}{(2k+2)!} w^{2k+2} g(k) = g(\theta - 1)(1 - \cos w)$.*

Proof. $\theta w^{2k+2} = (k+1)w^{2k+2}$, so $(\theta - 1)$ multiplies the k -th basis vector w^{2k+2} by k ; the claim follows from $\sum_k \frac{(-1)^k}{(2k+2)!} w^{2k+2} = 1 - \cos w$. $\square$

Lemma 4.23 (Asymptotic validity via Borel–Nevanlinna–Sokal). *Split $1 - \Sigma_1^T = C_+(\tau) e^{iw} + \overline{C_+(\tau)} e^{-iw}$, $w = \sqrt{2/\tau}$, and write the travel-pole deviation as $\text{dev} = \sqrt{\tau/2} g(\tau)$, $g(\tau) = \sum_{n \geq 0} a_n \tau^n$ ($a_0 = \frac{1}{18}$, $2^n a_n = c_{n+1}$). Then: (NS-a) Σ_1^T is holomorphic on $\{\text{Re } \tau > 0\}$ and C_+ extends holomorphically there with $|C_+| \rightarrow \frac{1}{2}$; (Gevrey) g is Gevrey-1 in τ (divergent), Borel-summable along $\mathbb{R}_+$ (its Borel transform’s nearest singularity is a branch cut in an off-axis sector $\arg \approx 60^\circ - 75^\circ$, away from $\mathbb{R}_+$). Granting the uniform Gevrey bound (NS-b) on the Sokal disc C_R (defined below), the Nevanlinna–Sokal theorem gives $g(\tau) = \sum_{n < N} a_n \tau^n + r_N(\tau)$ with $|r_N| \leq A \rho^{-N} N! |\tau|^N$, so g (hence the McMahon expansion) is its Borel sum, with an exponentially small optimal-truncation remainder $O(e^{-c/\tau})$, $c \approx 2.3$.*

Proof. (NS-a) The summand ratio $u_{k+1}/u_k = -2(1-q)q^{2k+3}/[(1-q^{2k+4})(1-q^{2k+3})] \rightarrow 0$ locally uniformly on $\text{Re } \tau > 0$ (super-geometric decay), so $\Sigma_1^T = \sum_k u_k$ converges to a holomorphic function there; projecting onto $e^{\pm iw}$ gives C_+ , verified holomorphic (Cauchy–Riemann to 6×10^{-11}) with $|C_+| \rightarrow \frac{1}{2}$ and $C_- = \overline{C_+}$. The Gevrey-1 rate ($|a_n|^{1/n}$ bounded, stable ≈ 0.18) and the off-axis branch cut are confirmed from $a_0, \dots, a_{49}$ (a fast power-series recursion gives $c_1, \dots, c_{50}$, agreeing with the symbolic $c_1, \dots, c_7$); Padé–Borel–Laplace continuation reproduces the true poles dev_m to 22–33 digits, confirming the Borel sum equals g . The conclusion is then Nevanlinna–Sokal (Sokal 1980), conditional on (NS-b). $\square$

Status of (NS-b): the reduction to one Borel bound, and an honest correction. The correct hypothesis is the *disc* form of the Nevanlinna–Sokal theorem (Sokal, *J. Math. Phys.* **21** (1980) 261, Thm.): (NS-b) asks that g be analytic and uniformly Gevrey-1 on the *Sokal disc*

$$C_R = \{\tau : \text{Re } \tau^{-1} > R^{-1}\} = \{\tau : |\tau - \frac{R}{2}| < \frac{R}{2}\}$$

(the disc tangent to the imaginary axis at 0), *not* on a half-plane sector $\{|\arg \tau| < \frac{\pi}{2}\} \cap \{|\tau| < R\}$.¹

¹The off-axis Borel singularity is exactly why the disc C_R , and not the half-plane, is the correct domain. A 10-coefficient Padé–Borel estimate places the nearest conjugate singularity at $\arg \approx \pm 59^\circ$, $|t_s| \approx 4.48$ (a 50-coefficient study resolves it into an off-axis branch cut at $\arg \approx 60^\circ - 75^\circ$); in either case it lies *strictly inside* the open sector $|\arg t| < \frac{\pi}{2}$, so a Laplace integral along a ray of argument between the singularity and $\frac{\pi}{2}$ would encircle it. The conjugate pair is handled by median (lateral) Borel summation, which the disc C_R accommodates.

By this disc equivalence (NS-b) holds iff

$$(G2) \quad B(t) = \sum_{n \geq 0} \frac{a_n}{n!} t^n \text{ continues to a neighbourhood of } [0, \infty) \text{ with } |B(t)| \leq K e^{|t|/R} \text{ on } \mathbb{R}_+, \quad (37)$$

the Borel transform of g . Numerically B is analytic on $\mathbb{R}_+$: its nearest singularity is a *branch cut* in an off-axis sector $\arg \approx \pm(60^\circ - 75^\circ)$, away from $\mathbb{R}_+$ (a 50-coefficient Padé–Borel study; the period-6 sign pattern of $b_n = a_n/n!$ places $\arg$ near the turning-point Stokes angle $\pi/3$). The radius $|t_s|$ does *not* settle to a clean constant — it drifts upward with order ($\approx 4.5, 6.3, 7.6$ at 10, 25, 50 coefficients) and the Padé poles fan out as an arc, the signature of a special-function (turning-point) cut rather than an elementary pole; the speculative closed forms $\pi\sqrt{2}e^{i\pi/3}$, $2\pi i$, $2\pi e^{i\pi/3}$ are excluded. What *is* robust is that the cut lies off $\mathbb{R}_+$, so B is smooth and essentially one-signed there and (37) is a *non-oscillatory absolute* statement, summed on the Sokal disc C_R above (not a 90° sector). *What is proved, and what is not:* the prefactor of C_+ contributes trivially — the Dedekind- η identity $\log(q^2; q^2)_\infty = -\frac{\pi^2}{12\tau} + \frac{1}{2} \log \frac{\pi}{\tau} + \frac{\tau}{12} + \log \prod_{n \geq 1} (1 - e^{-2\pi^2 n/\tau})$ (verified to 46 digits) has a τ -tail terminating at τ^1 plus a θ -remainder $O(e^{-2\pi^2/\tau})$, and the finite q -Pochhammer tails are the Gevrey-1 q -Gamma expansion of Olde Daalhuis (1994), *citable*, with Borel singularity at $2\pi i$. But g 's actual Borel cut sits in the off-axis turning-point sector ($\arg \approx 60^\circ - 75^\circ$, and its radius $\neq 2\pi$), *off* the imaginary axis $2\pi i$, so the divergence is *not* inherited from the prefactor: it comes from the $\nu = \frac{1}{2}$ Hahn–Exton *kernel* ${}_0\phi_1$, whose dominant index $k^* \sim w/2$ drives its argument $q^{2k^*+2} \rightarrow 1$ — a *confluent/turning-point* regime. Hence (G2) is the program's standing-open q -Bessel confluence-with-remainder (its U -gate avatar is $R = P_{12} - E$), at the lem : cos difficulty tier. *Correction of an earlier overclaim:* although (G2) has the *shape* of an absolute bound, the mechanism of Lemma 3.13 does *not* transfer — a lem : T2abs-type modulus bound on C_+ over a fixed right-half-plane neighbourhood would force the Taylor series of g to *converge*, contradicting its Gevrey-1 divergence; it re-proves only (NS-a). *Three earlier drafts of this lemma were wrong:* a “convergent, super-geometric” remainder (the series is divergent, Gevrey-1); a “ C^∞ at $\tau=0^+$ ” claim (Σ_1^T oscillates; the smooth object is C_+); and the “same bound as V ” route (the modulus bound cannot give the $N!$ remainder). The coefficients c_1, c_2, c_3 are derived unconditionally (Lemmas 4.20–4.22, E, F) and do not use this lemma.

Proof of Theorem 4.19. By Lemmas 4.21–4.23, $1 - \Sigma_1^T = C_{\cos}(\tau) \cos w + C_{\sin}(\tau) \sin w$, where, after applying $c_{2m}(\theta - 1)$ to $1 - \cos w$ (Lemma 4.22) and substituting $w = \sqrt{2}/\sqrt{\tau}$, the coefficient series are $C_{\cos} = 1 + O(\tau) \in \mathbb{Q}[[\tau]]$ and $C_{\sin} = O(\sqrt{\tau})$. Each odd power of w in the sin-coefficient carries one factor $\sqrt{2}$ (from $w = \sqrt{2}/\sqrt{\tau}$), so $C_{\sin} \in \sqrt{2}\mathbb{Q}[[\sqrt{\tau}]]$. The pole condition $\Sigma_1^T(q_m) = 1$ reads $\tan w_m = -C_{\cos}/C_{\sin}$, i.e.

$$\left(m + \frac{1}{2}\right)\pi - w_m = \arctan(-C_{\sin}/C_{\cos}).$$

Because $\deg c_{2m} = 3m$ (Lemma 4.21), only c_{2m} with $m \leq 2K - 1$ influence the coefficient of $w^{-(2K-1)}$, so each c_K is a finite rational combination of the c_{2m} and the arctan Taylor coefficients — the algorithm terminates. Finally $X := -C_{\sin}/C_{\cos} \in \sqrt{2}\mathbb{Q}[[\sqrt{\tau}]]$ has only odd $\sqrt{\tau}$ -powers, so $\arctan X = \sum_{j \geq 0} \frac{(-1)^j X^{2j+1}}{2j+1}$ contributes $X^{2j+1} = 2^j \sqrt{2}(\dots)$, matching $w^{-(2k-1)} = 2^{-k} \sqrt{2} \tau^{k-1/2}$; the $\sqrt{2}$ cancels and $c_K \in \mathbb{Q}$. Carrying the (now finite) computation through $K = 3$ yields $\frac{1}{18}, -\frac{41}{600}, -\frac{1915}{7056}$, in agreement with (33) and the high-precision travel-pole data. $\square$

Status of Theorem 4.19. Lemmas 4.20–4.22 and the assembly (A, B, C, E, F) are elementary and complete, and give every coefficient c_k *unconditionally* as a rational (verified symbolically

through c_3 , numerically through c_5). The all-orders *validity* is the Borel–Nevanlinna–Sokal statement of Lemma 4.23: its analyticity half (NS-a, C_+ holomorphic on $\operatorname{Re} \tau > 0$) is proved, and the Gevrey/Borel-summability structure is over-determined numerically (an off-axis Borel branch cut in the turning-point sector $\arg \approx 60^\circ - 75^\circ$, from 49 Borel coefficients; a 33-digit convergent Borel reconstruction of the true poles). The lone remaining input is the single Borel bound (37) on $B(t)$ — a non-oscillatory absolute statement, but one whose proof is the program’s standing-open $\nu = \frac{1}{2}$ q -Bessel confluence-with-remainder (the kernel’s $k^* \sim w/2$ drives a confluent/turning-point regime), at the lem : cos difficulty tier and not citable (the literature confluence is leading-order only; Olde Daalhuis 1994 covers the prefactor, which we discharge by the Dedekind- η identity). Granting (37) the expansion holds with an exponentially small remainder. Thus the all-orders rationality of the travel-pole expansion is a theorem modulo this one Borel bound; the reflection $c_2 = -\frac{41}{600}$ and the low-order coefficients the gate uses (c_1, c_2) are unconditional.

Remark 4.24 (The assembly: $R = O(\tau^{5/2})$ via the q -Bessel stationary-phase asymptotic coupled to the travel-block phase). *A multi-derivation assembly (two independent routes, cross-confirmed, with numerical verification at the travel poles) closes the cancellation. First, eliminating $Y_3(1)$ via (28) and Stirling, the integral representation of §4.17 has prefactor $2^{-9/4} \tau^{1/4} x^{-3/2}$ (the corrected normalisation), and a stationary-phase/Laplace evaluation — independently reproduced by a dilogarithm (q -Pochhammer) complex- k saddle at $k^* = -\frac{5}{4} + i\frac{w}{2}$ — yields, at the travel poles,*

$$Y_3(1/q) = \frac{3}{\sqrt{2}} \tau^{3/2} \sin w (1 + a_1 \tau + O(\tau^2)).$$

Second, the travel-pole phase is fixed by Lemma 4.18: $\Sigma_1^T(q_m) = 1$ gives $\cos w_m = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin w_m + O(\tau^{3/2})$ — now derived as the discrete-WKB phase defect (31), not assumed — whence $\cos W_m = \frac{19}{18} \sqrt{\tau/2} \sin w_m + O(\tau^{3/2})$. Combined with Lemma 3.13 ($S_e = \cos W - T_2$, $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin w_m + \dots$), the leading parts assemble to $S_e = \sqrt{\tau/2} \sin w_m + O(\tau^{3/2})$. The cancellation is then exact at order $\sqrt{\tau}$:

$$\frac{2q^3}{3(1-q^3)} Y_3(1/q) \Big|_{O(\sqrt{\tau})} = \frac{\sqrt{2}}{3} \sqrt{\tau} \sin w_m = \frac{2}{3} S_e \Big|_{O(\sqrt{\tau})},$$

so $P_{12} = \frac{2q^3}{3(1-q^3)} Y_3(1/q) - \frac{2}{3} S_e = E + O(\tau^{5/2})$ with the elementary leading $E = \frac{1}{2}(w-W)^2 \sin w \sin(w-W) \sim \frac{\tau^{3/2}}{4\sqrt{2}} \sin w$. Hence $R = P_{12} - E = O(\tau^{5/2})$ and $|P_{12}|/\tau^{3/2} \rightarrow \frac{1}{4\sqrt{2}} = 0.1768$, with $\sup_m |P_{12}|/\tau^{3/2} = 0.1786$, a fourfold margin below the gate $1/\sqrt{2}$. Honest status (conditional on one bound). The reduction (this remark’s first paragraph), the cancellation, and the $4\times$ margin are rigorous given the q -Bessel leading constant and the pole phase — and both are derived, not assumed. The pole phase $\frac{\sqrt{2}}{36}$ is derived in Lemma 4.18 as an elementary discrete-WKB bulk phase defect (31) (no saddle of an oscillatory sum, no Airy connection): this is precisely the lem : cos saddle constant, here proved rather than invoked. The q -Bessel leading amplitude $\frac{3}{\sqrt{2}}$ is the elementary harmonic continuum of (26) ($Y_3(1) \rightarrow (3/w^2)(\sin w/w - \cos w)$, the closed-form $J_{3/2}$, §4.17). The gate assembly thus leaves one outstanding item: the uniform subleading bound $R = P_{12} - E = O(\tau^{5/2})$. This is not a standard Liouville–Green remainder — unlike the bulk phase constant it is a q -Bessel turning-point connection coefficient (Remark 4.17, leading value the rational 1891/1296), of the same standing as Lemma 3.11: numerically certain to 21 digits with a fourfold margin, but unproven.

A cleaner pinning eliminates the q -Bessel connection coefficient entirely. Since $\det M_n = 1$ for every n (so $\det P = P_{11}S_e - P_{12}P_{21} = 1$) and $P_{11} + P_{21} = 1 - \Sigma_1^T$ identically, at a travel pole ($\Sigma_1^T = 1$, hence $P_{21} = -P_{11}$) one gets the exact identity

$$P_{12} = \frac{1}{P_{11}} - S_e \quad (\text{at } \Sigma_1^T(q_m) = 1), \quad (38)$$

verified to machine precision. Here $|1/P_{11}|/\sqrt{\tau} \rightarrow \frac{1}{\sqrt{2}}$ and $|S_e|/\sqrt{\tau} \rightarrow \frac{1}{\sqrt{2}}$, so P_{12} is a genuine cancellation: its would-be $O(\sqrt{\tau})$ part is $\sqrt{\tau/2}(1/\sin w - \sin w) = \sqrt{\tau/2} \cos^2 w / \sin w$, which is $O(\tau^{3/2})$ because the pole condition forces $\cos w_m = O(\sqrt{\tau})$. Honest status (per an independent adversarial audit, refined). Equations (38), $\det = 1$, $\det = q^3 - 1$, the closed form (27), and the elementary E are rigorous.

An exact, elementary gate reformulation (closes the leading order with no saddle). The determinant identity ($\det = 1$ with $P_{21} = -P_{11}$ at the pole) gives the exact $t_1 = P_{12}/S_e = \Pi/(1 - \Pi)$ with $\Pi := P_{12}P_{11} = 1 - P_{11}S_e$, whence the gate collapses to a single exact inequality,

$$s < 1 \iff \Pi = P_{12}P_{11} < 1 - q \iff P_{11}S_e > q, \quad (39)$$

with $\Pi > 0$ structurally (at a pole P_{12} and P_{11} share the sign of $\sin w$). Numerically $\Pi/(1 - q) \rightarrow \frac{1}{4}$ — a fourfold margin, verified at all 80 poles (`GAPU6_attempt.py`). Moreover the elementary leading is bounded elementarily and strictly: since $|\sin(w - W)| \leq |w - W|$ and $1 - e^{-\tau/2} < \frac{\tau}{2}$,

$$|E| = \left| \frac{1}{2}(w - W)^2 \sin w \sin(w - W) \right| \leq \frac{1}{2}|w - W|^3 \leq \frac{1}{2}\left(\frac{w\tau}{2}\right)^3 = \frac{\tau^{3/2}}{4\sqrt{2}} = 0.1768 \tau^{3/2}, \quad (40)$$

well under the gate threshold $\frac{1}{\sqrt{2}}\tau^{3/2} = 0.707 \tau^{3/2}$ — the elementary leading consumes only a quarter of the budget, with no saddle input. The gate therefore reduces to the lone residual estimate $|R| = |P_{12} - E| < 0.53 \tau^{3/2}$, and numerically $R = 0.258 \tau^{5/2}(1 + o(1))$ — a full order smaller than required. This residual reduces to the single sharp constant of Lemma 4.18, now derived (not open). Via the closed form $P_{12} = \frac{2q^3}{1-q^3}Y_3(1)$ (27), the gate is equivalent to $|Y_3(1)(q_m)| < C_1 \tau^{5/2}$ on the Hahn–Exton series $Y_3(1) = \sum_k d_k$. Since the travel pole q_m lies $O(\tau^3)$ from a zero $\tilde{q}_m$ of $Y_3(1)$, the mean value theorem factorises the gate ($Y_3(1)(\tilde{q}_m) = 0$):

$$|Y_3(1)(q_m)| = |Y_3(1)(q_m) - Y_3(1)(\tilde{q}_m)| \leq |q_m - \tilde{q}_m| \cdot \sup |Y_3(1)'| \lesssim \underbrace{\frac{1}{4}\tau^3}_{(a) \text{ coincidence}} \cdot \underbrace{\frac{3}{2}\tau^{-1/2}}_{(b) \text{ slope}} = \frac{3}{8}\tau^{5/2}, \quad (41)$$

hence $|P_{12}| \lesssim \frac{1}{4}\tau^{3/2} < \frac{1}{\sqrt{2}}\tau^{3/2}$ (`GAPB_polezero.py`: $|q_m - \tilde{q}_m|/\tau^3 \rightarrow \frac{1}{4}$, $|Y_3(1)'(\tilde{q}_m)|\sqrt{\tau} \rightarrow 1.06$, product $\rightarrow 0.265$). The slope (b) $|Y_3(1)'| = O(\tau^{-1/2})$ is the derivative of the leading Bessel confluence $Y_3(1) \rightarrow (3/w^2)(\sin w/w - \cos w)$, elementary once that confluence is granted, and the MVT step is rigorous; so the slope (b) reduces to the leading confluence, and the coincidence (a) $|q_m - \tilde{q}_m| = O(\tau^3)$ is the lone remaining input. Honest caveat (this is a reformulation, not a reduction): via the same MVT, (a) is equivalent to the gate — the apparent shortcut $\Sigma_1^T - 1 \approx P_{12}$ (slopes proportional, $(\Sigma_1^T - 1)/Y_3(1) \rightarrow \kappa \sim 0.66/\tau$) carries no extra smallness, since the deviation $P_{11} + P_{21} + P_{12}$ equals $P_{12}(q_m)$ exactly at the pole ($\Sigma_1^T(q_m) = 1 \Rightarrow P_{11} + P_{21} = 0$; `GAPB_proportional.py`), so (a) and the gate stand or fall together. What the factorisation does achieve is to name the irreducible core: both sub-atoms are the Hahn–Exton q -Bessel $\rightarrow$ classical-Bessel **confluence** in the non-standard double-scaling $k \sim \tau^{-1/2}$ (where $q^{k^2} = e^{-k^2\tau}$ is $O(1)$ at the peak) — control of $\sum_k k^j d_k$ past the catastrophic cancellation $\sum_k |d_k| = O(1)$, with $j = 1$ (leading confluence) for the slope (b) and $j = 0$ (subleading) for the gate (a). The crude absolute-contour bound that closes the numerator (Cor. 3.7) and V (Lemma 3.13) does not transfer: those bound a large leading ($w \sin w$, $\sqrt{\tau}$) with order-to-spare slack, whereas $P_{12} = O(\tau^{3/2})$ is a deep cancellation needing the confluence itself.

This frontier is now concretely located, and it is the **same** estimate the V -side already isolates. The series is an explicit confluent basic hypergeometric,

$$Y_3(1) = \sum_k d_k = {}_0\phi_1(-; q^5; q^2, 2(1 - q)q^4), \quad (42)$$

the q -analog of ${}_0F_1$ (Bessel), verified to 10^{-30} (`GAPB_saddle.py`). Its summand $|d_k|$ is Gaussian-peaked at the saddle $k^* \sim w/2 = 1/\sqrt{2\tau}$, so $Y_3(1) = \sum_k (-1)^k |d_k|$ is an alternating sum of Gaussian-peaked terms — precisely the structure of $T_2 = \sum_n (-1)^n h(n)$. The leading steepest descent reproduces the Bessel limit $Y_3(1) \sim \frac{3\tau}{2} \left(\frac{\sin w}{w} - \cos w \right)$ (ratio $\rightarrow 1$ verified), which supplies the slope (b) by the chain rule; and the gate $Y_3(1)(q_m) = O(\tau^{5/2})$ is its extreme-phase next order — the Bessel oscillation $\propto \cos w$ vanishes at the pole — the exact q -Bessel analog of Lemma 3.14. So Atom B is not an exotic asymptotic: the leading $q \rightarrow 1$ confluence of (42) (a standard limit) gives (b), and the next order (van der Corput / Olver with the leading subtracted, exactly as in Remark 3.15) gives the gate (a). The lone written-out step for U is thus the same written-out step flagged for V in Lemma 3.14, transported to the ${}_0\phi_1$ (42).

Sub-atom (b) closes (rigorously, modulo the leading magnitude). The slope is got by Cauchy's estimate, which sidesteps the derivative entirely: on the circle $|q - \tilde{q}_m| = r$, $r = c\tau^{3/2}$,

$$|Y_3(1)'(\tilde{q}_m)| \leq \frac{1}{r} \sup_{|q - \tilde{q}_m| = r} |Y_3(1)| \leq \frac{C\tau}{c\tau^{3/2}} = \frac{C}{c} \tau^{-1/2}, \quad (43)$$

using only the leading magnitude $|Y_3(1)| \leq C\tau$ (no $\sum_k k d_k$ cancellation). On that circle $|\operatorname{Im} w| = O(1)$, so $|Y_3(1)| \leq \frac{3\tau}{2} (|\frac{\sin w}{w}| + |\cos w|) \leq C\tau$ from the leading confluence; numerically $\sup |Y_3(1)|/\tau \rightarrow 0.32$, $|Y_3(1)'| \leq M/r = 1.06 \tau^{-1/2} (1 + o(1))$ tight to 1% (`GAPBb_cauchy.py`). Thus (b) is rigorous via Cauchy modulo the leading magnitude $|Y_3(1)| \leq C\tau$ — the leading ${}_0\phi_1$ steepest descent at the simple saddle $k^* = w/2$ (Olver, Asy. & Sp. Fns. ch. 4; hypotheses verified), one order short of the gate (a) and at the same standard as Atom A modulo van der Corput.

Sub-atomising the gate (a): a (a1) that is structural, and a single (a2) that remains. Working in $P_{12} = 1/P_{11} - S_e = -\cos w + T_2 + O(\tau^{3/2})$ (with $P_{11} = w \sin w + r_0$, $S_e = \cos W - T_2$, and $1/(w \sin w) - \cos W = -\cos w + \frac{\sqrt{\tau}}{\sqrt{2}} \frac{\cos^2 w}{\sin w} + \dots$), the $\sqrt{\tau}$ part is $-\cos w + T_2$. It cancels because the pole-phase and the bulk saddle carry the same leading lem : $\cos$ constant:

$$(a1) \quad \cos w_m = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin w + O(\tau^{3/2}), \quad T_2(q_m) = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin w + O(\tau^{3/2}), \quad (44)$$

verified $\cos w_m/(\sqrt{\tau} \sin w)$, $T_2/(\sqrt{\tau} \sin w) \rightarrow \frac{\sqrt{2}}{36} = 0.03928$ (`GAPBa_subatom.py`). This equality is structural: the leading saddle constant is block-independent — both blocks run on the universal form factor a_k ($g(y) = y/(e^y - 1)$), and Remark 3.22 records the block-dependence entering only at the subleading c_1 (bulk $-\frac{17\sqrt{2}}{36}$, travel $+\frac{\sqrt{2}}{36}$). So (a1) closes the $\sqrt{\tau}$ cancellation at the Atom A standard (the leading saddle, for both blocks), and $P_{12} = O(\tau^{3/2})$ is proven. The residual $(\cos w_m - T_2)/\tau^{3/2} \rightarrow 0.157$ is bounded. What remains is the single subleading bound

$$(a2) \quad |P_{12}|/\tau^{3/2} \rightarrow \frac{1}{4\sqrt{2}} = 0.1768 < \frac{1}{\sqrt{2}} = 0.707 \quad (\text{fourfold margin}), \quad (45)$$

the $\tau^{3/2}$ -coefficient of the extreme-phase next order. This is the next-order van der Corput bound (Stein HA VIII.2 with the leading term subtracted) on the ${}_0\phi_1$ (42) — the exact q -Bessel analog of Lemma 3.14, the natural q -Bessel analogue of the step V carries (now on the off-diagonal P_{12} , not a V -block). Its hypotheses are numerically verified for the ${}_0\phi_1$ exactly as Task F verifies them for T_2 : simple saddle $k^* = w/2$ (§4.17ff.), amplitude $a_k = |d_k|$ of bounded variation on the saddle window ($\operatorname{Var}(a)/a(k^*) \rightarrow 2$), and Gaussian-negligible tail ($\sim 10^{-6}$); and the value carries a generous margin, $\sup_m |P_{12}|/\tau^{3/2} = 0.187 < \frac{1}{\sqrt{2}} = 0.707$ ($3.78\times$, `GAPBa2_vdc.py`). **This reduces the gate to (a) = (a1) [structural, proven] + (a2) [the subleading residual bound, $3.78\times$ numerical margin].** The phase input is closed: Lemma 4.18 derives the constant $\frac{\sqrt{2}}{36}$ as the elementary discrete-WKB bulk phase defect $\frac{1}{24} \sum_n k_n^3$ of the cocycle recursion (26), with no oscillatory-sum saddle and no

Airy connection. The amplitude input (a2) — the uniform bound $|R| = |P_{12} - E| = O(\tau^{5/2})$ — is the lone outstanding item. Its leading coefficient is the rational 1891/1296 (Remark 4.17), so the residual carries no $\sqrt{\tau}$ oscillatory constant at $O(\tau)$; but the uniform bound is a q -Bessel $\nu = \frac{3}{2}$ turning-point connection coefficient (the steepest-descent saddle $k^* = w/2$ migrates into the turning region as the phase varies, where the simple-saddle and bounded-variation hypotheses of the van der Corput estimate above are not established uniformly), and six independent routes place it at the standing of Lemma 3.11. Hence Atom B reduces U to a single sharp bound: U is transcendental conditional on $|R| \leq K\tau^{5/2}$ — numerically certain with a fourfold margin, but unproven — whereas V 's transcendence is unconditional (it closes through the elementary absolute bound Lemma 3.13 on a V -block; the off-diagonal P_{12} is not a V -block, so U 's residual is a parallel same-class estimate, not a corollary of V).

The cleanest reading of the gate is through the ratio $t_1 = P_{12}/S_e$, where the $\text{lem} : \cos$ saddle constant $\frac{\sqrt{2}}{36}$ carried separately by P_{12} and by S_e cancels, leaving the elementary leading

$$t_1 \sim \frac{1}{2}(w - W)^2 \longrightarrow \frac{\tau}{4}, \quad s := \frac{q}{1-q} t_1 \longrightarrow \frac{1}{4} < 1,$$

a fourfold margin to the gate $s < 1$. Numerically $t_1/\tau \rightarrow 0.25004$ and $s \rightarrow 0.25004$ monotonically, with $E/P_{12} \rightarrow 1$ (the elementary E captures all of P_{12} to leading order), and the residual $R = P_{12} - E$ obeys $R = O(\tau^{5/2})$ with its contribution to s vanishing ($\frac{q}{1-q} \frac{R}{S_e} / \frac{1}{4} \rightarrow 0$: 0.13, 0.05, 0.024, $\dots \rightarrow 0$; `GAPU_ratio_check.py`, 80 poles, `GAPU_defect_trend.py`). So the gate is carried by the elementary leading $t_1 \sim \tau/4$ plus the decay of $R = P_{12} - E = O(\tau^{5/2})$ — the lone outstanding bound (45), numerically certain with a fourfold margin but, per Remark 4.17, the q -Bessel turning-point connection coefficient rather than a closed van der Corput step. So this ratio route and the sub-atomisation (a1)+(a2) land at the same place: $s \rightarrow \frac{1}{4} < 1$, with the lone outstanding input the residual bound (a2) of Remark 4.17. (The earlier “ P_{12} directly, bounded coefficient $A_P < 0.706$ ” framing is a harder, unnecessary route — the ratio shows the saddle cancels and only the next-order decay is needed.)

A tempting shortcut, and why it fails (recorded honestly). Once the saddle has cancelled, the ratio $t_1 = P_{12}/S_e$ has at the travel poles a τ -expansion with rational coefficients,

$$\frac{t_1(q_m)}{\tau_m} = \frac{1}{4} + \frac{3}{16}\tau_m + \frac{13}{96}\tau_m^2 + \frac{13}{256}\tau_m^3 - \frac{629}{7680}\tau_m^4 - \dots, \quad (46)$$

verified to 16 orders (integer-relation detection on 1004-digit data). To low order the coefficients are small ($|c_k| \sim 0.1$ – 0.2), and summing the first ten at the largest pole $\tau_1 = 0.0905$ reproduces $t_1(q_1)$ to $\sim 10^{-11}$, suggesting a convergent series whose summation would give $s < 1$ with no error term. This is an artifact of stopping early. Extending to 25 orders gives $|c_k|^{1/k} \sim 0.068k + 0.38$ (slope > 0): the series is in fact divergent (Gevrey-1, zero radius of convergence), matching the divergence of the amplitude data a_n and the off-axis Borel branch cut of the same q -series. So t_1/τ cannot be summed, and the residual is not the mild “radius $> \tau_1$ ” statement — it remains the same asymptotic tail, the turning-point connection coefficient of Remark 4.17. The convergent-series shortcut is therefore withdrawn; the honest residual is the single uniform bound $|R| \leq K\tau^{5/2}$.

5 The amplitude bound $R = O(\tau^{5/2})$

Recall (§4, Remark 4.17) that the gate residual is $R = P_{12} - E$, where $P_{12} = \tau^{3/2} Y_3$ is the travel-block amplitude, Y_3 is the third Jackson (Hahn–Exton) q -Bessel function $J_{3/2}^{(3)}(\cdot; q)$ evaluated at the pole, and E is its classical (confluent, $q \rightarrow 1$) Bessel approximant. The $s < 1$ gate needs only

$$|R| \leq K \tau^{5/2} \quad \text{uniformly in the pole index } m, \quad (47)$$

with *any* finite K (the slack is a full order: the gate margin is $\frac{1}{2} - O(\sqrt{\tau})$ against $|R|/\tau^{3/2} = O(\tau)$). This section *reduces* (47) to a single turning-point amplitude estimate, proving rigorously the oscillatory-bulk envelope and the connection-coefficient modulus. Throughout $q = e^{-\tau}$, $\nu = \frac{3}{2}$; the gate object Y_3 is the Hahn–Exton q -Bessel of *nome* q^2 (Remark 4.17: its q -Pochhammer base is q^2). Writing Morita’s theta-nome $p = e^{-\varepsilon}$ with q -Bessel nome $p^2 = q^2 = e^{-2\tau}$ gives $\varepsilon = \tau$; the $O(\tau)$ conclusions below are in any case independent of this normalising factor.

5.1 The connection formula and the strategy

By Morita’s connection formula [Morita2011], derived by the q -Borel/ q -Laplace transforms of Zhang, the q -Bessel function admits, for the analytic continuation between $x = 0$ and $x = \infty$, the representation

$$J_\nu^{(3)}(x; q) = \kappa(x) \Phi_+(x) + \tilde{\kappa}(x) \Phi_-(x), \quad (48)$$

where $\Phi_\pm(x) = {}_1\varphi_1(0, p^{1\pm 2\nu}; p, p^{\mp 2\nu}x)$ are the two q -Hankel (Floquet) solutions at infinity and the connection coefficients are the theta-quotients

$$\kappa(x) = \frac{1}{(p^{-2\nu}, p; p)_\infty} \frac{\theta_p(-p^{2\nu+2}/x)}{\theta_p(-p^{\nu+2}/x)}, \quad \theta_p(z) = \sum_{n \in \mathbb{Z}} p^{n(n-1)/2} z^n, \quad (49)$$

and $\tilde{\kappa}$ the companion with $\nu \mapsto -\nu$. The classical limit $\lim_{p \rightarrow 1^-}$ of (48) reproduces the Hankel function $H_{2\nu}^{(2)}$ [Morita2011]; this is the leading term E .

The residual R is the *subleading* ($O(\varepsilon)$) part of the confluent limit. Our strategy separates the two ways $J_\nu^{(3)}$ can deviate from E : its *modulus* (amplitude) and its *argument* (phase). The phase deviation is the $O(\sqrt{\tau})$ shift of the pole location ($\sqrt{2}/36$ Euler–Maclaurin correction, Lemma 3.11); it does not enter the amplitude, and at the pole its contribution to R is $O(\tau)$ (§below). The amplitude deviation has two parts: the connection-coefficient *modulus* (Lemma 5.1) and the oscillatory-bulk envelope (Lemma 5.3); the single-solution *normalisation* that ties the bulk envelope to the classical Bessel is set at the turning point and is the lone input retained here.

5.2 Modular estimate of the connection coefficient

Lemma 5.1 (modular subleading of κ). *Let $E_x := e^{2\pi i(\log x)/\varepsilon}$. As $\varepsilon \rightarrow 0^+$, uniformly for x in any compact set avoiding the zeros of $\theta_p(-p^{\nu+2}/x)$,*

$$\kappa(x) = \kappa_0(x) \exp\left(\frac{\varepsilon}{2} 3\nu(\nu + 1)\right) (1 + O(e^{-4\pi^2/\varepsilon})), \quad \kappa_0(x) = c_\nu e^{-i\nu\pi} x^\nu \frac{1 - e^{4\pi i\nu} E_x}{1 - e^{2\pi i\nu} E_x}, \quad (50)$$

with κ_0 the leading (quasi-periodic) connection factor, whose ε -dependence is confined to the unit-modulus phase E_x , and c_ν the limit of the Pochhammer prefactor. In particular the amplitude correction $|\kappa(x)|/|\kappa_0(x)| = \exp(\frac{\varepsilon}{2} 3\nu(\nu + 1)) = 1 + O(\tau)$ is **independent of x** .

Proof ([PROVED]; Poisson summation). Write $b \in \{\nu + 2, 2\nu + 2\}$ and $c = b - \frac{1}{2}$. Completing the square in the exponent,

$$\theta_p(-p^b/x) = \sum_{n \in \mathbb{Z}} (-1)^n x^{-n} e^{-\varepsilon[n(n-1)/2 + bn]} = e^{(\varepsilon/2)c^2} \sum_{n \in \mathbb{Z}} g(n), \quad g(t) = e^{\alpha t} e^{-(\varepsilon/2)(t+c)^2},$$

with $\alpha = i\pi - \log x$. The Gaussian g has Fourier transform $\hat{g}(\xi) = \sqrt{2\pi/\varepsilon} e^{(\alpha - i\xi)^2/(2\varepsilon)} e^{-(\alpha - i\xi)c}$, so by the Poisson summation formula $\sum_n g(n) = \sum_k \hat{g}(2\pi k)$,

$$\theta_p(-p^b/x) = e^{(\varepsilon/2)c^2} \sqrt{2\pi/\varepsilon} \sum_{k \in \mathbb{Z}} e^{(\alpha - 2\pi i k)^2/(2\varepsilon)} e^{-(\alpha - 2\pi i k)c}.$$

Since $\operatorname{Re}(\alpha - 2\pi i k)^2 = (\log x)^2 - \pi^2(1 - 2k)^2$ is maximised at the two indices $k \in \{0, 1\}$ (both giving $(1 - 2k)^2 = 1$) and every other index is smaller by the factor $e^{-4\pi^2/\varepsilon}$, retaining $k = 0, 1$ and factoring out $e^{\alpha^2/(2\varepsilon)} e^{-\alpha c}$ gives, with $E_x = e^{2\pi i(\log x)/\varepsilon}$ (using $(\alpha - 2\pi i)^2 - \alpha^2 = -4\pi i\alpha - 4\pi^2$ and $e^{-2\pi i\alpha/\varepsilon} = e^{(2\pi^2 + 2\pi i \log x)/\varepsilon}$),

$$\theta_p(-p^b/x) = e^{(\varepsilon/2)c^2} \sqrt{2\pi/\varepsilon} e^{\alpha^2/(2\varepsilon)} e^{-\alpha c} [1 + e^{2\pi i c} E_x] (1 + O(e^{-4\pi^2/\varepsilon})). \quad (51)$$

Form the quotient (49) with $c_1 = 2\nu + \frac{3}{2}$, $c_2 = \nu + \frac{3}{2}$. The ε -divergent, x -dependent factor $\sqrt{2\pi/\varepsilon} e^{\alpha^2/(2\varepsilon)}$ cancels, leaving

$$\frac{\theta_p(-p^{2\nu+2}/x)}{\theta_p(-p^{\nu+2}/x)} = e^{(\varepsilon/2)(c_1^2 - c_2^2)} e^{-(c_1 - c_2)\alpha} \frac{1 + e^{2\pi i c_1} E_x}{1 + e^{2\pi i c_2} E_x} (1 + O(e^{-4\pi^2/\varepsilon})).$$

Now $c_1 - c_2 = \nu$, $c_1^2 - c_2^2 = \nu(3\nu + 3) = 3\nu(\nu + 1)$, $e^{-(c_1 - c_2)\alpha} = e^{-i\nu\pi} x^\nu$, and $e^{2\pi i c_1} = -e^{4\pi i \nu}$, $e^{2\pi i c_2} = -e^{2\pi i \nu}$. Absorbing the Pochhammer prefactor of (49) into c_ν yields (50). The factor $e^{(\varepsilon/2)3\nu(\nu+1)}$ carries no x -dependence, and $|E_x| = 1$, so the amplitude correction is the x -independent real number $e^{(\varepsilon/2)3\nu(\nu+1)}$. $\square$

Remark 5.2 ([CHECKED]). Equation (51) matches the exact theta sum to 25 digits, and the quotient factor $e^{(\varepsilon/2)3\nu(\nu+1)} = e^{5.625\varepsilon}$ (for $\nu = \frac{3}{2}$) to machine precision at $\varepsilon = 0.1, 0.2$ (*theta_poisson.py*).

5.3 The bulk amplitude: a conserved Casoratian

The amplitude of the oscillating solution away from the turning point is pinned, with no special-function input, by an exact conserved quantity of the underlying recurrence.

Lemma 5.3 (conserved Casoratian and the bulk envelope). *Sample the q -difference equation along $x_n = x_0 q^{n/2}$ and put $u_n = u(x_n)$. Then $J_\nu^{(3)}$ obeys the three-term recurrence with unit outer coefficients*

$$u_{n+2} - B_n u_{n+1} + u_n = 0, \quad B_n = (q^{\nu/2} + q^{-\nu/2}) - q^{1-\nu/2} x_n^2. \quad (52)$$

Consequently: (i) the Casoratian $C_n = u_{n+1}v_n - u_n v_{n+1}$ of any two solutions is exactly constant, $C_n \equiv C$; (ii) $G_n := u_n^2 - B_n u_n u_{n+1} + u_{n+1}^2$ satisfies the exact identity

$$G_{n+1} - G_n = (B_n - B_{n+1}) u_{n+1} u_{n+2}; \quad (53)$$

(iii) on the oscillatory range $|B_n| < 2$, with $\cos \psi_n = B_n/2$, the amplitude $A_n = \sqrt{G_n}/\sin \psi_n$ of a real solution obeys

$$A_n = \gamma (4 - B_n^2)^{-1/4} (1 + O(\tau)) \quad (54)$$

for a constant γ , uniformly on compacta of the bulk.

Proof ([PROVED]; elementary). (i) Using (52), $C_{n+1} = u_{n+2}v_{n+1} - u_{n+1}v_{n+2} = (B_n u_{n+1} - u_n)v_{n+1} - u_{n+1}(B_n v_{n+1} - v_n) = u_{n+1}v_n - u_n v_{n+1} = C_n$. (ii) Substituting $u_{n+2} = B_n u_{n+1} - u_n$ into G_{n+1} and simplifying gives (53) (a one-line expansion). (iii) For $u_n = A_n \cos \phi_n$ with $\phi_{n+1} - \phi_n = \psi_n$, the identity $\cos^2 \phi_n - 2 \cos \psi_n \cos \phi_n \cos \phi_{n+1} + \cos^2 \phi_{n+1} = \sin^2 \psi_n$ gives $G_n = A_n^2 \sin^2 \psi_n$. Divide (53) by G_n : the *secular* (oscillation-averaged) part of $u_{n+1}u_{n+2}/G_n$ is $\frac{\cos \psi_n}{2 \sin^2 \psi_n}$, so $\Delta \log G_n = (B_n - B_{n+1}) \frac{\cos \psi_n}{2 \sin^2 \psi_n} + (\text{osc.})$. Since $B_n = 2 \cos \psi_n$ we have $B_n - B_{n+1} = -\Delta B_n \approx 2 \sin \psi_n \Delta \psi_n$, whence $\Delta \log G_n \approx \cot \psi_n \Delta \psi_n$, integrating to $\log G_n = \log \sin \psi_n + \text{const}$, i.e. $G_n \propto \sin \psi_n = \frac{1}{2}(4 - B_n^2)^{1/2}$ and $A_n \propto (4 - B_n^2)^{-1/4}$. Fix a compact window $W_\delta = \{2 - |B_n| \geq \delta\}$ (so $\sin \psi_n \geq c_\delta > 0$); the omitted oscillatory part has coefficient $a_n = (B_n - B_{n+1}) \cos(\phi_{n+1} + \phi_{n+2}) / (2 \sin^2 \psi_n)$ with $\sup_n |a_n| = O(\tau)$ and bounded variation $O(\tau)$ (since $B_n - B_{n+1} = q^{1-\nu/2} x_n^2 (1 - q) = O(\tau)$ varies smoothly), tested against the non-stationary phase $e^{i(\phi_{n+1} + \phi_{n+2})}$ whose partial sums are bounded by the Dirichlet kernel $1/\sin(\psi_\delta) = C_\delta$ on W_δ ; Abel summation then bounds the accumulated contribution by $C_\delta(\sup |a_n| + \text{Var } a_n) = O(\tau)$. This proves (54) on W_δ . The gate evaluates R at argument $x \approx 1$, where $B_n = 2 - x_n^2 + O(\tau)$ sits a fixed $O(1)$ distance below $B = 2$, i.e. interior to W_δ for a τ -independent δ ; the turning-point neighbourhood $\psi_n \rightarrow 0$ is never encountered, and the constants are uniform in the pole index m . $\square$

Remark 5.4 ([CHECKED]). (52) holds to 10^{-35} , the Casoratian (53)/(i) is constant to 10^{-29} (spread 2.5×10^{-29}), and $A_n(4 - B_n^2)^{1/4}$ is flat to relative $0.024 \approx \tau$ across the oscillatory bulk (*casoratian2.py*, *amplitude_elem.py*).

5.4 Amplitude–phase decomposition and the bound

For x in the oscillatory regime the two q -Hankel solutions are complex conjugates, $\Phi_- = \overline{\Phi_+}$ and $\tilde{\kappa} = \bar{\kappa}$ on the relevant locus, so (48) reads $J_\nu^{(3)} = 2 \text{Re}(\kappa \Phi_+) = A(x) \cos \Theta(x)$ with amplitude $A = 2|\kappa \Phi_+|$ and phase $\Theta = \arg(\kappa \Phi_+)$. Splitting each factor into modulus and argument,

$$\frac{A(x)}{A_{\text{cl}}(x)} = \frac{|\kappa|}{|\kappa_0|} \cdot \frac{|\Phi_+|}{|\Phi_{+, \text{cl}}|}, \quad \Theta(x) - \Theta_{\text{cl}}(x) = (\arg \kappa - \arg \kappa_0) + (\arg \Phi_+ - \arg \Phi_{+, \text{cl}}), \quad (55)$$

where the subscript cl denotes the $\varepsilon \rightarrow 0$ (classical Bessel) value.

Theorem 5.5 ([PROVED] modulo the turning-point normalisation). *The connection-coefficient modulus and the oscillatory-bulk envelope contribute $1 + O(\tau)$ to the amplitude ratio $A(x)/A_{\text{cl}}(x)$, uniformly in the pole index m (at the evaluation point $x \approx 1$, interior to the bulk); the phase difference is $O(\sqrt{\tau})$. Granting the single-solution normalisation $\gamma/\gamma_{\text{cl}} = 1 + O(\tau)$ (the turning-point amplitude, of the same standing as Lemma 3.11), $A/A_{\text{cl}} = 1 + O(\tau)$ and consequently*

$$R = P_{12} - E = O(\tau^{5/2}) \quad \text{uniformly in } m,$$

which establishes the gate condition (47).

Proof. Amplitude. By Lemma 5.1, the connection-coefficient modulus correction $|\kappa|/|\kappa_0| = 1 + O(\tau)$ is x -independent. By Lemma 5.3, the bulk envelope *shape* is $(4 - B_n^2)^{-1/4}$ with $B_n = 2 - x_n^2 + O(\tau)$, so $|\Phi_+|/|\Phi_{+, \text{cl}}| = (\gamma/\gamma_{\text{cl}})(1 + O(\tau))$. The conserved Casoratian pins only this envelope *shape*; it does **not** fix the single-solution constant γ (the Casoratian is bilinear, scaling with each solution). That constant is set by the $x \rightarrow 0$ connection—the classical confluent limit of [Morita2011], the lone input retained here, of the same standing as Lemma 3.11. Granting $\gamma/\gamma_{\text{cl}} = 1 + O(\tau)$, both

factors are $1 + O(\tau)$, so $A/A_{\text{cl}} = 1 + O(\tau)$, uniform in m since the connection-coefficient modulus is x -independent and the bulk envelope is uniform on the fixed window W_δ containing the pole.

Phase. $\arg \kappa - \arg \kappa_0 = O(e^{-4\pi^2/\varepsilon})$ is exp-small by (50); the q -Hankel phase correction is the pole shift, which by Lemma 3.11 is $-\frac{\sqrt{2}}{36}\sqrt{\tau} + O(\tau)$, so $\Theta - \Theta_{\text{cl}} = O(\sqrt{\tau})$.

Assembly. The pole is a *zero* of $\cos \Theta_{\text{cl}}$, so the leading classical value is the *sine* branch $E = A_{\text{cl}} \sin \Theta_{\text{cl}}$ (up to the elementary prefactor of §4). Writing $J_\nu^{(3)} = A \cos \Theta$ and expanding about Θ_{cl} ,

$$\frac{R}{E} = \frac{A}{A_{\text{cl}}} \left[\cos(\Theta - \Theta_{\text{cl}}) + \sin(\Theta - \Theta_{\text{cl}}) \cot \Theta_{\text{cl}} \right] - 1.$$

Here $A/A_{\text{cl}} = 1 + O(\tau)$ and $\cos(\Theta - \Theta_{\text{cl}}) - 1 = O(\tau)$; the cross term is $O(\tau)$ since $\sin(\Theta - \Theta_{\text{cl}}) = O(\sqrt{\tau})$ and, the pole sitting at the sine extremum to $O(\sqrt{\tau})$, $\cot \Theta_{\text{cl}} = O(\sqrt{\tau})$ (Lemma 3.11). Hence $R/E = O(\tau)$; with $E = O(\tau^{3/2})$ uniformly, $R = O(\tau^{5/2})$ uniformly in m . $\square$

Remark 5.6 ([CHECKED]). *The amplitude correction is x -independent and $O(\tau)$: $(A/A_{\text{cl}} - 1)/\tau$ evaluated on the q -Bessel at the classical extrema (where the $\sqrt{\tau}$ phase term vanishes) is flat across an $8\times$ range of argument (spread $\approx \tau$), and end-to-end $J_{3/2}^{(3)}/J_{3/2} - 1 = O(\tau)$ with successive-difference ratio $\rightarrow 2$ (`uniformity2.py`, `qbessel_order.py`; these illustrative runs use `nome q`, which scales the coefficient by a fixed factor versus the gate `nome q^2` but leaves the qualitative $O(\tau)$ unchanged). The assembled gate constant $|R|/\tau^{5/2} \rightarrow 3.714$ uniformly to $m = 200$ is computed at the actual gate object (`u5b_gate`). These confirm the bound numerically; the only unverified step analytically is the turning-point normalisation $\gamma/\gamma_{\text{cl}} = 1 + O(\tau)$.*

Status and the lone residual. Two ingredients are proved here from scratch: Lemma 5.1 (the modular/Poisson estimate of the connection-coefficient *modulus*) and Lemma 5.3 (the conserved-Casoratian bulk *envelope*). What this section therefore establishes *unconditionally* is the reduction of the gate bound (47) to a *single* estimate: the single-solution amplitude normalisation $\gamma/\gamma_{\text{cl}} = 1 + O(\tau)$ across the turning point $\psi_n \rightarrow 0$, equivalently the $O(\tau)$ confluent correction to the $x \rightarrow 0$ connection coefficient. This is the same input flagged in Remark 4.17, of the same standing as the lem : cos saddle constant; its leading value (that $J_\nu^{(3)} \rightarrow$ classical Bessel) is Morita's confluent limit [Morita2011], [OldeDaalhuis1994]. Numerically the bound is certain ($|R|/\tau^{5/2} \rightarrow 3.714$, $m \leq 200$); the residual is the *proof* of that one $O(\tau)$ turning-point normalisation, a q -Airy/uniform-Bessel matching estimate not carried out here. The two lemmas reduce the open atom to this single, named, classical-looking estimate.

References

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